

# Draft: the finite excess in transfinite nim multiplication

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July 2026

## Abstract

Conway's field  $\text{On}_2$  makes the ordinals below  $\omega^{\omega^\omega}$  an algebraic closure of  $\mathbb{F}_2$  under nim-arithmetic. Its multiplication is governed by a Kummer tower whose carries are Lenstra's *excess* values: for each odd prime  $p$ , the carry is  $\alpha_p = \kappa_{f(p)} + m_p$ , where the transfinite part  $\kappa_{f(p)}$  has a known closed shape (Lenstra, DiMuro) but the finite correction  $m_p$  does not. This note consolidates the state of the excess problem (docs/OPEN.md Problem 3 of ogdoad): the exact finite-field reformulation of the root test as a multiplicative-order criterion and its structural norm reduction; a complete analysis of the 3-power column via a half-angle splitting in the cyclotomic component fields, certifying  $m_r = 1$  for all primes  $r$  with  $f(r) \in \{3, 9, \dots, 3^6\}$  and forcing  $m_p \geq 4$  unconditionally on the  $f(p) = 2 \cdot 3^k$  column — now sharpened to  $m_p = 4$  *exactly* at every prime current factor tables reach (universally for  $k \leq 6$ ), via a corrected compositum norm  $(\kappa + 4)(\kappa + 5) = \kappa^2 + \kappa + \omega$  that collapses the  $m = 4$  test into the same trinomial field, together with a correction of the formerly cumulative  $D_k$  to the exact current factor  $D'_k$  and an all-level Capelli/resultant formulation, an exact-degree sparse projection, and a dominant-factor sieve; the empirically exact 0/1/4 candidate rule and the proof that the Lenstra component set alone cannot decide the excess; an exact transverse-norm formulation of every singleton-odd row; two embedded maximal-order problems (the Conway Fermat tower on the even side and explicit Gaussian-period primitivity on the 3-power side), with the latter now reduced to primitivity of the irreducible Dickson iterate  $D_{3^k}(X, 1) + 1$  and to an explicit recursive Singer-generator problem in a cubic norm-one torus, proved analytically through the first three prime-order kernel steps, equipped here with an unconditional partition-order bound exponential in the square root of the level and an exact short cyclotomic-period test, and reduced thereafter to character nonvanishing for a norm-coherent cyclotomic unit; a new Fibonacci-polynomial stratification of the Conway–Fermat arm, including universal countermodels proving that degree, trace, and norm cannot force maximal quotient order, and an exact resultant recursion for the remaining distinguished-factor conjecture, now realized as a Kummer symbol at a distinguished point of a fixed supersingular elliptic correspondence, together with an exact full-stratum-to-proper-stratum counterexample proving that one-step resultant induction does not preserve maximality for arbitrary predecessor factors, and a complete trace/norm ancestry criterion whose exact power trace-fibres collapse back to the same fibotomic test; and the fact that generic bounded-translate arguments cannot prove the conjecture, by a finite-window Kummer-cover theorem producing arbitrarily large exact-degree elements whose entire fixed translate window consists of  $p$ -th powers. The three distinguished arms are sharpened further: cubic failure saturates every cyclotomic secant although the natural Kummer cover is rational, while Stickelberger and Hilbert reciprocity are proved to vanish on precisely the simple exceptional eigenspace carrying its local Artin value; the Fermat obstruction is the ramified toric projection of a Miller unit in a three-point generalized Jacobian, whose complete extension calculation now has explicit Rosenlicht/Miller and rational five-torsion coordinates while proving their evident reciprocity collapse circular; and  $D'_k$  is equivalent both to an explicit translated-binomial irreducibility statement over  $\mathbb{F}_{16}$  and to a relative Dickson-composition criterion with a proved degree ceiling, while one bad translate saturates full six- and eight-dimensional additive spaces and the complete semiprimitive Gauss–Jacobi/Fourier table is proved compatible with that failure, now

sharpened to one extremal second Dirichlet coefficient whose functional equation, trace dual, and exact subspace polynomial are all proved compatible with the bad value. Finally, proving the candidate would settle Lenstra’s open boundedness question, while already requiring those two hard maximal-order inputs. Feasibility of the first open row,  $p = 719$ , is assessed: a Frobenius-orbit norm recurrence reduces it to arithmetic in  $\mathbb{F}_{2^{359}}$ , with the cost concentrated in tower-aware Frobenius composition. Each statement carries an explicit claim-level tag.

**Claim levels.** [**proved**] – proved here, machine-checked, or classically verified; [**certified**  $4 \leq k \leq 6$ ] – proved for the stated range, blocked beyond it only by unfactored cofactors; [**consistent**] – no counterexample found, not proved; [**conjectured**] – explicitly open in the literature or posed in this thread; [**open**] – neither confirmed nor refuted, no feasible attack on record.

## 1 Notation, implemented state, external data

### 1.1 Notation

For an odd prime  $p$ :

- $f(p) = \text{ord}_p(2)$ , the multiplicative order of 2 modulo  $p$ ;
- $\kappa_h$  is the Lenstra/DiMuro tower element indexed by  $h$ ;
- $\mathcal{Q}(h)$  is Lenstra’s set of prime-power components appearing in  $\kappa_h$ ;
- the *excess*  $m_p$  is the least finite  $m$  such that  $\kappa_{f(p)} + m$  has no  $p$ -th root in the relevant finite component field;
- the Kummer carry is  $\alpha_p = \kappa_{f(p)} + m_p$ .

For the ordinal tower, a row with  $\mathcal{Q}(f(p)) = \{q\}$  and finite excess  $m$  gives the ordinal sum corresponding to  $\kappa_q + m$ . Addition of numbers is XOR;  $\otimes$  is the nim product.

### 1.2 Implemented tower state

In the Rust tower (`src/scalar/big/ordinal/tower.rs`), as of 2026-06-13:

- the finite excess rows  $m_p$  are now sourced from OEIS A380496 [16], the b-file’s 126 known rows (odd primes  $3 \leq p \leq 709$ ); the first 14 rows reproduce DiMuro Table 1 + the formerly locally verified  $\alpha_{47} = \omega^{(\omega^7)} + 1$ ;
- the operational boundary is  $\alpha_{719}$ , the first OEIS-unknown row: a carry needing  $\alpha_{719}$  or beyond returns **None**;
- the table extends *reach*, not *feasibility*: each  $\alpha_p$  is reconstructed in code from  $\text{ord}_p(2)$ ,  $\mathcal{Q}(f(p))$ , and  $m_p$ , but for large  $p$  the  $\mathcal{Q}$ /finite-subfield reconstruction lives in the degree- $e_p$  component field ( $e_p$  in the millions; see the snapshot below), so only the small- $e_p$  rows are materializable end-to-end today.

### 1.3 External data snapshot (2026-06-09)

- OEIS A380496 [16] has 1417 extended rows: 799 known and 618 unknown; the b-file has 126 initial known rows.
- The first OEIS unknown is row  $n = 127$ , the 127th odd prime  $p = 719$ . For  $p = 719$ :  $f(719) = \text{ord}_{719}(2) = 359$  and  $\mathcal{Q}(359) = \{359\}$ .

- The transfinite-nim-calculator logs [18] record the direct component exponent as  $e_{719} = 1,258,230,380$ , the practical wall for direct exponentiation.

## 2 The order criterion and structural norm reduction

### 2.1 The exact criterion

Throughout,  $E$  denotes the degree of the component field containing  $\beta = \kappa_{f(p)} + m$ , so  $f = f(p) \mid E$  and hence  $p \mid 2^E - 1$ .

**Proposition 1** (power criterion). *[proved]  $\beta \in \mathbb{F}_{2^E}^\times$  has no  $p$ -th root in  $\mathbb{F}_{2^E}$  iff  $\beta^{(2^E-1)/p} \neq 1$ . Hence*

$$m_p = \min\{m \geq 0 : (\kappa_{f(p)} + m)^{(2^E-1)/p} \neq 1\}.$$

*Proof.*  $\mathbb{F}_{2^E}^\times$  is cyclic of order  $2^E - 1$  and  $p \mid 2^E - 1$ ; in a cyclic group of order  $N$ ,  $\beta$  is a  $p$ -th power iff  $\beta^{N/p} = 1$ .  $\square$

**Corollary 1** (order form). *[proved] If  $v_p(2^E - 1) = 1$ , then  $\beta$  has no  $p$ -th root in  $\mathbb{F}_{2^E}$  iff  $p \mid \text{ord}(\beta)$ , so  $m_p$  is the least  $m$  with  $p \mid \text{ord}(\kappa_{f(p)} + m)$ .*

**Remark 1** (Wieferich caveat). In general “no  $p$ -th root” is equivalent to  $v_p(\text{ord } \beta) = v_p(2^E - 1)$ , which collapses to  $p \mid \text{ord}(\beta)$  only when  $v_p(2^E - 1) = 1$ . By lifting the exponent,

$$v_p(2^E - 1) = v_p(2^{f(p)} - 1) + v_p(E/f(p)).$$

Thus the hypothesis can fail either at base-2 Wieferich-type primes with  $p^2 \mid 2^{f(p)} - 1$  or when  $p \mid E/f(p)$ . The only known base-2 Wieferich primes are  $p = 1093$  and  $p = 3511$ , both of which lie inside the extended A380496 range. For those rows, and for any row with  $p \mid E/f(p)$ , the exact test of Proposition 1 (or the full- $p$ -part order condition) must be used. The certifications of Section 3 compute orders exactly, with squarefreeness checks, and are unaffected. The research-note predecessor of this document stated only the order form; the power criterion is the correct general statement.

### 2.2 Norm reduction

**Proposition 2** (norm reduction). *[proved] For  $\beta \in \mathbb{F}_{2^E}^\times$  with  $f = f(p) \mid E$ ,*

$$\beta^{(2^E-1)/p} = N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^f}}(\beta)^{(2^f-1)/p}.$$

*Proof.*  $N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^f}}(\beta) = \beta^{(2^E-1)/(2^f-1)}$ , and  $((2^E - 1)/(2^f - 1)) \cdot ((2^f - 1)/p) = (2^E - 1)/p$ .  $\square$

So the  $p = 719$  test reduces to: (1) compute  $N = N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^{359}}}(\kappa_{359} + 1)$  in  $\mathbb{F}_{2^{359}}$ ; (2) test whether  $719 \mid \text{ord}(N)$  in  $\mathbb{F}_{2^{359}}^\times$ , a field of degree 359 in which  $719 \mid 2^{359} - 1$ .

**Proposition 3** (Galois-conjugacy invariance). *[proved] The order of  $\kappa_{f(p)} + m$  is constant on its Frobenius orbit in  $\mathbb{F}_{2^E}$ , and the norm  $N = \prod_{i=0}^{E/f-1} \text{Frob}^{if}(\kappa_f + m)$  is determined structurally by the tower equations without materialising the full orbit.*

**Proposition 4** (tower pinning). *[proved] The tower equations pin  $\kappa_{359}$  up to Frobenius conjugacy: each composed minimal polynomial along the component chain  $359 \rightarrow 179 \rightarrow 89 \rightarrow 11 \rightarrow 5 \rightarrow \text{finite}$  is irreducible (the non- $q$ -th-power property defining the lower excess guarantees this), so  $\kappa_{359}$  is a well-defined conjugacy class and  $N$  is a well-defined element of  $\mathbb{F}_{2^{359}}$ .*

### 2.3 Exact support and the general singleton-odd obstruction

For  $h > 1$ , write

$$\Pi_h := \prod_{\substack{\ell \text{ odd prime} \\ \text{ord}_\ell(2)=h}} \ell^{v_\ell(2^h-1)}.$$

This is the full prime-power part of  $2^h - 1$  supported at primes whose order of 2 is exactly  $h$ ; using the full valuations, rather than only the radical, incorporates the Wieferich cases.

**Proposition 5** (exact-order support). *[proved] Let  $x = \kappa_h$ , let  $E = d(x)$ , and put*

$$T_h = N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^h}}(x).$$

*Then  $m_p = 0$  for every odd prime  $p$  with  $f(p) = h$  if and only if*

$$\Pi_h \mid \text{ord}(T_h).$$

*For an individual such  $p$ ,  $m_p = 0$  iff  $v_p(\text{ord } T_h) = v_p(2^h - 1)$ .*

*Proof.* The definition of  $\kappa_h$  gives  $h \mid E$ . Proposition 2 with  $\beta = x$  identifies the  $p$ -power test in  $\mathbb{F}_{2^E}$  with the same test on  $T_h$  in  $\mathbb{F}_{2^h}$ . In the cyclic group  $\mathbb{F}_{2^h}^\times$ , failure to be a  $p$ -th power is exactly the presence of the full  $p$ -primary part of the group order in  $\text{ord}(T_h)$ . Taking the product over the distinct eligible primes proves the claim. No hypothesis on  $p \nmid E/h$  is hidden here: if  $c = v_p((2^E - 1)/(2^h - 1))$ , the norm exponent gives  $v_p(\text{ord } T_h) = \max\{0, v_p(\text{ord } x) - c\}$ , so equality with  $v_p(2^h - 1)$  is equivalent to  $v_p(\text{ord } x) = v_p(2^E - 1)$ .  $\square$

The singleton-odd case admits a sharper field square. Let  $r$  be an odd prime,  $q = r^a$ , and set  $b_r = d(\alpha_r)$ , where  $\kappa_q^q = \alpha_r$ . Since  $\alpha_r < \kappa_r$ , Lenstra's field description gives  $r \nmid b_r$ , while the successive Kummer equations have degree  $r$  [11, Proposition 1.8]; hence

$$d(\kappa_q) = qb_r, \quad \gcd(q, b_r) = 1.$$

**Proposition 6** (singleton transverse-norm criterion). *[proved] Suppose  $\mathcal{Q}(h) = \{q\}$  with  $q = r^a$  odd. Then  $h = qs$  for a divisor  $s \mid b_r$ . In the field square*

$$F = \mathbb{F}_{2^{qb_r}}, \quad B = \mathbb{F}_{2^{b_r}}, \quad K = \mathbb{F}_{2^{qs}}, \quad C = \mathbb{F}_{2^s},$$

*one has  $B \cap K = C$  and  $F = BK$ . Put  $x = \kappa_q$  and*

$$\Theta_{q,s} := N_{F/K}(x+1) \in K^\times.$$

*For every prime  $p$  with  $f(p) = h$ :*

(i)  *$x$  is a  $p$ -th power in  $F$ , so  $m_p \geq 1$ ;*

(ii)  *$m_p = 1$  iff*

$$v_p(\text{ord } \Theta_{q,s}) = v_p(2^{qs} - 1);$$

(iii) *the easy Kummer norm is*

$$N_{F/B}(x+1) = 1 + \alpha_r,$$

*but  $p \nmid |B^\times|$ , so this norm cannot see the required  $p$ -primary obstruction. The norm to  $K$  is genuinely transverse.*

*Proof.* Lenstra's component theorem gives  $\kappa_h = \sum_{t \in Q(h)} \kappa_t$  and, for each component  $t$ ,  $t \mid h$  and  $\gcd(t, h/t) = 1$  [11, Corollary 2.2]. Thus the singleton assumption identifies  $\kappa_h = \kappa_q$ . Since  $h \mid d(\kappa_h) = d(\kappa_q) = qb_r$ , one has  $h = qs$  with  $s \mid b_r$ . The intersection and compositum statements follow from the gcd/lcm rules for finite subfields and  $\gcd(q, b_r) = 1$ .

Since  $p > h \geq r$ , one has  $p \nmid q$ . Also  $f(p) = qs \nmid b_r$ , so  $p \nmid 2^{b_r} - 1$ . The relation  $x^q = \alpha_r \in B$  therefore gives  $p \nmid \text{ord}(x)$ , proving (i), including the full  $p$ -primary statement. Part (ii) is Proposition 2 applied to  $x + 1$ . Finally,  $X^q + \alpha_r$  is the minimal polynomial of  $x$  over  $B$ , and evaluating it at 1 gives (iii); the invisibility follows again from  $p \nmid 2^{b_r} - 1$ .  $\square$

Thus the singleton-one arm is not an unspecified root search. It is the uniform assertion that the explicitly defined transverse norm  $\Theta_{q,s}$  contains the full  $p$ -primary part for every prime  $p$  of exact order  $qs$ . The  $C_k$  norms below are the special case  $q = 3^k$ ,  $b_r = 2$ ,  $s = 1$ ; the  $m_{359}$  obstruction is the same square at  $q = 179$ ,  $b_r = 19580$ ,  $s = 1$ . Lenstra's use of Carlitz's primitive-root distribution theorem gives an effective bound at which *some* finite translate works [3, 11]; it does not evaluate this fixed first transverse norm.

There is a sharp reason that this effective argument cannot be replaced, for arbitrary finite-field elements, by any fixed initial translate window.

**Proposition 7** (finite-window Kummer no-go). *[proved] Let  $p$  be an odd prime and let*

$$A = \{b_1, \dots, b_r\} \subset \overline{\mathbb{F}_2}$$

*be a finite set of distinct elements. Choose  $k_0 = \mathbb{F}_{2^d}$  containing both  $A$  and  $\mu_p$ . There are arbitrarily large integers  $n$  and elements  $x_n \in \mathbb{F}_{2^{dn}}$  of exact degree  $dn$  over  $\mathbb{F}_2$  such that*

$$x_n + b_i \in (\mathbb{F}_{2^{dn}}^\times)^p \quad (1 \leq i \leq r).$$

*Consequently no absolute truncation of Lenstra's translate-generation lemma, nor any argument using only full degree, an order- $p$  character, and a fixed affine translate set, can prove an absolute bound for arbitrary elements. This does not decide the distinguished elements  $\kappa_{f(p)}$ : any uniform excess theorem must use their special Conway-tower identities.*

*Proof.* Over  $k_0(X)$  form the Kummer extension

$$\mathcal{K} = k_0(X)(Y_1, \dots, Y_r), \quad Y_i^p = X + b_i.$$

The classes of the  $X + b_i$  are independent in  $\overline{k_0(X)}^\times / \overline{k_0(X)}^{\times p}$ : in a relation  $\prod_i (X + b_i)^{e_i} = G^p$ ,  $0 \leq e_i < p$ , the valuation at  $X = -b_i$  forces  $e_i = 0$ . Thus the smooth projective curve  $C$  with function field  $\mathcal{K}$  is geometrically integral and

$$[\mathcal{K} : k_0(X)] = p^r, \quad \text{Gal}(\mathcal{K}/k_0(X)) \cong (\mathbb{Z}/p\mathbb{Z})^r.$$

The cover  $C \rightarrow \mathbb{P}_X^1$  is ramified at the  $r$  points  $-b_i$  and at infinity, each with inertia order  $p$ . Hence Riemann–Hurwitz gives

$$2g(C) - 2 = -2p^r + (r+1)p^r \left(1 - \frac{1}{p}\right).$$

Put  $Q = 2^d$ . Hasse–Weil, followed by deletion of the finitely many branch points, supplies

$$Q^n + O_{p,r}(Q^{n/2})$$

unramified  $\mathbb{F}_{Q^n}$ -points. Since the map to the  $X$ -line has degree  $p^r$ , the number of resulting  $X$ -coordinates is at least

$$\frac{2^{dn} - O_{p,r}(2^{dn/2})}{p^r}.$$

The union of all proper subfields of  $\mathbb{F}_{2^{dn}}$  has at most

$$\sum_{\substack{e|dn \\ e < dn}} 2^e \leq dn \cdot 2^{dn/2}$$

elements. For all sufficiently large  $n$  the former lower bound is larger. Choose an unramified  $X$ -coordinate  $x_n$  outside that union. It has exact degree  $dn$ , and its lifted coordinates satisfy  $x_n + b_i = Y_i^p \neq 0$  for every  $i$ , as required.  $\square$

## 2.4 Feasibility for $p = 719$

The component field degree is  $E = 2 \cdot 2 \cdot 5 \cdot 11 \cdot 89 \cdot 179 \cdot 359 = 1,258,230,380$ ; the norm requires  $E/f = 3,504,820$  Frobenius steps.

**[consistent]** The norm can in principle be computed by the binary-splitting Frobenius recurrence  $N_{2k} = N_k \cdot \text{Frob}^{f^k}(N_k)$  (iterated-Frobenius technique, von zur Gathen–Shoup [20], Kaltofen–Shoup [10]), requiring about  $\log_2(3,504,820) \approx 22$  squarings-with-Frobenius-composition at degree  $E$  over  $\mathbb{F}_2$ . Each Frobenius composition is a modular composition in  $\mathbb{F}_2[x]$  modulo a degree- $E$  polynomial; with dense polynomial arithmetic at  $E \approx 1.26 \times 10^9$  this is not locally feasible.

**[conjectured]** (as a cost model, not a theorem) Kummer-tower-aware Frobenius – standard-lattice arithmetic for compatibly embedded finite fields in the style of De Feo–Randriam–Rousseau [6], where  $q$ -power Frobenius acts near-monomially per tower level – is the likely 10–100× lever: the sparse factored structure  $E = 2 \cdot 2 \cdot 5 \cdot 11 \cdot 89 \cdot 179 \cdot 359$  admits a stepwise Frobenius representation in which each level contributes a small-degree modular composition. A *measured* cost model, not a run, is the correct first deliverable.

**[open]** Whether the Kummer-tower Frobenius model makes the  $m_{719}$  computation feasible in wall-clock hours rather than months on available hardware (a single consumer GPU; GF(2) arithmetic is XOR-sliced and GPU-friendly).

**[proved]** (2026-06-14 local rehearsal.) The first two dependency rows for the  $p = 719$  norm path are locally certified by `experiments/ordinal_excess_probe.py`:  $m_{89} = 1$  in the  $E = 220$  component field, and  $m_{179} = 1$  in the  $E = 19,580$  component field via the fixed-base power path (`--deep`, about one minute locally). The remaining local rehearsal row is  $m_{359} = 1$ , with  $E = 3,504,820$ . This row is already source-pinned by the imported A380496 table ( $359 \leq 709$ ); what remains open here is an independent certificate on the same code path needed for the genuinely unknown  $p = 719$  row.

## 2.5 The $m_{359}$ rehearsal obstruction

Set

$$d = 19,580, \quad E = 179d = 3,504,820, \quad B = \mathbb{F}_{2^d}, \quad F = \mathbb{F}_{2^E}, \quad L = \mathbb{F}_{2^{179}}.$$

For the  $m_{359} = 1$  rehearsal,  $f(359) = 179$ ,  $\mathcal{Q}(f(359)) = \mathcal{Q}(179) = \{179\}$ , and the candidate element is  $\beta = \kappa_{179} + 1$ . To represent  $\kappa_{179}$ , the tower uses the lower prime row  $f(179) = 178$ ,  $\mathcal{Q}(178) = \{89\}$ ,  $m_{179} = 1$ . Thus in the tower model  $F = B(x)$  with  $x = \kappa_{179}$  and

$$x^{179} = \kappa_{89} + 1 \in B,$$

because the lower row  $m_{179} = 1$  has already been certified.

**[proved]** The cheap Kummer norm down the top tower step is not the 359-test norm. Indeed,

$$N_{F/B}(x+1) = 1 + (\kappa_{89} + 1) = \kappa_{89}$$

(evaluate the monic polynomial  $T^{179} + (\kappa_{89} + 1)$  at  $T = 1$ , in characteristic 2). But  $359 \nmid |B^\times| = 2^{19580} - 1$ , since  $\text{ord}_{359}(2) = 179$  and  $179 \nmid 19580$ . Thus the structurally cheap norm lands in exactly the field where the 359-primary obstruction is invisible.

**[proved]** The norm forced by Proposition 2 is the transverse one

$$N_{F/L}(\beta) = \prod_{i=0}^{19579} \text{Frob}^{179i}(\beta) \in L.$$

Here  $\gcd(179, 19580) = 1$ , so  $B \cap L = \mathbb{F}_2$  and  $F = BL$ . The Frobenius  $\text{Frob}^{179}$  preserves  $B$  as a set but acts nontrivially on it; it is not the relative Kummer action  $x \mapsto \zeta x$  over  $B$ . Consequently the known lower certificates for  $m_{89}$  and  $m_{179}$  do not propagate to  $m_{359}$  by taking the easy top-step norm.

**[proved]** The Wieferich caveat is absent at the next two pressure points: direct modular checks give  $2^{179} \not\equiv 1 \pmod{359^2}$  and  $2^{359} \not\equiv 1 \pmod{719^2}$ . The order form of Corollary 1 is therefore equivalent to the full power criterion for the  $m_{359}$  rehearsal and the proposed  $m_{719}$  test.

**[consistent]** (2026-06-16 diagnostic.) Re-running the maintained probe confirmed the quick certificate table and the `--deep`  $m_{179} = 1$  certificate. A side norm experiment then computed the correct transverse norm in the current term basis for the feasible analogues: for  $p = 89$  the norm to  $\mathbb{F}_{2^{11}}$  has support 111/220, and for  $p = 179$  the norm to  $\mathbb{F}_{2^{178}}$  has support 9691/19580. Thus the target-subfield element is essentially half-dense in this representation; using it as the new fixed base makes the final root-test exponent slower than the direct fixed-base certificate. This is a representation-level diagnostic, not a theorem that no sparse description exists.

**[open]** The current pure-Python term algebra has no feasible route to a local  $m_{359}$  certificate. The direct fixed-base path would have to precompute multiplication by  $\beta$  for  $E = 3,504,820$  basis terms and then run an  $E$ -bit exponent. The norm path lowers the mathematical problem to a degree-179 target field, but computing that norm still requires the 19580-term transverse Frobenius orbit through  $F_{2^{3504820}}$ . A practical certificate now appears to require either (i) dense/sliced GF(2) arithmetic such as `gf2x/NTL`, or (ii) a tower-aware Frobenius representation that makes the transverse orbit cheap. This is the obstruction hit in the present pass.

**Remark 2** (shared abstraction). The structural primitive needed here –  $N_{E/K}(\beta) = \prod_i \text{Frob}^i(\beta)$  computed by a Frobenius-orbit recurrence – is the same primitive that the cyclic-algebra Brauer-invariant bridge (Bridge K, `docs/TBD.md`) requires for reduced norms over a Frobenius-generated cyclic extension. The existing `FieldExtension::relative_norm` does not apply to the term-algebra/transfinite setting, but the shape is identical; factoring out a reusable `relative_norm_over_frobenius_orbit` is an engineering opportunity. This is not a claim that the bounded `Fpn` norm certifies  $m_{719}$ .

### 3 The $3^k$ family

Write  $\zeta := \kappa_{3^k}$  and  $h := 3^k$  throughout this section.

### 3.1 Structure

**Proposition 8** (cyclotomic recognition). *[proved]  $\zeta^{3^k} = \kappa_2$ , which has order 3; hence  $\zeta$  is a primitive  $3^{k+1}$ -st root of unity. Since 2 is a primitive root modulo  $3^{k+1}$ , the cyclotomic polynomial  $\Phi_{3^{k+1}}(x) = x^{2^h} + x^h + 1$  is irreducible over  $\mathbb{F}_2$ , the component field is  $\mathbb{F}_2(\zeta) = \mathbb{F}_{2^{2h}}$ , and it carries the index-2 subfield  $L = \mathbb{F}_{2^h}$ .*

**Proposition 9** (half-angle splitting). *[proved] With  $s = (3^{k+1} + 1)/2$  (the inverse of 2 modulo  $3^{k+1}$ ),*

$$\kappa_{3^k} + 1 = \zeta^s (\zeta^s + \zeta^{-s}),$$

*where  $\zeta^s$  lies in the norm-one circle  $U$  (of order exactly  $3^{k+1}$  within the subgroup of order  $2^h + 1$ ) and  $(\zeta^s + \zeta^{-s})^2 = \zeta + \zeta^{-1} =: \gamma_k \in L^\times$ . Since  $\mathbb{F}_{2^{2h}}^\times = L^\times \times U$  with coprime orders,*

$$\text{ord}(\kappa_{3^k} + 1) = 3^{k+1} \cdot \text{ord}(\gamma_k), \quad \text{ord}(\gamma_k) \mid 2^{3^k} - 1.$$

*Machine-checked corollaries:  $(\kappa + 1)^{2^h - 1} = \zeta^{-1}$  and  $N_{F/L}(\kappa + 1) = \gamma_k$ .*

**Proposition 10** (translates  $m = 2, 3$ ). *[proved]  $\kappa_{3^k} + 2$  and  $\kappa_{3^k} + 3$  split as in Proposition 9 with  $\gamma_k$  replaced by a Galois conjugate. Therefore  $\text{ord}(\kappa_{3^k} + m) = 3^{k+1} \cdot \text{ord}(\gamma_k)$  for  $m = 1, 2, 3$ , and  $\text{ord}(\kappa_{3^k}) = 3^{k+1}$ .*

**Proposition 11** (the  $2 \cdot 3^k$  exception, unconditionally). *[proved] Any prime  $p$  with  $f(p) = 2 \cdot 3^k$  satisfies  $p \mid 2^{3^k} + 1$ , hence  $p$  divides neither  $3^{k+1}$  nor  $2^{3^k} - 1$ , hence  $p$  never divides  $\text{ord}(\kappa_{3^k} + m)$  for  $m \in \{0, 1, 2, 3\}$ . By the order criterion,  $m_p \geq 4$ . The full primitivity conjecture  $C_k$  below is not needed; the splitting alone forces the bound. The argument applies beyond the current tables, e.g.  $p = 87211$ ,  $f = 54$ .*

**Proposition 12** (the 3-power column). *[proved] For a prime  $r$  with  $f(r) = 3^k$ , the values  $m \in \{0, 2, 3\}$  are impossible:  $m_r = 0$  would require  $r \mid \text{ord}(\kappa_{3^k}) = 3^{k+1}$ , a power of 3, while  $r \equiv 1 \pmod{3^k}$  is a prime distinct from 3; and  $m = 2, 3$  share  $m = 1$ 's order class by Proposition 10. Any failure of  $C_k$  therefore gives  $m_r \geq 4$  directly.*

**Proposition 13** (norm tower). *[proved]  $\gamma_{k-1} = \gamma_k^3 + \gamma_k = N_{L_k/L_{k-1}}(\gamma_k)$  (minimal polynomial  $X^3 + X + \gamma_{k-1}$ ), so  $\text{ord}(\gamma_{k-1}) \mid \text{ord}(\gamma_k)$ . With lifting-the-exponent ( $v_r(2^{3^k} - 1) = v_r(2^{f(r)} - 1)$ ) for old primes  $r \neq 3$ , full-order parts propagate upward, and conjecture  $C_k$  reduces to  $C_{k-1}$  plus primitivity at the new primes  $r \mid \Phi_{3^k}(2)$ .*

**Proposition 14** (the norm-kernel seam). *[proved] Put  $q = 2^{3^{k-1}}$ , so  $L_k = \mathbb{F}_{q^3}$  and  $L_{k-1} = \mathbb{F}_q$ . There is a canonical direct product*

$$L_k^\times = L_{k-1}^\times \times U_k, \quad |U_k| = q^2 + q + 1 = \Phi_{3^k}(2),$$

*where  $U_k = \ker N_{L_k/L_{k-1}}$ . If  $\gamma_k = a_k b_k$  in this product, then*

$$\gamma_{k-1} = N(\gamma_k) = a_k^3,$$

*and cubing is an automorphism of  $L_{k-1}^\times$ . Consequently*

$$C_k \iff C_{k-1} \text{ and } b_k \text{ generates } U_k.$$

*In particular, the norm recursion determines the old component  $a_k$  exactly but supplies no order information about the new component  $b_k$ .*



*Proof.* The two displayed factor orders are coprime:

$$\gcd(q-1, q^2+q+1) = \gcd(q-1, 3) = 1,$$

because  $q = 2^{\text{odd}} \equiv 2 \pmod{3}$ . The cyclic group  $L_k^\times$  therefore splits as asserted. The relative norm is  $z \mapsto z^{1+q+q^2}$ ; it kills  $U_k$  and restricts on  $L_{k-1}^\times$  to  $a \mapsto a^3$ , which is bijective because  $3 \nmid q-1$ . The final equivalence follows from coprimality of the two factor orders.  $\square$

**Proposition 15** (primitive Dickson iterate). *[proved] Let  $D_n(X, 1)$  be the Dickson polynomial characterized by  $D_n(u + u^{-1}, 1) = u^n + u^{-n}$ , and put  $f(X) = X^3 + X$ . Then*

$$P_k(X) := D_{3^k}(X, 1) + 1 = f^{\circ k}(X) + 1$$

*is the minimal polynomial of  $\gamma_k$  over  $\mathbb{F}_2$ . In particular it is irreducible of degree  $3^k$ , and*

$$C_k \iff P_k(X) \text{ is a primitive polynomial over } \mathbb{F}_2.$$

*The polynomials also obey the coefficient recursion*

$$P_{k+1} = P_k^3 + P_k^2 + 1, \quad P_1 = X^3 + X + 1.$$

*Proof.* The Dickson composition law gives  $D_{3^k} = D_3^{\circ k} = f^{\circ k}$  in characteristic two. Since  $\zeta^{3^k}$  is a primitive cube root of unity,

$$D_{3^k}(\gamma_k, 1) = \zeta^{3^k} + \zeta^{-3^k} = 1.$$

On the other hand, the Frobenius orbit of  $\zeta + \zeta^{-1}$  has length  $3^k$ : the least  $j > 0$  with  $2^j \equiv \pm 1 \pmod{3^{k+1}}$  is  $3^k$ . Thus the displayed monic polynomial, which has degree  $3^k$ , is the minimal polynomial. Its root is primitive precisely when the polynomial is primitive. Finally, writing  $D_{3^k} = P_k + 1$  and using  $D_3(Y) = Y^3 + Y$  gives  $P_{k+1} = (P_k + 1)^3 + (P_k + 1) + 1 = P_k^3 + P_k^2 + 1$ .  $\square$

**Proposition 16** (shifted cyclotomic-unit form). *[proved] Put  $x_k = 1 + \zeta \in \mathbb{F}_{2^h}^\times$ , where  $h = 3^k$  and  $N = 3^{k+1}$ . Then*

$$x_k^{2^h-1} = \zeta^{-1}, \quad x_k^{2^h+1} = \gamma_k, \quad \text{ord}(x_k) = N \text{ ord}(\gamma_k).$$

*Consequently*

$$C_k \iff \text{ord}(1 + \zeta) = 3^{k+1}(2^{3^k} - 1).$$

*Thus the cubic arm is equivalently maximal order for the reduction at 2 of the cyclotomic unit  $1 + \zeta_{3^{k+1}}$ .*

*Proof.* Because  $2^h \equiv -1 \pmod{N}$ ,  $x_k^{2^h} = 1 + \zeta^{-1} = \zeta^{-1}x_k$ . Dividing and multiplying by  $x_k$  gives the first two identities. The factors  $2^h - 1$  and  $2^h + 1$  are coprime. In the corresponding direct-product decomposition of  $\mathbb{F}_{2^h}^\times$ , the first displayed power has exact order  $N$  and the second has order  $\text{ord}(\gamma_k)$ , so the order of  $x_k$  is their product.  $\square$

**Proposition 17** (norm-coherent cyclotomic lift). *[proved] Let  $\xi_k = \zeta_{3^{k+1}}$  in characteristic zero and  $u_k = 1 + \xi_k$ . Then 2 is inert in  $\mathbb{Q}(\xi_k)$  and*

$$N_{\mathbb{Q}(\xi_k)/\mathbb{Q}(\xi_{k-1})}(u_k) = u_{k-1}.$$

*The norm to the maximal real subfield is*

$$N_{\mathbb{Q}(\xi_k)/\mathbb{Q}(\xi_k + \xi_k^{-1})}(u_k) = 2 + \xi_k + \xi_k^{-1},$$

whose reduction modulo the unique prime above 2 is  $\gamma_k$ . Thus Proposition 16 is the residue-field shadow of a norm-coherent cyclotomic-unit system in the cyclotomic  $\mathbb{Z}_3$ -tower. Conditional on  $C_{k-1}$ ,  $C_k$  is exactly the assertion that, for every prime  $\ell \mid \Phi_{3^k}(2)$ , the reduction of  $u_k$  is not an  $\ell$ -th power.

*Proof.* The residue degree of 2 in  $\mathbb{Q}(\xi_k)$  is  $\text{ord}_{3^{k+1}}(2) = 2 \cdot 3^k = \varphi(3^{k+1})$ , proving inertness. The three relative conjugates of  $\xi_k$  are  $\xi_k, \xi_k\omega$ , and  $\xi_k\omega^2$ , so

$$\prod_{j=0}^2 (1 + \xi_k\omega^j) = 1 + \xi_k^3 = 1 + \xi_{k-1}.$$

Multiplication by the complex-conjugate factor gives the real norm. The final assertion follows from Proposition 16: in a cyclic group an element has the full  $\ell$ -primary part of the group order precisely when it is not an  $\ell$ -th power.  $\square$

**Proposition 18** (circular-unit and ray-class form). *[proved] Let*

$$F_k = \mathbb{Q}(\xi_k + \xi_k^{-1}), \quad \lambda_k = (1 - \xi_k)(1 - \xi_k^{-1}),$$

let  $\mathfrak{p}_k = (2)$  be the unique prime of  $F_k$  above 2, and let  $\mathcal{C}_k \subset \mathcal{O}_{F_k}^\times$  be the real circular-unit group. If  $\sigma(\xi_k) = \xi_k^2$ , then

$$c_k := \frac{\sigma(\lambda_k)}{\lambda_k} = (1 + \xi_k)(1 + \xi_k^{-1}) = 2 + \xi_k + \xi_k^{-1}$$

is a circular unit, its Galois conjugates generate a subgroup of 2-power index in  $\mathcal{C}_k$ , and its reduction modulo  $\mathfrak{p}_k$  is  $\gamma_k$ . Since the residue-field unit group has odd order, that 2-power index disappears on reduction. Consequently

$$\overline{\mathcal{C}_k} = \langle \gamma_k \rangle \subset (\mathcal{O}_{F_k}/\mathfrak{p}_k)^\times \simeq \mathbb{F}_{2^{3^k}}^\times, \quad (\clubsuit)$$

and hence

$$[\mathbb{F}_{2^{3^k}}^\times : \langle \gamma_k \rangle] = [(\mathcal{O}_{F_k}/\mathfrak{p}_k)^\times : \overline{\mathcal{C}_k}]. \quad (\spadesuit)$$

In particular,  $C_k$  is exactly surjectivity modulo 2 of the real circular units of conductor  $3^{k+1}$ .

More precisely, put  $E_k = \mathcal{O}_{F_k}^\times$  and let  $\text{Cl}_{\mathfrak{p}_k}(F_k)$  be the ray class group of finite modulus  $\mathfrak{p}_k$ . The ray class exact sequence gives

$$[\mathbb{F}_{2^{3^k}}^\times : \langle \gamma_k \rangle] = \frac{|\text{Cl}_{\mathfrak{p}_k}(F_k)|}{|\text{Cl}(F_k)|} [\overline{E_k} : \overline{\mathcal{C}_k}], \quad (\heartsuit)$$

where  $[\overline{E_k} : \overline{\mathcal{C}_k}] \mid [E_k : \mathcal{C}_k]$ . The circular-unit index theorem identifies the odd part of the latter index with the odd part of  $|\text{Cl}(F_k)|$ . Thus, for an odd prime  $\ell \nmid |\text{Cl}(F_k)|$ ,

$$v_\ell([\mathbb{F}_{2^{3^k}}^\times : \langle \gamma_k \rangle]) = v_\ell\left(\frac{|\text{Cl}_{\mathfrak{p}_k}(F_k)|}{|\text{Cl}(F_k)|}\right). \quad (\diamondsuit)$$

Equation  $(\heartsuit)$  names the obstruction but does not reduce its difficulty: the ray class group is defined through this residue-unit quotient, and the class-number exclusion in  $(\diamondsuit)$  is itself not known uniformly in  $k$ . Finally, with  $h = 3^k$ ,  $q = 2^{h/3}$ , and  $\tau = \sigma^{h/3}$ , the relative circular unit

$$\varepsilon_k = \frac{\tau(c_k)}{c_k}, \quad N_{F_k/F_{k-1}}(\varepsilon_k) = 1,$$

reduces to  $\eta_k$ . Thus the exact character obstruction of Proposition 24 is the new relative part of the same ray-class obstruction.

*Proof.* The class of 2 generates  $(\mathbb{Z}/3^{k+1}\mathbb{Z})^\times/\{\pm 1\}$ , so  $\sigma$  generates  $\text{Gal}(F_k/\mathbb{Q})$ . The squares of the standard real circular units are generated, up to  $-1$ , by the quotients  $\sigma^j(\lambda_k)/\lambda_k$ . Telescoping gives

$$\frac{\sigma^j(\lambda_k)}{\lambda_k} = \prod_{i=0}^{j-1} \sigma^i(c_k),$$

so the conjugates of  $c_k$  generate this square subgroup. Squaring is an automorphism of the odd-order residue-field unit group, so the full circular-unit group and this subgroup have the same reduction image. Reduction commutes with  $\sigma$ , which becomes Frobenius at the inert prime 2, and

$$\overline{c_k} = (1 + \zeta)(1 + \zeta^{-1}) = \zeta + \zeta^{-1} = \gamma_k.$$

The reductions of the conjugates are therefore  $\gamma_k, \gamma_k^2, \gamma_k^{2^2}, \dots$ , and they generate the single multiplicative subgroup  $\langle \gamma_k \rangle$ . This proves  $(\clubsuit)$  and  $(\spadesuit)$ .

The exact sequence

$$E_k \longrightarrow (\mathcal{O}_{F_k}/\mathfrak{p}_k)^\times \longrightarrow \text{Cl}_{\mathfrak{p}_k}(F_k) \longrightarrow \text{Cl}(F_k) \longrightarrow 1$$

proves  $(\heartsuit)$ . The divisibility of reduction indices is immediate from  $\mathcal{C}_k \subset E_k$ , and the circular-unit index theorem gives  $(\diamondsuit)$ ; see Washington [22, Ch. 8]. Finally  $\tau$  generates  $\text{Gal}(F_k/F_{k-1})$ , so the norm of its coboundary  $\tau(c_k)/c_k$  telescopes to 1. Modulo 2 it is  $\gamma_k^q/\gamma_k = \eta_k$ .  $\square$

**Proposition 19** (exceptional eigenspace and Hilbert-reciprocity no-go). *[proved] Put  $n = 3^k$ ,  $m = 3n$ , let*

$$F = \mathbb{Q}(\xi_m + \xi_m^{-1}), \quad \mathfrak{p} = (2),$$

*and write  $\Delta = \text{Gal}(F/\mathbb{Q}) = \langle \sigma \rangle$ , where  $\sigma(\xi_m) = \xi_m^2$ . Let  $\ell \mid \Phi_n(2)$ , and set*

$$V_\ell = (\mathcal{O}_F/\mathfrak{p})^\times / ((\mathcal{O}_F/\mathfrak{p})^\times)^\ell.$$

*Then  $V_\ell$  is one-dimensional over  $\mathbb{F}_\ell$ , and  $\sigma$  acts on it as the scalar 2. Let*

$$\text{ev}_2 : \mathbb{F}_\ell[\Delta] \longrightarrow \mathbb{F}_\ell, \quad \sum a_j \sigma^j \longmapsto \sum a_j 2^j.$$

*The following reciprocity operators vanish on  $V_\ell$ :*

$$\text{ev}_2(1 + \sigma + \dots + \sigma^{n-1}) = 0, \quad \text{ev}_2(2 - \sigma) = 0. \quad (\text{ER}_1)$$

*If  $q = 2^{n/3}$  and  $\tau = \sigma^{n/3}$ , then*

$$\text{ev}_2(1 + \tau + \tau^2) = 1 + q + q^2 = 0, \quad \text{ev}_2(\tau - 1) = q - 1 \neq 0. \quad (\text{ER}_2)$$

*Moreover the plus projection of the classical Stickelberger element of  $\mathbb{Q}(\xi_m)$  acts as zero on  $V_\ell$ . Its zero is simple: for*

$$\nu_n(X) = 1 + X + \dots + X^{n-1}$$

*one has*

$$\nu_n(2) = 0, \quad \nu'_n(2) = n2^{n-1} \not\equiv 0 \pmod{\ell}. \quad (\text{ER}_3)$$

*Let*

$$c = (1 + \xi_m)(1 + \xi_m^{-1}), \quad \gamma = \bar{c} \in \mathbb{F}_{2^n}^\times, \quad \mathcal{A}_{k,\ell} = \gamma^{(2^n-1)/\ell} \in \mu_\ell. \quad (\text{ER}_4)$$

Then cubic failure at  $\ell$  is exactly  $\mathcal{A}_{k,\ell} = 1$ . Although  $\mathcal{A}_{k,\ell}$  is a local Kummer/Artin symbol, global Hilbert reciprocity imposes no condition on this one distinguished value. More precisely, put  $E = F(\mu_\ell)$ . The prime  $\mathfrak{p}$  splits into  $\ell - 1$  primes in  $E/F$ . If  $\mathfrak{P}_0 \mid \mathfrak{p}$  is chosen so that

$$(c, 2)_{\mathfrak{P}_0, \ell} = \mathcal{A}_{k, \ell}$$

for the reciprocity convention sending the uniformizer 2 to arithmetic Frobenius, then the  $\ell - 1$  symbols above two are

$$(c, 2)_{a\mathfrak{P}_0, \ell} = \mathcal{A}_{k, \ell}^a, \quad a \in \mathbb{F}_\ell^\times. \quad (\text{ER}_5)$$

Every local symbol away from two is one, and Hilbert reciprocity reduces to

$$\prod_{a \in \mathbb{F}_\ell^\times} \mathcal{A}_{k, \ell}^a = \mathcal{A}_{k, \ell}^{\ell(\ell-1)/2} = 1, \quad (\text{ER}_6)$$

which holds for every possible value of  $\mathcal{A}_{k, \ell}$ . Thus norm coherence, the classical Stickelberger ideal, and global Kummer reciprocity cannot distinguish failure from nonfailure. A completion must compute a first-order unit regulator in the simple exceptional eigenspace, rather than another ideal factorization or product formula.

*Proof.* The standard cyclotomic-divisor criterion gives  $\text{ord}_\ell(2) = n$ . Since  $(\mathcal{O}_F/\mathfrak{p})^\times \simeq \mathbb{F}_{2^n}^\times$ , its quotient by  $\ell$ -th powers has order  $\ell$ , and Frobenius acts by  $[x] \mapsto [x^2] = 2[x]$ . This proves the module assertion. Equations (ER<sub>1</sub>) follow from  $2^n = 1 \pmod{\ell}$ . The element  $q = 2^{n/3}$  has exact order three modulo  $\ell$ , proving (ER<sub>2</sub>).

Let  $G = \text{Gal}(\mathbb{Q}(\xi_m)/\mathbb{Q})$ , let  $j$  be complex conjugation, and write

$$\theta_m = \frac{1}{m} \sum_{\substack{1 \leq a < m \\ (a, m) = 1}} a \sigma_a^{-1}$$

for the Stickelberger element. Pairing  $a$  with  $m - a$  gives

$$(1 + j)\theta_m = \sum_{g \in G} g.$$

On an even representation, the right side is twice the norm element of  $\Delta$ , whose evaluation at two is zero by (ER<sub>1</sub>). Since  $\ell \neq 2, 3$ , division by two and by  $m$  is valid modulo  $\ell$ . Hence  $\theta_m$ , and therefore every classical Stickelberger-ideal multiple controlling Gauss–Jacobi-sum ideals, acts trivially on  $V_\ell$ . The particularly relevant integrality factor  $2 - \sigma$  vanishes separately. Differentiating

$$(X - 1)\nu_n(X) = X^n - 1$$

at  $X = 2$  gives (ER<sub>3</sub>), since  $\ell \nmid 2n$ .

The reduction of  $c$  is  $\gamma$ . In the local field  $F_{\mathfrak{p}}$ , which contains  $\mu_\ell$ , adjoining an  $\ell$ -th root  $z$  of  $c$  is the unramified Kummer extension selected by the residue of  $c$ . Arithmetic Frobenius satisfies

$$\frac{\text{Frob}(z)}{z} = \bar{z}^{2^n - 1} = \gamma^{(2^n - 1)/\ell},$$

proving (ER<sub>4</sub>) and the failure criterion. Also, for the relative circular unit  $\varepsilon = \tau(c)/c$ ,

$$(\varepsilon, 2)_{\mathfrak{p}, \ell} = \mathcal{A}_{k, \ell}^{q-1};$$

because  $q - 1 \not\equiv 0 \pmod{\ell}$ , the relative coboundary carries exactly the same unknown symbol and supplies no new relation.

It remains to audit global reciprocity. The completion  $F_{\mathfrak{p}}$  is the unramified degree- $n$  extension of  $\mathbb{Q}_2$ . Since  $\mathbb{Q}_2(\mu_{\ell})/\mathbb{Q}_2$  is also unramified of degree  $\text{ord}_{\ell}(2) = n$ , it is already contained in  $F_{\mathfrak{p}}$ . The conductors  $m$  and  $\ell$  are coprime, so  $F \cap \mathbb{Q}(\mu_{\ell}) = \mathbb{Q}$ ; consequently  $[E : F] = \ell - 1$ , and  $\mathfrak{p}$  splits completely in  $E/F$ . Galois functoriality of the Hilbert symbol gives (ER<sub>5</sub>).

We now check every place above  $\ell$ , the only point at which wild local reciprocity might have supplied extra information. Let  $K_0$  be the maximal unramified subfield of an  $\ell$ -adic completion of  $E$ . That completion is  $K = K_0(\mu_{\ell})$ , with absolute ramification index  $e = \ell - 1$ , and both  $c$  and 2 belong to  $K_0^{\times}$ . For any unit  $a \in K_0^{\times}$ , write

$$a = [\bar{a}] \langle a \rangle, \quad \langle a \rangle \in 1 + \ell \mathcal{O}_{K_0}.$$

The Teichmüller factor has order prime to  $\ell$ , hence is an  $\ell$ -th power. If  $U_i = 1 + \mathfrak{m}_K^i$ , then

$$1 + \ell \mathcal{O}_{K_0} \subset U_{\ell-1}.$$

The standard Hilbert-filtration orthogonality for a field containing  $\mu_{\ell}$  is [9]

$$(U_i, U_j)_{\ell} = 1 \quad \text{when} \quad i + j > \frac{\ell e}{\ell - 1} = \ell. \quad (\text{ER}_7)$$

Taking  $i = j = \ell - 1$  proves  $(c, 2)_{\lambda, \ell} = 1$  at every  $\lambda \mid \ell$ . At every finite place away from  $2\ell$ , both entries are units and the tame Hilbert-symbol formula is one; archimedean places contribute nothing because  $\ell$  is odd. The global product formula is therefore precisely (ER<sub>6</sub>). Since  $\sum_{a=1}^{\ell-1} a \equiv 0 \pmod{\ell}$ , this relation is tautological and leaves the single symbol at  $\mathfrak{P}_0$  arbitrary.  $\square$

**Proposition 20** (semidirect ray extension forced by cubic failure). *[proved] Retain Proposition 18, put  $n = 3^k$ , and let  $\ell \mid \Phi_{3^k}(2)$ . Assume  $\ell \nmid |\text{Cl}(F_k)|$ . Then*

$$\gamma_k \in (\mathbb{F}_{2^n}^{\times})^{\ell} \quad (\text{RF})$$

*if and only if there exists a cyclic extension  $M/F_k$  of degree  $\ell$ , conductor exactly  $\mathfrak{p}_k = (2)$ , totally tamely ramified there and unramified at every other finite prime. This extension is Galois over  $\mathbb{Q}$ , with*

$$\text{Gal}(M/\mathbb{Q}) \cong C_{\ell} \rtimes_2 C_n, \quad \sigma \tau \sigma^{-1} = \tau^2. \quad (\text{RF}_1)$$

*Moreover, the unique prime of  $F_k$  above three splits completely in  $M/F_k$ . Both prescribed local behaviors are admissible: at two this is the standard tame extension with inertia  $C_{\ell}$ , residue degree  $n$ , and Frobenius action  $\tau \mapsto \tau^2$ ; at three the  $C_{\ell}$ -layer is split. Thus secant saturation and circular-unit reciprocity give no local contradiction. Without the class-number hypothesis, cubic failure yields the sharp alternative*

$$\ell \mid |\text{Cl}(F_k)| \quad \text{or the extension in (RF}_1\text{) exists.} \quad (\text{RF}_2)$$

*Proof.* Write  $E_k = \mathcal{O}_{F_k}^{\times}$  and  $\mathcal{C}_k$  for the circular units. Proposition 18 identifies the reduction of  $\mathcal{C}_k$  with  $\langle \gamma_k \rangle$ . If (RF) holds, then  $\mathcal{C}_k$  maps trivially to

$$V := \mathbb{F}_{2^n}^{\times} / (\mathbb{F}_{2^n}^{\times})^{\ell} \cong C_{\ell}.$$

The odd part of  $[E_k : \mathcal{C}_k]$  is the odd part of the class number. Under the stated hypothesis its  $\ell$ -part vanishes, so  $E_k$  also maps trivially to  $V$ . The ray sequence then supplies an order- $\ell$  quotient of

$\text{Cl}_{\mathfrak{p}_k}(F_k)$ . Because the ordinary class group has no  $\ell$ -part, the quotient lies entirely in tame inertia at  $\mathfrak{p}_k$ . Class field theory gives the asserted  $M/F_k$ .

The quotient is Galois-stable. The generator  $\sigma$  of  $\text{Gal}(F_k/\mathbb{Q})$  acts on the residue field as  $x \mapsto x^2$ , hence on  $V$  by exponent two. Since  $\text{ord}_\ell(2) = n$ , this action is faithful; Schur–Zassenhaus, or the coprimality of  $n$  and  $\ell$ , gives (RF<sub>1</sub>). Conversely, tame inertia in any such extension gives the unique quotient  $\mathbb{F}_{2^n}^\times \twoheadrightarrow C_\ell$ . Global units lie in its kernel, so in particular  $\gamma_k$  is an  $\ell$ -th power.

Finally let

$$\lambda_k = (1 - \xi_k)(1 - \xi_k^{-1}), \quad c_k = (1 + \xi_k)(1 + \xi_k^{-1}).$$

The element  $\lambda_k$  generates the unique prime  $\mathfrak{r}_k$  above three, and  $\lambda_k/c_k \equiv 1 \pmod{\mathfrak{p}_k}$ , because both reduce to  $(1 + \zeta)(1 + \zeta^{-1}) = \gamma_k$ . Hence  $\mathfrak{r}_k = (\lambda_k/c_k)$  is trivial in the full ray class group and splits in  $M/F_k$ . The local statements now follow from  $\ell \mid 2^n - 1$  and the displayed semidirect action. If the class-number hypothesis fails, the first alternative in (RF<sub>2</sub>) holds; otherwise the preceding construction proves the second.  $\square$

**Proposition 21** (explicit cubic kernel tower). *[proved] Define  $\eta_0 = 0$ . For  $k \geq 1$  put*

$$q_k = 2^{3^{k-1}}, \quad \eta_k = \gamma_k^{q_k-1}, \quad U_k = \ker\left(N_{\mathbb{F}_{q_k^3}/\mathbb{F}_{q_k}}\right).$$

Then  $\eta_k \in U_k$ ,  $|U_k| = q_k^2 + q_k + 1 = \Phi_{3^k}(2)$ , and, with  $\omega = \zeta^{3^k} \in \mathbb{F}_4$ ,

$$q_k \equiv 3^k - 1 \pmod{3^{k+1}}, \quad \eta_k = \frac{\omega + \omega^2 \zeta^2}{1 + \zeta^2}, \quad \zeta^2 = \frac{\eta_k + \omega}{\eta_k + \omega^2}.$$

Moreover  $\eta_k$  has the irreducible cubic equation over  $\mathbb{F}_{q_k} = \mathbb{F}_{2^{3^{k-1}}}$

$$X^3 + \eta_{k-1}X^2 + (\eta_{k-1} + 1)X + 1 = 0. \quad (*)$$

Equivalently,

$$N(\eta_k) = 1, \quad \text{Tr}(\eta_k) = \eta_{k-1}, \quad e_2(\eta_k) = \eta_{k-1} + 1.$$

It also satisfies

$$\eta_k^{q_k+1} + \eta_k + 1 = 0, \quad \eta_k + 1 = \eta_k^{q_k+1}, \quad \text{ord}(\eta_k + 1) = \text{ord}(\eta_k). \quad (\dagger)$$

Thus  $\eta_k$  lies in the planar Singer difference set

$$\mathcal{H}_k = \{y \in U_k : y^{q_k+1} + y + 1 = 0\}$$

of size  $q_k + 1$  inside  $U_k$  [14]. Finally,

$$C_k \iff C_{k-1} \text{ and } \text{ord}(\eta_k) = q_k^2 + q_k + 1. \quad (**)$$

(Here  $C_0$  is the tautological assertion that 1 generates  $\mathbb{F}_2^\times$ .) Thus the new-prime obstruction is the single assertion that the distinguished root of  $(*)$  is a generator of the cubic norm-one torus (equivalently, multiplication by it is a Singer cycle).

*Proof.* The congruence starts with  $2 \equiv 3-1 \pmod{9}$  and lifts by cubing: if  $a \equiv -1+3^k \pmod{3^{k+1}}$ , then  $a^3 \equiv -1 + 3^{k+1} \pmod{3^{k+2}}$ . Hence  $\zeta^{q_k} = \omega\zeta^{-1}$ . Dividing  $\gamma_k^{q_k}$  by  $\gamma_k$  now gives the stated Möbius formula; solving it gives the inverse formula.

Put  $t = \zeta^2$  and  $M(T) = (\omega + \omega^2 T)/(1 + T)$ . The same formula one level down says  $\eta_{k-1} = M(t^3)$  (also for  $k = 1$ , when it gives  $\eta_0 = 0$ ). If  $a = \eta_{k-1}$  and  $X = M(t)$ , then

$$\left( \frac{X + \omega}{X + \omega^2} \right)^3 = \frac{a + \omega}{a + \omega^2}.$$

Cross-multiplication and  $\omega^2 + \omega + 1 = 0$  reduce exactly to  $(*)$ . Also  $\eta_k^{1+q_k+q_k^2} = \gamma_k^{q_k^3-1} = 1$ , so  $\eta_k \in U_k$ . It cannot lie in  $\mathbb{F}_{q_k}$ : otherwise its norm-one condition would give  $\eta_k^3 = 1$ , hence  $\eta_k = 1$  because  $3 \nmid q_k - 1$ ; the inverse Möbius formula would then give  $t = \omega$ , contradicting  $\text{ord}(t) = 3^{k+1} > 3$ . Thus  $(*)$  is irreducible, and its coefficients give the three symmetric identities.

The earlier minimal polynomial  $X^3 + X + \gamma_{k-1}$  gives  $\text{Tr}_{\mathbb{F}_{q_k^3}/\mathbb{F}_{q_k}}(\gamma_k) = 0$ . Dividing  $\gamma_k + \gamma_k^{q_k} + \gamma_k^{q_k^2} = 0$  by  $\gamma_k$  yields  $1 + \eta_k + \eta_k^{q_k+1} = 0$ , proving the first two parts of  $(\dagger)$ . Since  $\gcd(q_k + 1, q_k^2 + q_k + 1) = 1$ , exponentiation by  $q_k + 1$  is an automorphism of  $U_k$ , proving the equality of orders. The cited difference-set identification is the norm-one realization of the planar Singer set.

In the direct product of Proposition 14, raising to  $q_k - 1$  kills the old-field component and is an automorphism on  $U_k$ , since  $\gcd(q_k - 1, q_k^2 + q_k + 1) = 1$ . Therefore the kernel component of  $\gamma_k$  generates  $U_k$  exactly when  $\eta_k$  does; Proposition 14 gives  $(**)$ .  $\square$

**Proposition 22** (Singer-orbit sieve). *[proved] Let  $k \geq 1$ . Put  $n = 3^k$ ,  $q = 2^{n/3}$ , and  $N = q^2 + q + 1 = |U_k|$ . If a prime  $\ell \mid N$  satisfies*

$$\ell > \frac{N}{n(n-1) + 1},$$

*then  $\eta_k \notin U_k^\ell$ . Consequently, if  $\ell^a \parallel N$ , then*

$$v_\ell(\text{ord } \eta_k) = a.$$

*Unconditionally,*

$$\text{ord}(\eta_k) \geq n(n-1) + 1, \quad [U_k : \langle \eta_k \rangle] \leq \frac{N}{n(n-1) + 1}.$$

*Proof.* Let

$$D = \{y \in U_k : y^{q+1} + y + 1 = 0\}.$$

This is the cyclic  $(N, q+1, 1)$  Singer difference set: every nonidentity element of  $U_k$  has a unique expression  $xy^{-1}$  with distinct  $x, y \in D$ . Since the minimal polynomial of  $\eta_k$  over  $\mathbb{F}_2$  has degree  $n$ , its Frobenius orbit

$$A = \{\eta_k^{2^i} : 0 \leq i < n\}$$

has  $n$  elements. The equation defining  $D$  is Frobenius-stable, so  $A \subset D$ .

Suppose  $\eta_k \in H := U_k^\ell$ . The subgroup  $H$  is characteristic in the cyclic group  $U_k$ , hence Frobenius-stable, and  $A \subset H$ . The  $n(n-1)$  quotients  $xy^{-1}$  for ordered distinct pairs  $(x, y) \in A^2$  are nonidentity elements of  $H$  and are all distinct by the Singer difference-set property. Therefore

$$|H| - 1 \geq n(n-1).$$

Since  $|H| = N/\ell$ , this contradicts the displayed strict inequality. In a cyclic group of order  $\ell^a m$ ,  $\ell \nmid m$ , lying outside the subgroup of  $\ell$ -th powers is equivalent to having the full  $\ell^a$  part in one's order, proving the first assertion.

Finally the same distinct quotients all lie in  $\langle \eta_k \rangle$ , whether or not an  $\ell$  is fixed. Thus  $\text{ord}(\eta_k) - 1 \geq n(n-1)$ , which gives the last two inequalities.  $\square$

**Remark 3** (why the translated orbit does not double the sieve). The identity  $y + 1 = y^{q+1}$  puts the translated Frobenius orbit  $A + 1 = A^{q+1}$  in the same subgroup as  $A$ . It supplies no second quotient family: modulo  $N$  one has  $(q + 1)^2 \equiv q$  and  $q(q + 1) \equiv -1$ , while the quotient set of  $A$  is stable under both  $q$ -th powers and inversion. Hence the internal quotient sets of  $A$  and  $A + 1$  coincide. Cross quotients already collide, for example  $(y + 1)/y = y^q$ . Improving Proposition 22 therefore requires cross-correlation or character information beyond the  $(N, q + 1, 1)$  difference-set axiom.

**Proposition 23** (partition-order bound). *[proved] Retain Proposition 22. Let  $B(n)$  be the number of partitions into distinct positive parts not divisible by 3 whose total is at most  $n = 3^k$ . Then*

$$\text{ord}(\eta_k) \geq B(n).$$

*In particular, if  $p_3^{\text{dist}}(n)$  counts such partitions of  $n$  itself, then*

$$\text{ord}(\eta_k) \geq p_3^{\text{dist}}(n) = \exp\left(\left(\frac{\sqrt{2}\pi}{3} + o(1)\right)\sqrt{n}\right). \quad (\text{B})$$

*If a prime  $\ell \mid N$  satisfies*

$$\ell > \frac{N}{B(n)},$$

*then  $\eta_k \notin U_k^\ell$ , so the order of  $\eta_k$  contains the full  $\ell$ -primary part of  $N$ .*

*Proof.* Put  $r = 3n$ ,  $t = \zeta^2$ , and retain  $\omega = \zeta^n$ , so  $t^n = \omega^2$  and

$$\eta_k = M(t), \quad M(X) = \frac{\omega + \omega^2 X}{1 + X}.$$

The element 2 generates  $(\mathbb{Z}/r\mathbb{Z})^\times$ , of order  $2n$ . For every positive  $j$  not divisible by 3, choose  $z_j$  with  $2^{z_j} \equiv j \pmod{r}$ . Frobenius gives

$$\eta_k^{2^{z_j}} = \omega^j \frac{1 + \omega^j t^j}{1 + t^j}. \quad (6)$$

For a finite set  $I$  of such  $j$ , put

$$s_I = \sum_{j \in I} j, \quad E_I = \sum_{j \in I} 2^{z_j}, \quad A_I(X) = \prod_{j \in I} (1 + \omega^j X^j), \quad B_I(X) = \prod_{j \in I} (1 + X^j).$$

Thus

$$\eta_k^{E_I} = \omega^{s_I} \frac{A_I(t)}{B_I(t)}. \quad (7)$$

All denominators are nonzero when  $s_I \leq n < r$ .

Suppose  $s_I, s_J \leq n$  and  $\eta_k^{E_I} = \eta_k^{E_J}$ . Cross-multiplication gives

$$F(X) := \omega^{s_I} A_I(X) B_J(X) + \omega^{s_J} A_J(X) B_I(X), \quad F(t) = 0. \quad (8)$$

Now  $M(t^{-1}) = M(t) + 1 = \eta_k + 1$ , and the Singer equation  $\eta_k + 1 = \eta_k^{q+1}$  shows that the assumed equality also holds after replacing  $\eta_k$  by  $M(t^{-1})$ . Hence  $F(t^{-1}) = 0$ . Over  $\mathbb{F}_4$ , the two elements  $t, t^{-1}$  have the distinct degree- $n$  minimal polynomials

$$X^n + \omega^2, \quad X^n + \omega,$$



because  $\text{ord}_r(4) = n$ . Therefore

$$X^{2n} + X^n + 1 \mid F(X). \quad (9)$$

But  $\deg F \leq s_I + s_J \leq 2n$ . In the only possible equality case  $s_I = s_J = n$ , the two leading coefficients are both  $\omega^{2n}$  and cancel in characteristic two. Thus  $\deg F < 2n$ , and (9) forces  $F = 0$ .

The constant coefficient first gives  $\omega^{s_I} = \omega^{s_J}$ . Cancel that scalar and all factors belonging to  $I \cap J$ . If  $j_0$  is the least remaining part, comparison of the  $X^{j_0}$ -coefficients gives  $\omega^{j_0} = 1$ : on one side  $j_0$  occurs in an  $A$ -factor, with coefficient  $\omega^{j_0}$ , and on the other in a  $B$ -factor, with coefficient 1; larger parts cannot contribute. This is impossible because  $3 \nmid j_0$ . Hence  $I = J$ .

The  $B(n)$  powers in (7) are therefore pairwise distinct, proving the order bound. Meinardus's partition theorem [15], applied to the Euler product

$$G(x) := \prod_{3 \nmid j} (1 + x^j) = \frac{\prod_{j \geq 1} (1 + x^j)}{\prod_{j \geq 1} (1 + x^{3j})}, \quad \log G(e^{-t}) \sim \frac{\pi^2}{18t},$$

gives (B). Finally, if  $\eta_k \in U_k^\ell$ , all  $B(n)$  displayed powers lie in the subgroup of order  $N/\ell$ . The strict inequality rules this out; the cyclic-group argument of Proposition 22 then gives the full  $\ell$ -primary assertion.  $\square$

**Remark 4** (ceiling of the height-budgeted amplification). The preceding construction is scale-optimal within the following precise class of degree arguments. For  $J = (\mathbb{Z}/3n\mathbb{Z})^\times / \{\pm 1\}$ , assign an atom indexed by  $j \in J$  the height

$$h(j) := \min\{|b| : b \equiv \pm j \pmod{3n}\}.$$

Suppose a family of integer exponent vectors is certified pairwise distinct only by bounding the Laurent span after cross-multiplication with the triangle estimate

$$w(e - e') := \sum_{j \in J} h(j) |e_j - e'_j| \leq 2n$$

and then invoking the two degree- $n$  minimal polynomials as above (including the verified leading-term cancellation at the boundary). Then the family has cardinality  $\exp(O(\sqrt{n}))$ .

Indeed, fix one vector  $e_0$ . Every difference  $e - e_0$  lies in the displayed weighted  $\ell^1$ -ball. Each positive height occurs only  $O(1)$  times, so the number of such lattice points is bounded by the coefficients through degree  $2n$  of a fixed power of

$$\prod_{m \geq 1} (1 + 2x^m + 2x^{2m} + \cdots) \preceq \prod_{m \geq 1} (1 - x^m)^{-2}.$$

Colored-partition bounds give  $\exp(O(\sqrt{n}))$ . Polynomially many Frobenius or cyclic shifts do not change that scale, whereas  $|U_k| = \exp((2 \log 2/3 + o(1))n)$ . Thus more triangle-budgeted subset encodings cannot prove primitivity. This does *not* rule out systematic factor cancellations or exact reduction modulo  $\Phi_{3n}$ ; those require new arithmetic input rather than the generic degree budget.

**Proposition 24** (no transfer and the exact character test). *[proved] The kernel orders are pairwise coprime: for  $1 \leq i < k$ ,*

$$\gcd(\Phi_{3^i}(2), \Phi_{3^k}(2)) = 1.$$

*For each prime  $\ell \mid |U_k|$ , let  $\chi_{k,\ell} : U_k \rightarrow \mu_\ell$  be a character of exact order  $\ell$ . Then*

$$\text{Hom}\left(\mathbb{F}_{2^{3k-1}}^\times, \mu_\ell\right) = 1; \quad (\dagger_0)$$

no multiplicative character or power-residue datum from a previous level can transfer an  $\ell$ -part to level  $k$ . Moreover,

$$C_k \iff C_{k-1} \text{ and } \chi_{k,\ell}(\eta_k) \neq 1 \text{ for every prime } \ell \mid |U_k|. \quad (\ddagger)$$

Equivalently, with

$$\rho_{k,\ell}(A) = \chi_{k,\ell}(A^{q_k-1}) \quad (A \in \mathbb{F}_{q_k}^\times),$$

where  $\rho_{k,\ell}$  is trivial on  $\mathbb{F}_{q_k}^\times$ , the missing assertion is  $\rho_{k,\ell}(\gamma_k) \neq 1$  for every such  $\ell$ . This is the smallest power-residue form of the cubic obstruction.

*Proof.* If a prime  $r$  divided two displayed cyclotomic values, then  $r \neq 3$  and the standard cyclotomic-divisor criterion would give both  $\text{ord}_r(2) = 3^i$  and  $\text{ord}_r(2) = 3^k$ , a contradiction. Thus every  $\ell \mid |U_k|$  is coprime to  $|\mathbb{F}_{2^{3k-1}}^\times| = \prod_{j < k} \Phi_{3^j}(2)$ , proving  $(\ddagger_0)$ . In the cyclic group  $U_k$ , the kernel of an exact order- $\ell$  character is  $U_k^\ell$ . If  $|U_k| = \ell^a m$  with  $\ell \nmid m$ , then an element has the full  $\ell^a$  in its order exactly when it is not in  $U_k^\ell$ . Applying this for every prime divisor and using  $(**)$  proves  $(\ddagger)$ . The formula for  $\rho_{k,\ell}$  and  $\eta_k = \gamma_k^{q_k-1}$  give the final equivalence.  $\square$

**Proposition 25** (short cyclotomic-period form of the cubic obstruction). *[proved] Let  $k \geq 1$ , put  $n = 3^k$ ,  $q = 2^{n/3}$ , and let  $\ell$  be a prime divisor of  $\Phi_{3^k}(2)$ . Let  $\rho = \rho_{k,\ell}$  be the exact  $\ell$ -character from Proposition 24, realized with values in the complex roots of unity. In  $\mathbb{F}_{2^{2n}}$  let  $\mu_{3n}$  denote the roots of unity of order dividing  $3n = 3^{k+1}$ , and set*

$$\mathcal{S}_{k,\ell} := \sum_{\substack{t \in \mu_{3n} \\ t \neq 1}} \rho(t + t^{-1}). \quad (\text{S})$$

Every argument of  $\rho$  in this sum is a nonzero element of  $\mathbb{F}_{2^n}$ . If  $\rho(\gamma_k) = \zeta_\ell^a$ , then

$$\mathcal{S}_{k,\ell} = (n-1) + 2 \sum_{i=0}^{n-1} \zeta_\ell^{a2^i}. \quad (\text{P})$$

Consequently

$$\chi_{k,\ell}(\eta_k) = 1 \iff \mathcal{S}_{k,\ell} = 3n - 1. \quad (\text{C})$$

Thus the surviving datum is exactly a short Gauss period over the order- $n$  subgroup generated by 2 in  $(\mathbb{Z}/\ell\mathbb{Z})^\times$ .

*Proof.* Since  $2^n \equiv -1 \pmod{3n}$ , every  $t + t^{-1}$  in (S) is fixed by  $2^n$  and lies in  $\mathbb{F}_{2^n}$ ; it is nonzero unless  $t = 1$ . The  $n-1$  nonidentity elements of  $\mu_n$  give real cyclotomic elements in the lower field  $\mathbb{F}_{2^{n/3}}^\times$ , on which  $\rho$  is trivial.

The remaining  $2n$  elements are the primitive  $3n$ -th roots. Pairing them by inversion leaves the  $n$  Frobenius conjugates  $\gamma_k^{2^i}$ , each twice. This proves (P). Since  $q-1$  is invertible modulo  $\ell$ ,  $\chi_{k,\ell}(\eta_k) = 1$  exactly when  $\rho(\gamma_k) = 1$ , which gives the forward implication in (C). Conversely every summand in (S) has complex absolute value one. Equality with the number of summands  $3n-1$  forces equality in the triangle inequality and hence forces every summand to be 1, in particular  $\rho(\gamma_k) = 1$ .  $\square$

**Remark 5** (why a full Singer character sum is too coarse). The Singer difference-set character sum controls a set of size  $q+1$  and has the usual square-root magnitude. Equation (P) instead localizes to only  $3n-1$  cyclotomic terms, with the unknown part supported on the order- $n$  subgroup  $\langle 2 \rangle$  modulo  $\ell$ . Generic Weil or difference-set bounds do not evaluate this localized scalar; an all-level proof must rule out its extremal value in (C).

**Proposition 26** (cyclotomic-secant saturation). *[proved] Retain  $n = 3^k$ , put  $m = 3n$ , and let  $K = \mathbb{F}_{2^{2n}}$  contain a primitive  $m$ -th root  $\zeta$ . For every prime  $\ell \mid \Phi_{3^k}(2)$ , the following are equivalent:*

- (i)  $\eta_k \in U_k^\ell$ ;
- (ii)  $\gamma_k \in (\mathbb{F}_{2^n}^\times)^\ell$ ;
- (iii)  $1 + \zeta \in (K^\times)^\ell$ ;
- (iv) for every  $a \not\equiv b \pmod{m}$ ,  $\zeta^a + \zeta^b \in (K^\times)^\ell$ .

Thus, for an exact order- $\ell$  character  $\chi$  of  $K^\times$ , failure is also equivalent to the extremal identity

$$\sum_{\substack{t \in \mu_m \\ t \neq 1}} \chi(1 - t) = m - 1. \quad (\text{CS})$$

It saturates every cyclotomic secant, not only the Frobenius orbit of the distinguished Gaussian period.

*Proof.* The first equivalence follows from

$$\eta_k = \gamma_k^{2^{n/3}-1},$$

because  $\ell \nmid 2^{n/3} - 1$ . Next,

$$\gamma_k = \zeta^{-1}(1 + \zeta)^2.$$

The order of  $\zeta$  is coprime to  $\ell$ , and squaring is invertible on the  $\ell$ -primary quotient. Moreover the inclusion  $\mathbb{F}_{2^n}^\times \subset K^\times$  preserves that quotient, since  $\ell \nmid 2^n + 1$ . This proves the equivalence with (iii).

Write  $d = b - a$ . The factor  $\zeta^a$  is automatically an  $\ell$ -th power. If  $3 \nmid d$ , then (2) generates  $(\mathbb{Z}/m\mathbb{Z})^\times$ , so  $d \equiv 2^j \pmod{m}$  and

$$1 + \zeta^d = (1 + \zeta)^{2^j}.$$

If  $3 \mid d$ , then  $1 + \zeta^d \in \mathbb{F}_{2^{2n/3}}^\times$ . This group has order coprime to  $\ell$ , because  $\text{ord}_\ell(2) = n$ , so all its elements are  $\ell$ -th powers in  $K$ . This proves (iii) $\Rightarrow$ (iv); the converse is the case  $a = 0, b = 1$ .

Finally, all terms in (CS) have complex absolute value one. Part (iv) makes them all one; conversely equality with the positive real number  $m - 1$  forces every term to be one by equality in the triangle inequality.  $\square$

**Remark 6** (the natural secant Kummer cover is rational). Over  $\mathbb{F}_4$ , the Möbius function

$$M(T) = \frac{\omega + \omega^2 T}{1 + T}$$

which expresses the cyclotomic secant has Kummer cover

$$Z^\ell = M(T), \quad T = \frac{Z^\ell + \omega}{Z^\ell + \omega^2}.$$

Its function field is therefore  $\mathbb{F}_4(Z)$ , of genus zero. A positive-genus Weil estimate on this cover cannot rule out the distinguished torsion fibre; the remaining information is its actual Artin value.

**Proposition 27** (Capelli form of the surviving lemma). *[proved] Put*

$$A(X) = X^3 + X + 1, \quad B(X) = X^2 + X, \quad R(X) = \frac{A(X)}{B(X)}.$$

Let  $Q_0(X) = X$  and, for  $k \geq 1$ , define the monic binary polynomial

$$Q_k(X) = B(X)^{3^{k-1}} Q_{k-1}(R(X)). \quad (\star)$$

Then  $Q_k$  is the minimal polynomial of  $\eta_k$  over  $\mathbb{F}_2$  and is irreducible of degree  $3^k$ . Its first four values are

$$\begin{aligned} Q_1(X) &= X^3 + X + 1, \\ Q_2(X) &= X^9 + X + 1, \\ Q_3(X) &= X^{27} + X^{25} + X^{24} + X^{19} + X^{18} + X^{16} + X^{10} + X^9 + X^3 + X + 1, \\ Q_4(X) &= X^{81} + X^{64} + X^{16} + X + 1. \end{aligned}$$

For every prime  $\ell \mid \Phi_{3^k}(2)$ , the following are equivalent:

- (i)  $\chi_{k,\ell}(\eta_k) \neq 1$ ;
- (ii)  $\eta_k$  is not an  $\ell$ -th power in  $\mathbb{F}_{2^{3^k}}^\times$ ;
- (iii)  $X^\ell - \eta_k$  is irreducible over  $\mathbb{F}_{2^{3^k}}$ ;
- (iv)  $Q_k(X^\ell)$  is irreducible over  $\mathbb{F}_2$ .

Thus the surviving character lemma is equivalently the assertion that all the explicitly recursive compositions  $Q_k(X^\ell)$ , for  $\ell \mid \Phi_{3^k}(2)$ , are irreducible. At the first composite level  $k = 4$ , the three targets are the monomial compositions of the displayed pentanomial  $Q_4$  by  $\ell \in \{2593, 71119, 97685839\}$ .

*Proof.* Proposition 21 gives  $R(\eta_k) = \eta_{k-1}$  and an irreducible cubic for  $\eta_k$  over  $\mathbb{F}_{2^{3^{k-1}}}$ . Since  $\eta_{k-1} \in \mathbb{F}_2(\eta_k)$ , induction gives  $[\mathbb{F}_2(\eta_k) : \mathbb{F}_2] = 3^k$ . Clearing the denominator in the minimal polynomial of  $\eta_{k-1}$  gives  $(\star)$ , a monic polynomial of degree  $3^k$  having  $\eta_k$  as a root: its leading term is  $A(X)^{3^{k-1}} = X^{3^k} + \dots$ . It is therefore the minimal polynomial. Direct substitution in  $(\star)$  gives the four displayed polynomials.

Here  $\Phi_{3^k}(2) = q_k^2 + q_k + 1 \equiv 1 \pmod{3}$ , so a prime  $\ell \mid \Phi_{3^k}(2)$  is not 3. The cyclotomic-divisor criterion therefore gives  $\text{ord}_\ell(2) = 3^k$ , and hence  $\mu_\ell \subset \mathbb{F}_{2^{3^k}}$ . Since  $\ell$  is prime, Kummer theory says that  $X^\ell - \eta_k$  either splits off a linear factor or is irreducible, according as  $\eta_k$  is or is not an  $\ell$ -th power. This proves the equivalence of (ii) and (iii), while Capelli's lemma gives the equivalence of (iii) and (iv). The direct product  $\mathbb{F}_{2^{3^k}}^\times \simeq \mathbb{F}_{q_k}^\times \times U_k$  and  $\ell \nmid (q_k - 1)$  show that being an  $\ell$ -th power in the full field is equivalent, for  $\eta_k \in U_k$ , to lying in  $U_k^\ell$ . This is the equivalence with (i).  $\square$

**Remark 7** (Singer membership is not primitivity). The difference-set identification cannot replace the character test. Indeed, exponentiation

$$s_k : U_k \longrightarrow U_k, \quad s_k(y) = y^{q_k+1},$$

is an automorphism of order 6: one has  $\gcd(q_k + 1, |U_k|) = 1$  and  $(q_k + 1)^3 \equiv -1 \pmod{|U_k|}$ , while  $(q_k + 1)^2 \equiv q_k \not\equiv 1$ . The Singer equation is simply  $s_k(y) = y + 1$ . Thus its consequence  $\text{ord}(y + 1) = \text{ord}(y)$  is automatic order preservation by an automorphism and supplies no order divisor by itself.

For example, take  $q = 32$  and let  $y$  be a root of  $X^3 + X^2 + 1$  over  $\mathbb{F}_2$ . Then  $y$  has order 7 and  $q + 1 \equiv 5 \pmod{7}$ ; the defining cubic gives  $y^5 + y + 1 = 0$ . Hence  $y \in \{x : x^{q+1} + x + 1 = 0\}$ , but  $y$  does not generate the norm-one group of order  $q^2 + q + 1 = 7 \cdot 151$ . What must be used in the Conway tower is the recursive coefficient  $\eta_{k-1}$  in  $(*)$ , not Singer membership alone.

**Proposition 28** (prime-step closure). *[proved] If  $\Phi_{3^k}(2)$  is prime, then*

$$C_k \iff C_{k-1}.$$

*Consequently  $C_1, C_2, C_3$  hold analytically: the three new kernel orders are respectively*

$$7, \quad 73, \quad 262657,$$

*and all are prime.*

*Proof.* Proposition 21 proves that  $\eta_k$  has degree 3 over  $\mathbb{F}_{q_k}$ , so in particular  $\eta_k \neq 1$ . A nonidentity element of a group of prime order generates it, and  $(**)$  gives the first claim. The primality of 7 and 73 is immediate. For completeness,  $262657 - 1 = 2^9 \cdot 3^3 \cdot 19$ , and base 5 gives the Pocklington certificate

$$5^{262656} \equiv 1 \pmod{262657}, \quad \gcd\left(5^{262656/r} - 1, 262657\right) = 1 \quad (r = 2, 3, 19),$$

which proves the third primality assertion. □

**Conjecture 1** ( $C_k$ :  $\gamma$ -primitivity).  $\gamma_k$  is primitive in  $L^\times = \mathbb{F}_{2^{3^k}}^\times$ .

**Proposition 29** (equivalence). *[proved] For primes  $r$  with  $f(r) \mid 3^k$  a 3-power:*

$$C_k \iff m_r = 1 \text{ for every prime } r \text{ with } f(r) \in \{3, 9, \dots, 3^k\}.$$

*The family formula is the 0/1/4 candidate of Section 4 restricted to the 3-power column.*

### 3.2 Certification status

**[proved]** Proposition 28 proves  $C_k$  for  $k = 1, 2, 3$  without an order computation.

**[certified]**  $4 \leq k \leq 6$   $C_k$  holds for  $k = 4, 5, 6$ :  $\gamma_k$  is primitive in  $\mathbb{F}_{2^{3^k}}^\times$ , and  $\text{ord}(\kappa_{3^k} + 1) = 3^{k+1}(2^{3^k} - 1)$  is verified exactly – product reconstruction, squarefreeness, primality by deterministic Miller–Rabin below  $3.3 \times 10^{24}$  and MR-64 probable-primality above, an independent sieve re-deriving the small factors, cross-validation against the independent term algebra for  $k = 1, 2$  and against the calculator rows  $m_{2593} = 1$ ,  $m_{487} = 1$  (`experiments/cyclotomic_3k_family.py`, `experiments/excess/`).

**[consistent]**  $k = 7, 8$ : every known prime factor of  $\Phi_{3^7}(2)$  and  $\Phi_{3^8}(2)$  divides  $\text{ord}(\gamma_k)$ . Full certification is blocked by the unfactored cofactors (FactorDB status composite-no-factor), including their unknown prime multiplicities; no failure has been found.

**[open]** Whether ECM or GNFS can factor the  $\Phi_{3^7}(2)$  and  $\Phi_{3^8}(2)$  cofactors within a realistic budget, converting  $k = 7, 8$  from CONSISTENT to CERTIFIED.

### 3.3 Certified excess rows produced

The following  $m_r = 1$  rows are certified by the order criterion independently of the cofactor issue (analysis-level results; they are not new Rust carries):

| $f$  | primes $r$ with $m_r = 1$                            |
|------|------------------------------------------------------|
| 27   | 262657                                               |
| 81   | 71119, 97685839                                      |
| 243  | 16753783618801, 192971705688577, 3712990163251158343 |
| 729  | 80191, 97687, 379081, P42, P90                       |
| 2187 | 39367, 7606246033, 263196614521, 529063556041        |
| 6561 | 209953, 1299079, 70063267397606709277393             |

(P42, P90 denote the certified/probable primes of 42 and 90 decimal digits dividing  $\Phi_{3^6}(2)$ .)

## 4 The 0/1/4 candidate

**Conjecture 2** (the 0/1/4 rule). [**consistent**] (*not proved*)

$$m_p = \begin{cases} 0 & \text{if } \mathcal{Q}(f(p)) \text{ is not a singleton odd prime-power,} \\ 4 & \text{if } f(p) = 2 \cdot 3^k, k \geq 1, \\ 1 & \text{otherwise (singleton odd prime-power } \mathcal{Q}(f(p))). \end{cases}$$

The rule matches all 950 calculator records with known  $\mathcal{Q}$ -sets and every OEIS-known row covered by those  $\mathcal{Q}$ -sets.

### 4.1 Two maximal-order problems already contained in the rule

The candidate contains strong explicit maximal-order assertions even before the non-cyclotomic singleton chains are reached. This prevents a proof from following from field degree or irreducibility alone.

**Proposition 30** (the singleton-even arm is Conway–Fermat maximality). [**proved**] For  $n \geq 0$ , put

$$h_n = 2^{n+1}, \quad c_n = \kappa_{h_n} = 2^{2^n}, \quad E_n = \mathbb{F}_{2^{h_n}}, \quad F_n = 2^{2^n} + 1.$$

With  $E_{-1} = \mathbb{F}_2$ , let

$$\delta_n = \text{ord}(c_n E_{n-1}^\times) \quad \text{in} \quad E_n^\times / E_{n-1}^\times.$$

Then  $1 < \delta_n \mid F_n$ , and the  $m = 0$  arm of Conjecture 2 for every prime  $p \mid F_n$  is equivalent to

$$\delta_n = F_n.$$

*Proof.* The Conway finite-nimber tower satisfies

$$c_n^2 + c_n = c_{n-1} c_{n-2} \cdots c_0$$

(with the empty product 1). Thus  $E_n/E_{n-1}$  is quadratic and  $c_n \notin E_{n-1}$ , so the coset order is nontrivial. Its ambient quotient has order

$$\frac{2^{2^{n+1}} - 1}{2^{2^n} - 1} = F_n,$$

which proves  $1 < \delta_n \mid F_n$ ; equivalently the relative norm is  $c_n^{F_n} = c_{n-1} \cdots c_0$ .

If  $p \mid F_n$ , then  $2^{2^n} \equiv -1 \pmod{p}$ , hence  $f(p) = 2^{n+1} = h_n$ . Fermat numbers are pairwise coprime, so the entire  $p$ -primary part of  $|E_n^\times|$  lies in the quotient by  $E_{n-1}^\times$ . The full power criterion therefore gives  $m_p = 0$  iff  $v_p(\delta_n) = v_p(F_n)$ . Requiring this for every prime divisor of  $F_n$  is exactly  $\delta_n = F_n$ .  $\square$

**Proposition 31** (Fibonacci-polynomial criterion). *[proved] Fix  $n \geq 0$ , put  $m = 2^n$ ,  $q = 2^m$ , and write  $K = E_{n-1} = \mathbb{F}_q$ ,  $L = E_n = \mathbb{F}_{q^2}$  and  $a = a_{n-1} = \prod_{j < n} c_j$ . If*

$$S_0(X) = 0, \quad S_1(X) = 1, \quad S_{r+2}(X) = S_{r+1}(X) + XS_r(X),$$

then

$$\delta_n = \min\{r > 0 : S_r(a) = 0\},$$

and hence

$$\delta_n = F_n \iff S_{F_n/\ell}(a) \neq 0 \quad \text{for every prime } \ell \mid F_n. \quad (\text{F}_n)$$

More precisely, for odd  $d > 1$  define the fibotomic polynomial

$$\mathfrak{F}_d(X) = \prod_{\substack{\{v, v^{-1}\} \\ \text{ord}(v)=d}} \left( X - \frac{v}{(v+1)^2} \right),$$

where the product takes one representative of each inverse-pair of primitive  $d$ -th roots. Then  $\mathfrak{F}_d \in \mathbb{F}_2[X]$ ,

$$S_r(X) = \prod_{\substack{d \mid r \\ d > 1}} \mathfrak{F}_d(X) \quad (r \text{ odd}), \quad (\text{B}_r)$$

and, if  $A_{n-1}$  is the minimal polynomial of  $a_{n-1}$  over  $\mathbb{F}_2$ ,

$$\delta_n = d \iff A_{n-1} \mid \mathfrak{F}_d.$$

Every irreducible factor of every  $\mathfrak{F}_d$  with  $1 < d \mid F_n$  has the same degree  $m$ . Thus the Conway–Fermat conjecture is exactly the assertion  $A_{n-1} \mid \mathfrak{F}_{F_n}$ ; field degree cannot distinguish that stratum from any proper-divisor stratum.

*Proof.* Put  $c = c_n$  and  $u = (c+1)/c$ . Since  $c^2 + c = a$  and  $c^q = c+1$ , the element  $u$  lies in the norm-one group of order  $q+1 = F_n$ , and  $\delta_n = \text{ord}(u)$ . The two roots  $c, c+1$  of  $T^2 + T + a$  give the Binet identity

$$S_r(a) = c^r + (c+1)^r = \frac{1+u^r}{(1+u)^r}. \quad (\text{B})$$

This proves the first criterion. In a cyclic group of order  $F_n$  an element has full order precisely when its  $F_n/\ell$ -th power is nontrivial for every prime  $\ell \mid F_n$ , proving  $(\text{F}_n)$ .

For odd  $r$ , the nontrivial  $r$ -th roots  $v$  occur in inverse pairs, and the common value  $v/(v+1)^2$  of a pair is, by (B), a root of  $S_r$ . Frobenius permutes the pairs, so the displayed factors have coefficients in  $\mathbb{F}_2$ . Both sides of  $(\text{B}_r)$  are monic of degree  $(r-1)/2$ , which proves the factorization and the divisibility criterion.

Finally  $2^m \equiv -1 \pmod{d}$  for every nontrivial  $d \mid F_n$ . Because  $m$  is a power of two,  $\text{ord}_d(2) = 2m$ , and the least  $j > 0$  with  $2^j \equiv \pm 1 \pmod{d}$  is  $m$ . This is exactly the Frobenius orbit length of  $v/(v+1)^2$ : indeed

$$\frac{v}{(v+1)^2} = \frac{w}{(w+1)^2} \iff (v+w)(vw+1) = 0,$$

so only the inverse pair  $\{v, v^{-1}\}$  has the same image. This proves the equal-degree assertion.  $\square$

**Proposition 32** (coarse top-step data cannot force maximality). *[proved] For every divisor  $d > 1$  of  $F_n$  there are elements  $a_d \in \mathbb{F}_q$  and  $c_d \in \mathbb{F}_{q^2}$  such that*

$$[\mathbb{F}_2(a_d) : \mathbb{F}_2] = m, \quad [\mathbb{F}_2(c_d) : \mathbb{F}_2] = 2m, \quad c_d^q = c_d + 1, \quad c_d^2 + c_d = a_d,$$

$$\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_2}(a_d) = 1, \quad \mathrm{N}_{\mathbb{F}_{q^2}/\mathbb{F}_q}(c_d) = a_d,$$

*but the order of  $c_d \mathbb{F}_q^\times$  is exactly  $d$ . In particular all proper-divisor strata obey the same degree, trace, Artin–Schreier and relative-norm data as the Conway top step.*

*Proof.* Choose  $v$  of order  $d$  in the norm-one subgroup of  $\mathbb{F}_{q^2}^\times$  and put

$$c_d = \frac{1}{v+1}, \quad a_d = \frac{v}{(v+1)^2}.$$

Then  $v^q = v^{-1}$ , so  $c_d^q = c_d + 1$ ; the remaining displayed identities follow directly. The orbit calculation in the proof of Proposition 31 gives the two field degrees. Consequently  $T^2 + T + a_d$  is irreducible over  $\mathbb{F}_q$ , which is equivalent to the stated absolute trace, and  $(c_d + 1)/c_d = v$  gives quotient order  $d$ .  $\square$

**Proposition 33** (the exact recursively selected factor). *[proved] Set  $a_{-1} = 1$ ,  $a_i = \prod_{j \leq i} c_j$ , and let  $A_i$  be its minimal polynomial over  $\mathbb{F}_2$ , with  $A_{-1}(Y) = Y + 1$ . Then*

$$a_i^2 + a_{i-1}a_i + a_{i-1}^3 = 0$$

*and*

$$A_i(X) = \mathrm{Res}_Y(A_{i-1}(Y), X^2 + YX + Y^3). \quad (\mathbf{R}_i)$$

*Hence the unresolved Conway–Fermat lemma is the precise recursive claim that the factor selected by  $(\mathbf{R}_i)$  lies in the full-order fibotomic stratum, rather than in any equal-degree  $\mathfrak{F}_d$  with  $d \mid F_{i+1}$  and  $d < F_{i+1}$ .*

*Proof.* Multiplying  $c_i^2 + c_i = a_{i-1}$  by  $a_{i-1}^2$  gives the first identity. We prove the degree assertion and the resultant formula inductively from  $a_{-1} = 1$ . The resultant in  $(\mathbf{R}_i)$  is monic of degree  $2 \deg A_{i-1}$  and vanishes at  $a_i$ . Since  $c_i = a_i/a_{i-1} \notin E_{i-1}$ , one has  $a_i \notin E_{i-1}$ . But  $E_i/\mathbb{F}_2$  has degree  $2^{i+1}$ , and every proper subfield of  $E_i$  is contained in its unique index-two subfield  $E_{i-1}$ . Therefore  $a_i$  has full degree  $2^{i+1} = 2 \deg A_{i-1}$ , so the resultant is its minimal polynomial.  $\square$

**Proposition 34** (one-step maximality is not an inductive invariant). *[proved] Let*

$$C(Y) = Y^{16} + Y^{15} + Y^{12} + Y^{11} + Y^{10} + Y^8 + 1$$

*and apply the exact Conway resultant operator of Proposition 33:*

$$\begin{aligned} B(X) &= \mathrm{Res}_Y(C(Y), X^2 + YX + Y^3) \\ &= X^{32} + X^{31} + X^{30} + X^{27} + X^{26} + X^{24} + X^{23} + X^{22} \\ &\quad + X^{19} + X^{15} + X^{14} + X^{11} + X^8 + X^7 + 1. \end{aligned} \quad (\mathbf{NI}_1)$$

*Then  $C$  is an irreducible factor of the full-order fibotomic stratum  $\mathfrak{F}_{F_4}$ , where  $F_4 = 65537$ , whereas  $B$  is an irreducible factor of the proper-order stratum*



$$\mathfrak{F}_{6700417}, \quad 6700417 = F_5/641. \quad (\text{NI}_2)$$

Thus the exact one-step resultant can send a full-order predecessor factor to a proper-order target factor, already at the first composite Fermat level. In particular, predecessor maximality plus the transition formula alone is not an induction theorem. This is not a counterexample to the distinguished Conway path:  $C$  is an alternative factor at the  $A_3$  level and need not have the earlier ancestry of the actual  $A_3$ .

*Proof.* First, exact modular squaring gives

$$Y^{2^{16}} \equiv Y \pmod{C},$$

and

$$Y^{2^{56}} + Y \equiv R := Y^{15} + Y^{14} + Y^{13} + Y^9 + Y^8 + Y^2 + Y \pmod{C}.$$

The Bézout identity  $UC + VR = 1$  holds for

$$\begin{aligned} U &= Y^{14} + Y^{13} + Y^{12} + Y^{11} + Y^7 + Y^6 + Y^5 + Y^4 \\ &\quad + Y^3 + Y + 1, \\ V &= Y^{15} + Y^{14} + Y^{12} + Y^{11} + Y^9 + Y^8 + Y^5 + Y^3 + Y + 1. \end{aligned}$$

Rabin's criterion, for which two is the only prime divisor of sixteen, therefore proves that  $C$  is irreducible.

The Fibonacci polynomials satisfy, in characteristic two,

$$S_{2r} = S_r^2, \quad S_{2r+1} = S_{r+1}^2 + Y S_r^2,$$

and hence

$$S_{2^m+1}(Y) = 1 + \sum_{j=0}^{m-1} Y^{2^j}. \quad (\text{NI}_3)$$

For a root  $y$  of  $C$ , the coefficient of  $Y^{15}$  gives  $\text{Tr}_{\mathbb{F}_{2^{16}}/\mathbb{F}_2}(y) = 1$ . Equation (NI<sub>3</sub>) therefore gives  $S_{65537}(y) = 0$ . Since 65537 is prime,  $y$  belongs to the full-order stratum  $\mathfrak{F}_{65537}$ .

Now choose  $z$  with  $z^2 + z = y$  and put  $x = yz$ . The trace just computed makes  $Z^2 + Z + y$  irreducible over  $\mathbb{F}_{2^{16}}$ . Thus  $x \notin \mathbb{F}_{2^{16}}$ ; because every proper subfield of  $\mathbb{F}_{2^{32}}$  lies in  $\mathbb{F}_{2^{16}}$ ,  $x$  has degree thirty-two over  $\mathbb{F}_2$ . Moreover

$$x^2 + yx + y^3 = y^2(z^2 + z + y) = 0. \quad (\text{NI}_4)$$

Modulo the relation in (NI<sub>4</sub>), direct reduction gives

$$B(X) = C(Y)(Q_1(Y)X + Q_0(Y)), \quad (\text{NI}_5)$$

where

$$\begin{aligned}
Q_1 &= Y^{29} + Y^{28} + Y^{25} + Y^{24} + Y^{21} + Y^{18} + Y^{17} + Y^{16} \\
&\quad + Y^{15} + Y^{14} + Y^{11} + Y^{10} + Y^9 + Y^6, \\
Q_0 &= Y^{32} + Y^{31} + Y^{29} + Y^{28} + Y^{27} + Y^{26} + Y^{25} + Y^{23} \\
&\quad + Y^{20} + Y^{14} + Y^{12} + Y^{11} + Y^{10} + Y^9 + 1.
\end{aligned}$$

Thus  $B(x) = 0$ . Its displayed degree is thirty-two, so it is the minimal polynomial of  $x$ , and consequently the stated resultant.

It remains to identify the target stratum. From a pair  $(f, g) = (S_r, S_{r+1})$ , put

$$A = f^2, \quad D = g^2 + Xf^2.$$

Reading one further binary digit sends the pair to  $(A, D)$  for a zero and to  $(D, g^2)$  for a one. Repeated reduction modulo  $B$  along the binary word

$$6700417 = (11001100011110110000001)_2$$

ends at the exact residue pair

$$(S_{6700417}, S_{6700418}) \equiv (00000000, 1875a343) \pmod{B}, \quad (\text{NI}_6)$$

where a hexadecimal bit in position  $j$  is the coefficient of  $X^j$ . This is an  $O(\log 6700417)$  polynomial-identity certificate, not field sampling. Hence  $S_{6700417}(x) = 0$ . The integer 6700417 is prime, so  $x$  belongs to the exact order-6700417 fibotomic stratum. Finally, if  $v = (z + 1)/z$ , then

$$y = \frac{v}{(v+1)^2}, \quad x = yz = \frac{v}{(v+1)^3},$$

showing directly that this is the same fibotomic/elliptic transition as the Conway resultant.  $\square$

**Proposition 35** (complete ancestry and exact trace-fibre exhaustion). *[proved] For  $i \geq 0$  and  $-1 \leq j < i$ , the distinguished elements satisfy*

$$\text{Tr}_{E_i/E_j}(a_i) = a_j, \quad \text{Norm}_{E_i/E_j}(a_i) = a_j^{3^{i-j}}. \quad (\text{AN}_1)$$

Conversely, fix  $i \geq 0$  and  $x_i \in E_i$ . Put

$$x_j = \text{Tr}_{E_i/E_j}(x_i) \quad (-1 \leq j < i).$$

If  $x_{-1} = 1$  and

$$\text{Norm}_{E_j/E_{j-1}}(x_j) = x_{j-1}^3 \quad (0 \leq j \leq i), \quad (\text{AN}_2)$$

then  $x_i$  is an  $\mathbb{F}_2$ -Frobenius conjugate of  $a_i$ . Thus  $(\text{AN}_1)$ – $(\text{AN}_2)$  are a necessary-and-sufficient ancestry invariant for the actual resultant factor.

Now fix  $n \geq 1$ , put

$$F = E_{n-1} = \mathbb{F}_q, \quad K = E_n = \mathbb{F}_{q^2},$$

and let  $\ell \mid q+1$ . Set

$$H = (K^\times)^\ell, \quad U = \ker \text{Norm}_{K/F}.$$

For every  $t \in F^\times$ ,

$$\#\{h \in H : \text{Tr}_{K/F}(h) = t\} = \frac{q+1}{\ell} - 1. \quad (\text{AN}_3)$$

If  $t = a_{n-1}$ , every element counted in (AN<sub>3</sub>) has absolute trace one and is a normal, hence full-degree, element of  $K/\mathbb{F}_2$ . Thus, whenever  $q + 1$  is composite, full degree, normality, the complete Frobenius-orbit length, and the actual relative-trace value all occur among  $\ell$ -th powers.

The normalized relative norms of the elements in (AN<sub>3</sub>) form exactly

$$\mathcal{B}_\ell = \left\{ \frac{u}{(u+1)^2} : u \in U^\ell, u \neq 1 \right\}, \quad |\mathcal{B}_\ell| = \frac{1}{2} \left( \frac{q+1}{\ell} - 1 \right). \quad (\text{AN}_4)$$

More precisely, an element  $h$  of relative trace  $t$  is an  $\ell$ -th power and has relative norm  $t^3$  if and only if  $t \in \mathcal{B}_\ell$ . For  $t = a_{n-1}$  this is equivalent to

$$S_{(q+1)/\ell}(a_{n-1}) = 0. \quad (\text{AN}_5)$$

Finally  $[a_n] = [c_n]$  in  $K^\times/K^{\times\ell}$ . Consequently the entire earlier ancestry is recoverable exactly, but after the last quadratic step its power-residue content is precisely the original fibotomic membership (AN<sub>5</sub>). Trace, separate norm, normality, and Frobenius-orbit invariants give no additional obstruction.

*Proof.* The relation

$$a_i^2 + a_{i-1}a_i + a_{i-1}^3 = 0$$

is the minimal polynomial of  $a_i$  over  $E_{i-1}$ . It gives immediate trace  $a_{i-1}$  and norm  $a_{i-1}^3$ ; transitivity gives (AN<sub>1</sub>).

For the converse, transitivity of trace gives  $\text{Tr}_{E_j/E_{j-1}}(x_j) = x_{j-1}$ . Together with (AN<sub>2</sub>) this says that  $x_j$  is a root of

$$X^2 + x_{j-1}X + x_{j-1}^3.$$

Starting from  $x_{-1} = a_{-1} = 1$ , induct on  $j$ . If  $x_{j-1}$  is a Frobenius conjugate of  $a_{j-1}$ , the displayed polynomial is the corresponding Frobenius conjugate of the polynomial for  $a_j$ ; its two roots are the two lifts of that conjugacy to  $E_j$ . Hence  $x_j$ , and finally  $x_i$ , is a Frobenius conjugate of the distinguished element.

Since  $\gcd(\ell, q-1) = 1$ , one has  $F^\times \subset H$ . Multiplication by any  $t \in F^\times$  bijects the trace-one and trace- $t$  fibres in  $H$ . In characteristic two,  $\text{Tr}_{K/F}(h) = 0$  is equivalent to  $h^q = h$ , so the nonzero trace-zero elements are exactly  $F^\times$ , all of which lie in  $H$ . If the common nonzero-fibre size is  $M$ , then

$$\frac{q^2 - 1}{\ell} = |H| = (q-1) + (q-1)M,$$

which gives (AN<sub>3</sub>).

The distinguished  $a_{n-1}$  has absolute trace one by (AN<sub>1</sub>). Thus every  $h$  in its relative-trace fibre also has absolute trace one. To justify normality, let  $N = [K : \mathbb{F}_2]$ , a power of two, and let  $\rho$  generate its cyclic Galois group. After choosing one normal generator,  $K$  is the regular module over

$$\mathbb{F}_2[\langle \rho \rangle] \simeq \mathbb{F}_2[X]/((X+1)^N),$$

a local ring. An element is a normal generator exactly when its group-ring coordinate is a unit, equivalently when multiplication by  $(\rho+1)^{N-1}$  does not kill it. Since

$$(\rho+1)^{N-1} = 1 + \rho + \cdots + \rho^{N-1},$$

this last value is its absolute trace. Hence absolute trace one is equivalent to normality.

Write  $h = tz$  in a nonzero trace fibre. Then  $\text{Tr}_{K/F}(z) = 1$ , so  $z^q = z + 1$  and

$$\text{Norm}_{K/F}(h) = t^2 z(z + 1).$$

Put  $u = (z + 1)/z \in U$ ; then  $z = 1/(u + 1)$  and

$$z(z + 1) = \frac{u}{(u + 1)^2}. \quad (\text{AN}_6)$$

We claim  $z \in H$  if and only if  $u \in U^\ell$ . The forward implication follows from  $u = z^{q-1}$ . Conversely, if  $u \in U^\ell$ , then  $u \in H$ . The right side of (AN<sub>6</sub>) lies in  $F^\times \subset H$ , so in  $K^\times/H$  one has

$$[u] = 2[u + 1].$$

Because  $\ell$  is odd, multiplication by two is invertible; hence  $[u] = 0$  forces  $[u + 1] = 0$ , and  $z = (u + 1)^{-1} \in H$ . The map  $u \mapsto u/(u + 1)^2$  identifies precisely the inverse pair  $\{u, u^{-1}\}$ , proving (AN<sub>4</sub>).

The equality  $\text{Norm}(h) = t^3$  is, after division by  $t^2$ , exactly  $t = z(z + 1)$ . This proves the first assertion following (AN<sub>4</sub>). The set  $U^\ell$  has order  $(q + 1)/\ell$ , and the Fibonacci-polynomial factorization identifies its nontrivial inverse pairs with the roots of  $S_{(q+1)/\ell}$ , proving (AN<sub>5</sub>). Finally  $a_n = a_{n-1}c_n$  and  $a_{n-1} \in F^\times \subset H$ , so  $[a_n] = [c_n]$ .  $\square$

**Proposition 36** (supersingular elliptic form of the Conway transition). *[proved] For  $i \geq 0$  put*

$$u_i := \frac{c_i + 1}{c_i}.$$

*Fix  $n \geq 1$ , set  $Q = 2^{2^{n-1}}$ ,  $v = u_{n-1}$  and  $u = u_n$ . Then the exact Conway recursion is*

$$\frac{u}{(u + 1)^2} = \frac{v}{(v + 1)^3}, \quad (\text{E})$$

*or equivalently*

$$u(v + 1)^3 = v(u + 1)^2, \quad u + u^{-1} = v^2 + v + 1 + v^{-1}.$$

*The normalization of this correspondence is the fixed supersingular elliptic curve over  $\mathbb{F}_2$*

$$\mathcal{E} : \quad y^2 + y = x^3 + x^2. \quad (\text{W})$$

*The maps to the  $v$ - and  $u$ -lines have degrees two and three, respectively; in particular the transition has no rational section recovering  $u$  from  $v$ .*

*On the original  $(u, v)$  correspondence, the rational map*

$$\mathcal{A}(u, v) = \left( \frac{u + v}{uv + 1}, v^{-1} \right)$$

*extends to an automorphism of order four, with*

$$\mathcal{A}^2(u, v) = (u^{-1}, v).$$

*For the distinguished Conway point  $P_n = (u_n, u_{n-1})$  one has*

$$\text{Frob}_Q(P_n) = \mathcal{A}(P_n) \quad \text{or} \quad \mathcal{A}^{-1}(P_n). \quad (\text{Q})$$

*Thus, at  $P_n$ , relative Frobenius is one of the two quarter-turns  $\mathcal{A}^{\pm 1}$  on the elliptic correspondence.*

*Proof.* Write  $a_i = \prod_{j \leq i} c_j$ . The identities

$$a_{n-1} = \frac{u}{(u+1)^2}, \quad a_{n-1} = a_{n-2}c_{n-1} = \frac{v}{(v+1)^2} \frac{1}{v+1}$$

give (E); inversion gives the second displayed form.

Put  $U = u + 1$ ,  $V = v + 1$ , and blow up the affine singularity  $(U, V) = (0, 0)$  by  $X = U/V$ ,  $Y = V$ . After division by  $V^2$  the equation becomes

$$XY^2 + (X^2 + 1)Y + X^2 = 0.$$

Its homogenization

$$H = XY^2 + (X^2 + Z^2)Y + X^2Z$$

has partial derivatives  $(Y^2, X^2 + Z^2, X^2)$  and is therefore smooth. The point  $[1 : 1 : 0]$  is a flex. The linear change

$$(x, y, z) = (Z, X, X + Y + Z)$$

takes  $H = 0$  to  $y^2z + yz^2 = x^3 + x^2z$ , which is (W). This curve has five  $\mathbb{F}_2$ -points, hence trace  $-2$ , and is supersingular. The original equation has degrees two in  $u$  and three in  $v$ , proving the degrees of the two projections. A rational section of the nontrivial degree-two finite cover would induce a retraction of its degree-two function-field extension, impossible for a proper finite field extension.

Direct substitution in (E) gives

$$\frac{\mathcal{A}_u}{(\mathcal{A}_u + 1)^2} = v \frac{u}{(u + 1)^2} = \frac{v^{-1}}{(v^{-1} + 1)^3},$$

so  $\mathcal{A}$  preserves the correspondence. A second substitution gives  $\mathcal{A}^2(u, v) = (u^{-1}, v)$ , proving the order assertion. Thus this birational self-map and its birational inverse extend over the smooth projective normalization.

Finally  $v^Q = v^{-1}$  and

$$\left( \frac{u}{(u+1)^2} \right)^Q = \left( \frac{v}{(v+1)^3} \right)^Q = v \frac{u}{(u+1)^2}.$$

The fibres of  $z \mapsto z/(z+1)^2$  are the inverse-pairs  $\{z, z^{-1}\}$ . The first coordinate of  $\text{Frob}_Q(P_n)$  is therefore either  $\mathcal{A}_u(P_n)$  or its inverse, while the second coordinate is  $v^{-1}$ , proving (Q).  $\square$

**Proposition 37** (new-prime character neutrality and the exact Kummer target). *[proved] Retain Proposition 36 and let  $\ell$  be a prime divisor of  $Q^2 + 1 = F_n$ . If  $\chi : E_n^\times = \mathbb{F}_{Q^4}^\times \rightarrow \mu_\ell$  has exact order  $\ell$ , and*

$$\alpha = \chi(u), \quad \beta = \chi(u+1),$$

then

$$\alpha = \beta^2, \quad \beta^{Q^2} = \beta^{-1}. \tag{K}$$

The second identity is tautological because  $Q^2 \equiv -1 \pmod{\ell}$ ; in particular the exact lower transition and the quarter-turn Frobenius symmetry do not by themselves exclude  $\beta = 1$ .

The following are equivalent:

- (i) the order  $\delta_n$  contains the full  $\ell$ -primary part of  $F_n$ ;
- (ii)  $u_n \notin U_n^\ell$ , where  $U_n = \ker N_{E_n/E_{n-1}}$ ;

(iii)  $c_n \notin (E_n^\times)^\ell$ ;

(iv)  $X^\ell - c_n$  is irreducible over  $E_n$ .

Consequently the all-level Conway–Fermat theorem is exactly the nontriviality, for every  $\ell \mid F_n$ , of this new Kummer symbol at the distinguished point of (W). No lower-field multiplicative character can supply its value.

*Proof.* The lower field has order  $Q^2$ , and  $\gcd(\ell, Q^2 - 1) = 1$ , so  $\chi$  is trivial on all of  $E_{n-1}^\times$ , in particular on every expression in  $v$ . Applying  $\chi$  to (E) gives  $\alpha = \beta^2$ . Since  $u^{Q^2} = u^{-1}$ ,

$$(u+1)^{Q^2} = u^{-1} + 1 = \frac{u+1}{u},$$

which gives the other identity in (K). It reduces to  $\beta^{-1} = \beta^{-1}$  modulo  $\ell$ .

In the cyclic group  $U_n$  of order  $F_n$ , nonmembership in  $U_n^\ell$  is equivalent to having the full  $\ell$ -primary order. Here

$$U_n^\ell = U_n \cap (E_n^\times)^\ell. \quad (\text{K}_0)$$

Indeed, if  $u = x^\ell \in U_n$ , then  $N_{E_n/E_{n-1}}(x)^\ell = 1$ ; because  $\gcd(\ell, Q^2 - 1) = 1$ , this forces  $N(x) = 1$ , so  $x \in U_n$ . The reverse inclusion is immediate. Moreover (E) lies in the lower field, hence is an  $\ell$ -th power in  $E_n$ . In  $E_n^\times/(E_n^\times)^\ell$  it therefore says

$$[u] = 2[u+1].$$

Since  $\ell$  is odd,  $u$  is an  $\ell$ -th power exactly when  $u+1$  is. But  $u+1 = c_n^{-1}$ , proving the equivalence of the first three assertions. Finally  $\mu_\ell \subset E_n$ , so the prime-degree Kummer criterion proves the equivalence with irreducibility of  $X^\ell - c_n$ .  $\square$

**Proposition 38** (four-point saturation of a Fermat failure). *[proved] Fix  $n \geq 2$ , put*

$$c = c_n, \quad d = c_{n-1}, \quad b = a_{n-2}, \quad Q = |E_{n-2}| = 2^{2^{n-1}}.$$

*Then, for one  $\epsilon \in \mathbb{F}_2$ ,*

$$c^Q = c + d + \epsilon, \quad c^{Q^2} = c + 1. \quad (\text{FP}_1)$$

*Consequently, if  $\ell \mid F_n$  and  $c$  is an  $\ell$ -th power in  $E_n$ , the complete affine plane*

$$c + \mathbb{F}_2 + \mathbb{F}_2 d = \{c, c+1, c+d, c+d+1\} \quad (\text{FP}_2)$$

*consists of  $\ell$ -th powers. Its product is nevertheless the old-field element*

$$N_{E_n/E_{n-2}}(c) = b^3, \quad (\text{FP}_3)$$

*which is automatically an  $\ell$ -th power. Thus the exact four-point saturation and its norm are compatible with failure; they do not supply the missing nonresidue.*

*Proof.* The tower equations give

$$d^2 + d = b, \quad c^2 + c = bd = d^3 + d^2, \quad d^Q = d + 1.$$

For  $s = c^Q + c$ , subtraction of the equation for  $c$  from its  $Q$ -th power gives  $s^2 + s = b$ . Hence  $s \in \{d, d+1\}$ , proving the first identity in (FP<sub>1</sub>); applying  $Q$ -Frobenius once more gives the second.

Frobenius preserves the subgroup of  $\ell$ -th powers, so one power value forces all four values in  $(\mathbb{F}_2)$ . Finally,

$$c(c+1) = bd, \quad (c+d)(c+d+1) = b(d+1),$$

and their product is  $b^2d(d+1) = b^3$ . Since  $\ell \mid Q^2 + 1$  is coprime to  $|E_{n-2}^\times| = Q - 1$ , every old-field element is an  $\ell$ -th power in  $E_n$ .  $\square$

**Proposition 39** (CM form and the Kummer-cover obstruction). *[proved] On  $\mathcal{E} : y^2 + y = x^3 + x^2$ , let  $O$  be the origin and  $T = (0, 0)$ . Then  $\mathcal{E}(\mathbb{F}_2) = \langle T \rangle \cong \mathbb{Z}/5\mathbb{Z}$ , with*

$$-T = (0, 1), \quad 2T = (1, 1), \quad -2T = (1, 0).$$

On the normalized Conway correspondence, the function  $\mathfrak{c} = 1/(u+1)$  is

$$\mathfrak{c} = \frac{x}{x^2 + x + y}, \quad \text{div}(\mathfrak{c}) = [-T] + 2[O] - 3[-2T]. \quad (\text{EC}_1)$$

The automorphism  $\mathcal{A}$  satisfies

$$\mathcal{A}^2(P) = T - P, \quad \mathfrak{c}(\mathcal{A}^2P) = \mathfrak{c}(P) + 1. \quad (\text{EC}_2)$$

If  $\pi$  is 2-power Frobenius and  $i = \pi + 1$ , then  $i^2 = -1$ ,  $i(T) = 2T$ , and, after possibly replacing  $\mathcal{A}$  by its inverse,

$$\mathcal{A} = \tau_{2T} \circ i.$$

Both orientations fix  $R_0 = -2T$ .

For  $n \geq 4$ , set

$$Q = 2^{2^{n-1}}, \quad R = \sqrt{Q} = 2^{2^{n-2}},$$

and let  $P_n$  be the distinguished Conway point. If  $X_n = P_n - R_0$ , then

$$\text{Frob}_Q(X_n) = \pm iX_n, \quad X_n \in \ker([R] \mp i). \quad (\text{EC}_3)$$

Each kernel is cyclic of order  $R^2 + 1 = Q + 1$ .

For every prime  $\ell \mid Q^2 + 1$ , the Kummer cover

$$\mathcal{C}_\ell : \quad z^\ell = \mathfrak{c}$$

is geometrically connected, ramified exactly above  $-T, O, -2T$ , and has genus

$$g(\mathcal{C}_\ell) = \frac{3\ell - 1}{2}. \quad (\text{EC}_4)$$

The automorphism  $\mathcal{A}$  does not lift to  $\mathcal{C}_\ell$ . Hence the quarter-turn on the base supplies no action on the Kummer fibre above  $P_n$ ; the surviving Conway condition is exactly

$$\mathfrak{c}(P_n) = c_n \notin (\mathbb{F}_{Q^4}^\times)^\ell. \quad (\text{EC}_5)$$

It is the Artin-symbol problem for one distinguished fibre of this ramified cover, not a consequence of the CM action on  $\mathcal{E}$ .

*Proof.* In the blow-up coordinates of Proposition 36,  $U = u + 1$ ,  $V = v + 1$ ,  $X = U/V$ ,  $Y = V$ , and  $(x : y : z) = (Z : X : X + Y + Z)$ . On  $z = 1$ , homogeneous representatives give

$$V = \frac{1 + x + y}{x}, \quad U = \frac{y(1 + x + y)}{x^2}.$$

Since  $y(1 + x + y) = xy + x^3 + x^2 = x(x^2 + x + y)$ , this proves the formula for  $\mathfrak{c} = 1/U$ . With  $d = x^2 + x + y$ , direct intersection gives

$$\operatorname{div}(x) = [T] + [-T] - 2[O], \quad \operatorname{div}(d) = [T] + 3[-2T] - 4[O].$$

Indeed  $d = 0$  gives  $y = x^2 + x$ , whence the curve equation gives  $x(x + 1)^3 = 0$ , and  $d$  has pole order four at  $O$ . Subtraction proves (EC<sub>1</sub>).

The function  $v = (y + 1)/x$  has pole divisor  $[O] + [T]$ , so the deck involution of its degree-two map is  $J(P) = T - P$ . Since  $\mathcal{A}^2$  fixes  $v$  and is nontrivial, it equals  $J$ . Also  $\mathcal{A}^2$  sends  $u$  to  $u^{-1}$ , and therefore

$$\mathfrak{c}(\mathcal{A}^2 P) = \frac{1}{u(P)^{-1} + 1} = \frac{u(P)}{u(P) + 1} = \mathfrak{c}(P) + 1,$$

proving (EC<sub>2</sub>).

The Frobenius polynomial of  $\mathcal{E}/\mathbb{F}_2$  is  $\pi^2 + 2\pi + 2 = 0$ . Thus  $i = \pi + 1$  satisfies  $i^2 = -1$ , and it acts on  $\mathcal{E}(\mathbb{F}_2)$  as multiplication by two. Write  $\mathcal{A} = \tau_S \circ \alpha$ , with  $\alpha(O) = O$ . As  $\mathcal{A}$  is defined over  $\mathbb{F}_2$ ,  $\alpha$  commutes with  $\pi$ . The centralizer of  $\pi$  in the supersingular endomorphism algebra is  $\mathbb{Q}(i)$ , whose integral units are  $\{\pm 1, \pm i\}$ . The linear part of  $\mathcal{A}^2$  is  $[-1]$ , so  $\alpha = \pm i$ . Comparing translation parts gives

$$\mathcal{A} = \tau_{2T} \circ i \quad \text{or} \quad \mathcal{A} = \tau_{-T} \circ (-i);$$

these are inverse and both fix  $R_0 = -2T$ .

For  $n \geq 4$ ,  $2^{n-1}$  is divisible by eight. Since  $\pi^8 = [16]$ ,

$$\pi^{2^{n-1}} = [2^{2^{n-2}}] = [R].$$

Translation of  $R_0$  to the origin makes  $\mathcal{A}^{\pm 1}$  the linear maps  $\pm i$ , proving (EC<sub>3</sub>). The separable endomorphisms  $[R] \mp i$  have degree  $R^2 + 1$ . For every odd prime  $p$ , the Tate module  $T_p(\mathcal{E})$  is free of rank one over  $\mathbb{Z}_p[i]$  (the polynomial  $X^2 + 1$  is separable away from 2). Hence the  $p$ -primary kernel is the additive quotient  $\mathbb{Z}_p[i]/(R \mp i)$ . The matrix of  $R \mp i$  over  $\mathbb{Z}_p$  has entries with gcd one and determinant  $R^2 + 1$ , so its first Smith invariant is one. Combining the odd primary parts shows that the geometric kernel is cyclic of order  $R^2 + 1$ .

Every prime  $\ell \mid Q^2 + 1$  is coprime to 15. Thus the valuations 1, 2, -3 in (EC<sub>1</sub>) are nonzero modulo  $\ell$ . Kummer theory makes  $z^\ell = \mathfrak{c}$  geometrically connected and totally ramified at exactly those three points. Riemann–Hurwitz gives

$$2g(\mathcal{C}_\ell) - 2 = 3(\ell - 1),$$

which proves (EC<sub>4</sub>).

It remains to prove non-lifting. Choose  $\mathcal{A} = \tau_{2T} \circ i$ , put  $B_j = \mathcal{A}^j(O)$ , and retain  $R_0 = -2T$ . Then

$$(B_0, B_1, B_2, B_3) = (O, 2T, T, -T).$$

For  $D_j = \operatorname{div}(\mathfrak{c} \circ \mathcal{A}^{-j})$ , pullback gives

$$D_j = 2[B_j] + [B_{j+3}] - 3[R_0]. \tag{EC_6}$$



If  $\sum_{j=0}^3 r_j D_j \equiv 0 \pmod{\ell}$ , comparison at the four  $B_j$ 's gives  $2r_j + r_{j+1} = 0$  cyclically. Iteration gives  $15r_0 = 0$ , and  $\ell \nmid 15$  forces every  $r_j = 0$ . Hence

$$\mathbf{c}, \quad \mathbf{c} \circ \mathcal{A}, \quad \mathbf{c} \circ \mathcal{A}^2, \quad \mathbf{c} \circ \mathcal{A}^3$$

are linearly independent in

$$\overline{\mathbb{F}}_2(\mathcal{E})^\times / \overline{\mathbb{F}}_2(\mathcal{E})^{\times \ell}.$$

A lift of  $\mathcal{A}$  would normalize the cyclic deck group, hence send  $z$  to  $gz^r$  for  $r \in (\mathbb{Z}/\ell\mathbb{Z})^\times$ , and would require  $\mathbf{c} \circ \mathcal{A} = g^\ell \mathbf{c}^r$ , contradicting this independence.

Finally, set

$$\chi(t) = t^{(Q^4-1)/\ell}, \quad \tau = \chi(\mathbf{c}(P_n)).$$

Since  $\text{Frob}_Q^2(P_n) = \mathcal{A}^2(P_n)$ ,  $(\text{EC}_2)$  gives  $\mathbf{c}(P_n)^{Q^2} = \mathbf{c}(P_n) + 1$ . Because  $Q^2 \equiv -1 \pmod{\ell}$ , the four CM/Frobenius character values are only

$$\tau, \quad \tau^{\pm Q}, \quad \tau^{-1}, \quad \tau^{\mp Q}.$$

All norm identities among them are tautological. This proves neither  $\tau = 1$  nor  $\tau \neq 1$ : the desired assertion remains  $(\text{EC}_5)$ .  $\square$

**Proposition 40** (Miller-unit and generalized-Jacobian location). *[proved] Retain Proposition 39, put  $S = -2T$ ,  $q = Q^2$ , and  $K_n = \mathbb{F}_{Q^4} = \mathbb{F}_{q^2}$ . Then*

$$\mathbf{c}^{-1} = f_{3,S}$$

*up to a nonzero scalar, where  $f_{3,S}$  is the Miller tripling function with divisor*

$$\text{div}(f_{3,S}) = 3[S] - [3S] - 2[O]. \tag{GJ_1}$$

*For every prime  $\ell \mid q + 1$ ,*

$$\#\mathcal{E}(K_n) = (q - 1)^2, \quad \mathcal{E}(K_n)[\ell] = 0, \quad [\ell] : \mathcal{E}(K_n) \xrightarrow{\sim} \mathcal{E}(K_n). \tag{GJ_2}$$

*Let  $B = \{-T, O, -2T\}$  and  $U = \mathcal{E} \setminus B$ . The Kummer class of  $\mathbf{c}$  on  $U$  has nonzero tame residue vector*

$$(1, 2, -3) \pmod{\ell}. \tag{GJ_3}$$

*It is therefore ramified and does not come from an unramified  $\mu_\ell$ -class on  $\mathcal{E}$ . In the generalized Jacobian*

$$1 \longrightarrow \mathbb{G}_m^2 \longrightarrow J_B \longrightarrow \mathcal{E} \longrightarrow 0, \tag{GJ_4}$$

*Hilbert 90 and  $(\text{GJ}_2)$  give*

$$J_B(K_n)/\ell J_B(K_n) \cong (K_n^\times / K_n^{\times \ell})^2. \tag{GJ_5}$$

*Thus the residue vector has a genuinely ramified toric projection; the ordinary  $K_n$ -rational elliptic quotient cannot detect that projection. Ordinary  $K_n$ -rational Tate–Lichtenbaum and formal-group data cannot by themselves decide  $(\text{EC}_5)$ . This does not assert that the ramified class evaluates nontrivially at the distinguished point, nor does it exclude a calculation using non-rational  $\mathcal{E}[\ell]$  or the full extension structure of  $J_B$ . An elliptic completion must evaluate that generalized-Jacobian Miller-unit residue.*

*Proof.* The point  $T$  has order five, so  $3S = 3(-2T) = -T$ . Equation (EC<sub>1</sub>) therefore gives

$$\operatorname{div}(\mathfrak{c}^{-1}) = 3[S] - [3S] - 2[O],$$

which proves (GJ<sub>1</sub>).

For  $n \geq 4$ , the Frobenius relation used in Proposition 39 gives

$$\pi^{2^{n+1}} = [q].$$

This is Frobenius on  $K_n = \mathbb{F}_{2^{2^{n+1}}}$ , so

$$\#\mathcal{E}(K_n) = \deg([q] - [1]) = (q - 1)^2.$$

As  $\ell \mid q + 1$  is odd, it is coprime to  $q - 1$ . This proves (GJ<sub>2</sub>); multiplication by  $\ell$  is likewise an automorphism of the characteristic-two formal group because its linear coefficient is a unit.

The divisor (EC<sub>1</sub>) gives the residue vector (GJ<sub>3</sub>), which is nonzero because  $\ell \nmid 6$ . The standard generalized-Jacobian sequence for the three-point reduced rational modulus is (GJ<sub>4</sub>). Its torus is split; Hilbert 90 makes the map on  $K_n$ -points surjective, and the kernel created on passing modulo  $\ell$  vanishes because  $\mathcal{E}(K_n)[\ell] = 0$ . This proves (GJ<sub>5</sub>) and the stated scope of the toric obstruction.  $\square$

**Proposition 41** (exact extension character and cohomological exhaustion). *[proved] Retain Proposition 40 and put*

$$D_1 = [-T] - [-2T], \quad D_2 = [O] - [-2T]$$

*in  $\operatorname{Div}^0(B)$ . The character lattice of the split torus in  $J_B$  is*

$$X^*(\mathbb{G}_m^2) = \operatorname{Div}^0(B) = \mathbb{Z}D_1 \oplus \mathbb{Z}D_2,$$

*and the extension is classified, under  $\mathcal{E} \cong \operatorname{Pic}^0(\mathcal{E})$ , by*

$$\lambda(aD_1 + bD_2) = (a + 2b)T. \tag{GJ_6}$$

*Thus its entire extension class is rational five-torsion. Moreover*

$$r := \operatorname{div}(\mathfrak{c}) = D_1 + 2D_2, \quad \lambda(r) = 5T = O,$$

*so  $r$  defines a character  $\varphi_r : J_B \rightarrow \mathbb{G}_m$ . If  $j_T : U \rightarrow J_B$  is the generalized Abel map normalized by  $j_T(T) = 0$ , then*

$$\varphi_r(j_T(P)) = \mathfrak{c}(P). \tag{GJ_7}$$

*For every prime  $\ell \mid q + 1$ , even the non-rational geometric  $\mathcal{E}[\ell]$  supplies no further rational Kummer class:*

$$H^1(K_n, \mu_\ell^2) \xrightarrow{\sim} H^1(K_n, J_B[\ell]), \quad J_B(K_n)/\ell J_B(K_n) \cong (K_n^\times/K_n^{\times\ell})^2. \tag{GJ_8}$$

*In torus coordinates dual to  $(D_1, D_2)$ , the character is  $(x, y) \mapsto xy^2$ , hence the nonzero linear functional  $(x, y) \mapsto x + 2y$  modulo  $\ell$ . Naturality of Kummer theory gives*

$$\varphi_{r,*}(\delta_J(j_T(P_n))) = P_n^*[\mathfrak{c}] = [\mathfrak{c}(P_n)] \in K_n^\times/K_n^{\times\ell}. \tag{GJ_9}$$

*Consequently (EC<sub>5</sub>) is exactly the assertion that the normalized Abel class of  $P_n$  avoids the hyperplane  $x + 2y = 0$ . Torus translation above any fixed elliptic point realizes every value of this functional. Hence the CM point, the full extension class, and non-rational elliptic  $\ell$ -torsion do not force (EC<sub>5</sub>); the remaining datum is the explicit toric coordinate of this one distinguished normalized Abel lift.*

*Proof.* For a reduced three-point rational modulus, the generalized-Jacobian torus is  $(\mathbb{G}_m^3)/\mathbb{G}_m$ , with character lattice  $\text{Div}^0(B)$ . The standard extension-class map sends a boundary divisor to its class in  $\text{Pic}^0(\mathcal{E})$ . Here

$$(-T) - (-2T) = T, \quad O - (-2T) = 2T,$$

which proves (GJ<sub>6</sub>). Equation (EC<sub>1</sub>) gives  $r = D_1 + 2D_2$ , and its image is  $T + 4T = O$ . Characters of  $J_B$  are precisely the torus characters whose pushout extension class vanishes, so  $r$  yields  $\varphi_r$ . The generalized-Albanese universal property gives (GJ<sub>7</sub>) after normalization at  $T$ . Indeed,  $T \notin B$  and  $\mathfrak{c}(T) \in \mathbb{F}_2^\times = \{1\}$ .

The  $K_n$ -Frobenius on  $\mathcal{E}$  is  $[q]$ , and  $q \equiv -1 \pmod{\ell}$ . It therefore acts as  $[-1]$  on  $\mathcal{E}[\ell]$ . Since  $\ell$  is odd,

$$\mathcal{E}[\ell](K_n) = 0, \quad H^1(K_n, \mathcal{E}[\ell]) = \mathcal{E}[\ell]/(\text{Frob} - 1)\mathcal{E}[\ell] = 0.$$

The cohomology sequence of

$$0 \longrightarrow \mu_\ell^2 \longrightarrow J_B[\ell] \longrightarrow \mathcal{E}[\ell] \longrightarrow 0$$

then proves the first isomorphism in (GJ<sub>8</sub>). Lang's theorem and Kummer theory prove the second. The natural Kummer square for  $\varphi_r$ , together with (GJ<sub>7</sub>), gives (GJ<sub>9</sub>). Finally  $\mathbb{G}_m^2(K_n)/\ell \rightarrow J_B(K_n)/\ell$  is an isomorphism: surjectivity uses bijectivity of  $[\ell]$  on  $\mathcal{E}(K_n)$ , and injectivity uses  $\mathcal{E}(K_n)[\ell] = 0$ . The character  $(x, y) \mapsto xy^2$  is surjective on the  $\ell$ -quotient, proving the fixed-fibre statement and the asserted exact boundary.  $\square$

**Proposition 42** (explicit Rosenlicht coordinate and its reciprocity boundary). *[proved] Retain Proposition 41, write  $S = -2T$ ,  $P = P_n$ , and put  $X = P - S$ . For  $\ell \mid q + 1$ , let  $Z \in \mathcal{E}(K_n)$  be the unique point with*

$$[\ell]Z = P - T, \quad W = T + Z. \tag{RC_1}$$

*If  $e\ell \equiv 1 \pmod{5(Q+1)}$ , then*

$$Z = [e]X + [2e]T, \quad W = [e]X + [2e + 1]T, \tag{RC_2}$$

*so the division point remains an explicit translate of the same cyclic CM kernel as  $X$ . Choose a Miller function  $F = F_{\ell, Z}$  with*

$$\text{div}(F) = \ell[Z] - [[\ell]Z] - (\ell - 1)[O].$$

*In the Rosenlicht coordinates dual to  $D_1 = [-T] - [S]$ ,  $D_2 = [O] - [S]$ , the normalized Abel Kummer class of  $P$  is*

$$\rho_1 = \frac{F(2T)}{F(S)}, \quad \rho_2 = \frac{F(2T)}{F(-T)}, \tag{RC_3}$$

*and its Miller-unit character is exactly*

$$[\mathfrak{c}(P)] = [\rho_1 \rho_2^2] \quad \text{in } K_n^\times / K_n^{\times \ell}. \tag{RC_4}$$

*There is also a formula avoiding the  $\ell$ -division point. On  $\mathcal{E} : y^2 + y = x^3 + x^2$ , set*

$$A_5 = 1 + x + x^2 + y + xy, \quad B_5 = 1 + x + xy, \quad g_1 = \frac{A_5}{B_5}, \quad g_2 = \frac{1}{B_5}.$$

*Then*

$$\operatorname{div}(g_1) = 5D_1, \quad \operatorname{div}(g_2) = 5D_2, \quad \mathfrak{c}^5 = g_1 g_2^2. \quad (\text{RC}_5)$$

If  $5\bar{5} \equiv 1 \pmod{\ell}$ , the full normalized Abel class in  $J_B(K_n)/\ell J_B(K_n)$  has coordinates

$$([g_1(P)]^{\bar{5}}, [g_2(P)]^{\bar{5}}). \quad (\text{RC}_6)$$

More explicitly, for the consecutive Conway variables  $u = u_n$ ,  $v = u_{n-1}$ , put

$$U = u + 1, \quad V = v + 1, \quad D = V^2 + V + U.$$

Then the normalized point has  $x = V/D$ ,  $y = U/D$ , and

$$g_1(P) = \frac{D^2 + D(U + V) + V^2 + UV}{D^2 + VD + UV}, \quad g_2(P) = \frac{D^2}{D^2 + VD + UV}, \quad (\text{RC}_7)$$

with  $g_1(P)g_2(P)^2 = U^{-5}$ .

These formulas compute the missing toric coordinate exactly, but do not force it to be nonzero modulo  $\ell$ . Indeed, Weil reciprocity gives

$$\mathfrak{c}(P) = \mathfrak{c}(W)^\ell \frac{F(2T)^3}{F(S)F(-T)^2}. \quad (\text{RC}_8)$$

Applying reciprocity instead to  $G(Y) = \mathfrak{c}(Y + T)$ , whose divisor is  $[S] + 2[-T] - 3[2T]$ , returns precisely the inverse identity

$$\frac{F(S)F(-T)^2}{F(2T)^3} = \frac{\mathfrak{c}(W)^\ell}{\mathfrak{c}(P)}. \quad (\text{RC}_9)$$

Thus the boundary Miller values and the rational five-torsion splitting exhaust the Rosenlicht coordinate, but collapsing them by the evident reciprocity law is exactly equivalent to the original Kummer value  $[\mathfrak{c}(P)]$ ; it supplies no independent nonresidue constraint.

*Proof.* Since  $P = X + 3T$ , one has  $P - T = X + 2T$ . Multiplication by  $\ell$  is invertible both on the cyclic kernel containing  $X$  and on  $\langle T \rangle$ , which proves  $(\text{RC}_2)$ .

Put  $h(Y) = F(Y - T)$ . Translation of the displayed divisor of  $F$  gives

$$\operatorname{div}(h) = \ell[W] - [P] - (\ell - 1)[T].$$

The Rosenlicht evaluation of this principal relation against  $D_1, D_2$  is

$$\frac{h(S)}{h(-T)} = \frac{F(2T)}{F(S)}, \quad \frac{h(S)}{h(O)} = \frac{F(2T)}{F(-T)},$$

which proves  $(\text{RC}_3)$ . Weil reciprocity between  $h$  and  $\mathfrak{c}$ , using

$$\operatorname{div}(\mathfrak{c}) = [-T] + 2[O] - 3[S], \quad \mathfrak{c}(T) = 1,$$

gives  $(\text{RC}_8)$ , and hence  $(\text{RC}_4)$ .

Direct intersection with the Weierstrass equation, including the local orders at the indicated five-torsion points, gives

$$\operatorname{div}(A_5) = 5[-T] - 5[O], \quad \operatorname{div}(B_5) = 5[S] - 5[O].$$

Thus  $g_1, g_2$  have the divisors asserted in (RC<sub>5</sub>). Both sides of  $\mathfrak{c}^5 = g_1 g_2^2$  have the same divisor and value one at  $T$ , proving the exact identity. Since five is invertible modulo  $\ell$ , this proves (RC<sub>6</sub>). Substitution of  $x = V/D$ ,  $y = U/D$  gives (RC<sub>7</sub>); here  $\mathfrak{c}(P) = 1/U$ , so its last identity is also immediate from (RC<sub>5</sub>).

Finally, the divisor of  $G$  follows by translating that of  $\mathfrak{c}$ . Weil reciprocity between  $F$  and  $G$  is (RC<sub>9</sub>), proving the exact neutrality statement.  $\square$

When  $F_n$  is prime, irreducibility and  $\delta_n > 1$  force  $\delta_n = F_n$ , recovering Lenstra's Fermat-prime rule. When  $F_n$  is composite, irreducibility forces only one nontrivial divisor: it does not force every prime-power factor. The condition  $\delta_n = F_n$  is exactly the maximal-coset-order hypothesis studied for Conway's quadratic tower by Popovych [19]. It is strictly weaker than primitivity of  $c_n$ , which already fails at  $n = 2$  since  $\text{ord}(c_2) = 5 \cdot 17 = 85 < 255$ ; Popovych uses the all-level hypothesis instead to prove that products such as  $c_0 c_n$  are primitive. Wiedemann's related iterated quadratic tower poses the analogous full-Fermat-order problem [21]. Thus the apparently easy "singleton even implies zero" clause already contains a longstanding explicit maximal-coset-order problem. Popovych verified the Conway condition only through  $n = 11$ ; the recent analysis of Cagliero–Herman–Szechtman still retains the quotient order as an undetermined divisor [4]. No all-level proof or counterexample is presently known.

**Proposition 43** (the 3-power arm is Gaussian-period primitivity). *[proved] Fix  $k \geq 1$ . The  $m = 1$  arm of Conjecture 2 for every prime  $p$  with  $f(p) \in \{3, 9, \dots, 3^k\}$  is equivalent to  $C_k$ : the Gaussian period*

$$\gamma_k = \zeta + \zeta^{-1} \in \mathbb{F}_{2^{3k}}^\times, \quad \zeta^{3^{k+1}} = 1,$$

*is primitive.*

*Proof.* Propositions 9 and 13 show that the old prime-power factors of  $2^{3^k} - 1$  propagate from lower levels. For a prime  $p$  with  $f(p) = 3^j$ , the full power criterion says  $m_p = 1$  iff  $v_p(\text{ord } \gamma_j) = v_p(2^{3^j} - 1)$ . Combining these statements for all  $j \leq k$  gives every prime-power factor of  $2^{3^k} - 1$ , which is equivalent to primitivity; the converse is immediate.  $\square$

The two arms are instances of the same projective statement. The quadratic arm asks that the class of  $c_n$  generate  $E_n^\times / E_{n-1}^\times$ . In the cubic arm the natural map

$$U_k \longrightarrow \mathbb{F}_{q_k^3}^\times / \mathbb{F}_{q_k}^\times$$

is an isomorphism, and the increment from  $C_{k-1}$  to  $C_k$  asks that the class of  $\gamma_k$  generate this quotient. Equivalently, the projective class of the multiplication-by- $\gamma_k$  matrix generates a Singer cycle in  $\text{PGL}_3(q_k)$ . The 0/1/4 rule therefore contains a quadratic and a cubic maximal-projective-order conjecture, rather than two unrelated order coincidences.

**Proposition 44** (cyclotomic-type residue reduction). *[proved] Let  $h = 3^k$  and let  $\ell$  be any prime divisor of  $2^h - 1$ . Then*

$$\gamma_k = \zeta^{-1}(1 + \zeta)^2$$

*is an  $\ell$ -th power in  $\mathbb{F}_{2^h}$  if and only if  $1 + \zeta$  is an  $\ell$ -th power in  $\mathbb{F}_{2^{2h}}$ . The same equivalence holds at every power  $\ell^a \mid 2^h - 1$ ; however, non- $\ell$ -power already implies that  $\gamma_k$  has the full  $\ell$ -primary part of  $2^h - 1$  in its order.*

*Proof.* Because  $\ell \mid 2^h - 1$ , one has  $\ell \nmid 2^h + 1$ . Extension from  $\mathbb{F}_{2^h}^\times$  to  $\mathbb{F}_{2^{2h}}^\times$  therefore preserves the  $\ell$ -primary quotient. Also  $\ell \neq 3$ , so exponentiation by  $\ell$  is an automorphism of the  $3^{k+1}$ -group generated by  $\zeta$ ; hence  $\zeta^{-1}$  is an  $\ell$ -th power. Finally, squaring is an automorphism of every  $\ell$ -primary quotient for odd  $\ell$ . The displayed factorization then proves both assertions.  $\square$

Even in the standard prime-conductor case, general results on multiplicative orders of Gaussian periods provide large lower bounds but not maximal order [1]. For the present prime-power conductor, the norm recursion  $\gamma_k^3 + \gamma_k = \gamma_{k-1}$  propagates the old factors only. At level  $k$ , a proof must still show that every new prime-power factor of  $\Phi_{3^k}(2)$  divides  $\text{ord}(\gamma_k)$ . That is the precise new-prime obstruction; degree  $3^k$  guarantees at most that some exact-order factor occurs, not that all of them do.

The Dickson formulation cleanly separates what the existing theory does and does not supply. Mullin–Mahalanobis give the corresponding irreducibility criterion for  $D_{3^k}(X, 1)+1$  [13]; irreducibility is not primitivity. For prime conductor, Gao–Vanstone’s analogous Gaussian-period primitivity statement remains conjectural [8], while Breuer shows that small divisors of the period index can reflect units and class numbers in an intermediate real quadratic field [2]. The real 3-power cyclotomic tower here has only odd-degree intermediate fields, so that quadratic norm argument does not apply. Proposition 21 supplies the appropriate cubic replacement and places the root on a Singer difference set, but neither the three symmetric coefficients nor the difference-set property proves that this distinguished member generates  $U_k$ ; the explicit  $q = 32$  example in Remark 7 rules out that shortcut.

## 4.2 The component set alone is insufficient

[proved]  $\mathcal{Q}(f(p))$  alone does not determine  $m_p$ :

| $\mathcal{Q}$ | $m = 4$             | $m = 1$            |
|---------------|---------------------|--------------------|
| $\{9\}$       | $p = 19, f = 18$    | $p = 73, f = 9$    |
| $\{81\}$      | $p = 163, f = 162$  | $p = 2593, f = 81$ |
| $\{243\}$     | $p = 1459, f = 486$ | $p = 487, f = 243$ |

[proved] The split for  $\mathcal{Q} = \{9\}$  is explained by the order criterion:  $\text{ord}(\kappa_9 + 1) = 3^3 \cdot (2^9 - 1) = 27 \cdot 511 = 13797$  and  $511 = 7 \cdot 73$ , so  $73 \mid \text{ord}(\kappa_9 + 1)$  but  $19 \nmid \text{ord}(\kappa_9 + 1)$ ; adding  $m = 4$  changes the order and introduces the factor 19.

[proved] The lower bound of the exception column is unconditional:  $m_p \geq 4$  for all  $p$  with  $f(p) = 2 \cdot 3^k$  (Proposition 11).

[conjectured] The matching upper bound  $m_p \leq 4$  on that column; its exact current-level formulation is  $D'_k$  in Conjecture 3, not the Gaussian-period conjecture  $C_k$ .

## 4.3 The $m = 4$ upper bound: settled at every visible prime

The  $m = 4$  translate  $\kappa_{3^k} + 4$  involves the number  $4 \in \mathbb{F}_{16}$ , which does not lie in  $\mathbb{F}_2(\zeta)$  (as  $4 \nmid 2 \cdot 3^k$ ); the translate lives in the degree-4  $\cdot 3^k$  compositum  $\mathbb{F}_{2^{4h}}$ ,  $h = 3^k$ . (June 2026 pass; probe `experiments/exception_column_m4.py`.)

**Proposition 45** (the corrected norm). [proved] Let  $\sigma = \text{Frob}^{2h}$  generate  $\text{Gal}(\mathbb{F}_{2^{4h}}/\mathbb{F}_{2^{2h}})$ . Then  $\sigma(4) = 5$ , and with  $\omega := \zeta^h = \kappa_2$ ,

$$N := N_{\mathbb{F}_{2^{4h}}/\mathbb{F}_{2^{2h}}}(\kappa_{3^k} + 4) = (\kappa + 4)(\kappa + 5) = \kappa^2 + \kappa + \omega.$$

*Proof.*  $\mathbb{F}_{16} \cap \mathbb{F}_{2^{2h}} = \mathbb{F}_4$  because  $\gcd(4, 2 \cdot 3^k) = 2$ , so  $\sigma|_{\mathbb{F}_{16}}$  is the nontrivial element of  $\text{Gal}(\mathbb{F}_{16}/\mathbb{F}_4)$ , which swaps the two roots 4, 5 of the Artin–Schreier minimal polynomial  $y^2 + y + \omega$  of 4 over  $\mathbb{F}_4$ . (Equivalently:  $2h \equiv 2 \pmod{4}$  and the Frobenius orbit of 4 is  $4 \rightarrow 6 \rightarrow 5 \rightarrow 7$ , so  $\sigma(4) = 4^{2^2} = 5$ .) Then  $(\kappa + 4)(\kappa + 5) = \kappa^2 + \kappa + (4 \otimes 5)$  and  $4 \otimes 5 = 4^2 + 4 = \omega$  by the same Artin–Schreier relation.  $\square$

**Remark 8** (erratum). The 2026-06-10 draft of this note stated the norm as  $(\kappa+4)(\kappa+\text{Frob}^2(4)) = (\kappa+4)(\kappa+6)$ . The conjugate  $6 = 4^2$  is  $\text{Frob}^1(4)$ , one squaring short of  $\text{Frob}^2(4) = 5$ . The slip is visible without computation:  $4 \otimes 6 = 14 \notin \mathbb{F}_4$ , so  $(\kappa+4)(\kappa+6) \notin \mathbb{F}_{2^{2h}}$  and cannot be a norm.

The corrected norm collapses the  $m = 4$  test into the *same* sparse trinomial model  $\mathbb{F}_2[x]/(x^{2h} + x^h + 1)$  as the rest of the 3-power analysis: no compositum arithmetic is needed.

**Corollary 2** (in-field criterion). *[proved] For a prime  $p$  with  $f(p) = 2 \cdot 3^k$  and  $v_p(2^{2h} - 1) = 1$ , write  $\overline{N} = N^{2^h}$  and  $M_k := \overline{N}/N = N^{2^h-1} \in U$ . Then*

$$m_p = 4 \iff p \mid \text{ord}(M_k) \iff M_k^{(2^h+1)/p} \neq 1.$$

*Proof.*  $m_p \geq 4$  is Proposition 11, so  $m_p = 4$  iff  $\kappa + 4$  has no  $p$ -th root. By Propositions 1 and 2 with  $E = 4h$ ,  $f = 2h$  (and  $v_p(2^{4h} - 1) = v_p(2^{2h} - 1) = 1$  by lifting the exponent, since  $p \nmid 2$ ), that holds iff  $N^{(2^{2h}-1)/p} \neq 1$ . Since  $\mathbb{F}_{2^{2h}}^\times = L^\times \times U$  with coprime orders,  $p \mid 2^h + 1$ , and  $(2^{2h} - 1)/p = (2^h - 1) \cdot ((2^h + 1)/p)$ , the test equals  $M_k^{(2^h+1)/p} \neq 1$ .  $\square$

**Proposition 46** (exact current-level form of the exception conjecture). *[proved] Let  $k \geq 1$ . Retain the notation above, put  $q = 2^h$ ,  $n = 3^{k+1} = 3h$ , and write*

$$\zeta := \kappa_{3^k} \in K := \mathbb{F}_{2^{2h}}, \quad \tilde{K} := \mathbb{F}_{2^{4h}}, \quad \beta := \zeta + 4.$$

*Thus  $\zeta$  is a primitive  $n$ -th root of unity. Define*

$$N := N_{\tilde{K}/K}(\beta) = \zeta^2 + \zeta + \zeta^h, \quad M_k := N^{q-1}, \quad \Psi_k := \frac{\Phi_{2 \cdot 3^k}(2)}{3},$$

*and*

$$R_k(Y) := \text{Res}_X \left( X^{2h} + X^h + 1, Y + X^h + X^2 + X \right) \in \mathbb{F}_2[Y].$$

*Then  $R_k$  is the irreducible minimal polynomial of  $N$  over  $\mathbb{F}_2$  and has degree  $2h$ . The prime divisors of  $\Psi_k$  are exactly the primes  $\ell \neq 3$  with  $\text{ord}_\ell(2) = 2 \cdot 3^k$ , with full valuations*

$$v_\ell(\Psi_k) = v_\ell(2^{2h} - 1) = v_\ell(2^h + 1).$$

*Moreover the following are equivalent:*

- (a)  $m_\ell = 4$  for every prime  $\ell$  with  $f(\ell) = 2 \cdot 3^k$ ;
- (b)  $\Psi_k \mid \text{ord}(M_k)$ ;
- (c)  $N \notin (K^\times)^\ell$  for every prime  $\ell \mid \Psi_k$ ;
- (d)  $R_k(X^\ell)$  is irreducible over  $\mathbb{F}_2$  for every prime  $\ell \mid \Psi_k$ .

*Proof.* The order identity  $\text{ord}_{3^{k+1}}(2) = 2h$  makes  $X^{2h} + X^h + 1$  the irreducible minimal polynomial of  $\zeta$ . We first prove that  $N = \zeta^h + \zeta^2 + \zeta$  also has degree  $2h$ . Suppose  $N^{2^j} = N$ , and put  $a \equiv 2^j \pmod{3h}$ . The resulting vanishing polynomial, of degree less than  $3h$ , has support equal to the symmetric difference of

$$\{ah, a, 2a\} \quad \text{and} \quad \{h, 1, 2\}. \tag{1}$$

Every multiple of  $X^{2h} + X^h + 1$  of degree less than  $3h$  has support a disjoint union of triples  $\{r, r+h, r+2h\}$ . If  $a \equiv 2 \pmod{3}$ , the support in (1) contains  $h$  and  $2h$  but not  $0$ , a contradiction.

If  $a \equiv 1 \pmod{3}$ , the  $h$ -terms cancel and each residue class modulo  $h$  occurs at most twice; a nonempty union of such triples is again impossible. Hence the support is empty, so  $\{a, 2a\} = \{1, 2\}$  modulo  $3h$ . Since  $h \geq 3$ , this forces  $a = 1$ . Thus the Frobenius orbit of  $N$  has length  $2h$ , and the displayed resultant is its irreducible minimal polynomial.

Writing  $A = 2^{3^{k-1}}$  gives  $\Phi_{2 \cdot 3^k}(2) = A^2 - A + 1 \equiv 3 \pmod{9}$ , so division by 3 removes exactly the prime dividing the cyclotomic index. The standard cyclotomic-divisor criterion now says that the prime divisors of  $\Psi_k$  are precisely the  $\ell$  with  $\text{ord}_\ell(2) = 2h$ . Their entire primary contribution to  $2^{2h} - 1$  lies in this cyclotomic factor and, since  $\ell \nmid 2^h - 1$ , in  $2^h + 1$ . This proves the valuation statement.

Fix  $\ell \mid \Psi_k$ . Propositions 10 and 11 make the translates  $\zeta, \zeta + 1, \zeta + 2, \zeta + 3$  into  $\ell$ -th powers, so  $m_\ell \geq 4$ . The corrected norm and the exact power criterion give

$$\beta^{(2^{4h}-1)/\ell} = N^{(2^{2h}-1)/\ell},$$

and therefore

$$m_\ell = 4 \iff N \notin (K^\times)^\ell. \quad (2)$$

This uses the full  $\ell$ -primary valuation and has no squarefreeness or non-Wieferich hypothesis. Since  $q - 1 \equiv -2 \pmod{\ell}$ , raising to  $q - 1$  is invertible on  $K^\times / (K^\times)^\ell$ , so

$$N \in (K^\times)^\ell \iff M_k \in (K^\times)^\ell. \quad (3)$$

The entire  $\ell$ -primary part of  $K^\times$  lies in its norm-one factor of order  $q + 1$ . In a cyclic group, nonmembership in the subgroup of  $\ell$ -th powers is equivalent to the order containing the full  $\ell$ -primary part. Equations (2)–(3) prove the equivalence of the first three assertions.

Finally  $\mu_\ell \subset K$ . Kummer theory says that  $X^\ell - N$  is irreducible over  $K$  exactly when  $N \notin (K^\times)^\ell$ . Since  $K = \mathbb{F}_2(N)$ , Capelli's lemma says in turn that this is equivalent to irreducibility of  $R_k(X^\ell)$  over  $\mathbb{F}_2$ , completing the proof.  $\square$

**Conjecture 3** ( $D'_k$ : the exact current exception factor). *[conjectured] One has*

$$D'_k : \quad \Psi_k \mid \text{ord}(M_k).$$

*By Proposition 46, this is exactly the  $m = 4$  upper bound for the primes at level  $f = 2 \cdot 3^k$ .*

**Proposition 47** (exact current-layer projection and dominant-factor sieve). *[proved] Let  $k \geq 2$ , put*

$$s = 3^{k-1}, \quad h = 3s, \quad a = 2^s, \quad q = a^3 = 2^h, \quad \Lambda_k = a^2 - a + 1 = 3\Psi_k,$$

*and retain  $N, M_k, \zeta, \omega = \zeta^h$  from Proposition 46. Define*

$$P_k := M_k^{a+1}.$$

*Then  $P_k^{\Lambda_k} = 1$ , and for every prime  $\ell \mid \Psi_k$ ,*

$$v_\ell(\text{ord } P_k) = v_\ell(\text{ord } M_k). \quad (\text{D}_1)$$

*Moreover*

$$P_k = \frac{\zeta^4 + \zeta + \omega}{\omega^2 \zeta^4 + \zeta^3 + 1}. \quad (\text{D}_2)$$

$$[\mathbb{F}_2(P_k) : \mathbb{F}_2] = 2h. \quad (\text{D}_3)$$



and

$$v_3(\text{ord } P_k) = \begin{cases} 1, & k \text{ odd}, \\ 0, & k \text{ even}. \end{cases} \quad (\text{D}_4)$$

Set  $c_k = 3$  for odd  $k$  and  $c_k = 1$  for even  $k$ . Then

$$\text{ord}(P_k) \geq c_k(2h+1). \quad (\text{D}_5)$$

Consequently, if  $\ell \mid \Psi_k$  and

$$\ell > \frac{\Lambda_k}{c_k(2h+1)}, \quad (\text{D}_6)$$

then  $P_k$ , and hence  $M_k$ , contains the full  $\ell$ -primary part of  $\Lambda_k$  in its order. In particular  $D'_k$  follows whenever  $\Psi_k$  is prime; this proves  $D'_2$  and  $D'_3$  analytically, since  $\Psi_2 = 19$  and  $\Psi_3 = 87211$ .

*Proof.* The factorization

$$q+1 = (a+1)(a^2 - a + 1) = (a+1)\Lambda_k$$

and  $M_k^{q+1} = 1$  give  $P_k^{\Lambda_k} = 1$ . Also  $\gcd(\Lambda_k, a+1) = 3$ . Since  $k \geq 2$ , one has  $a \equiv -1 \pmod{9}$ , so  $\Lambda_k \equiv 3 \pmod{9}$  and  $\gcd(\Psi_k, a+1) = 1$ . Exponentiation by  $a+1$  therefore preserves all the primary valuations belonging to  $\Psi_k$ , proving (D<sub>1</sub>).

The lifted congruence

$$2^{3^{k-1}} \equiv 3^k - 1 \pmod{3^{k+1}}$$

gives, modulo  $3h$ ,

$$a \equiv h-1, \quad a^2 \equiv h+1, \quad a^3 \equiv -1, \quad a^4 \equiv 2h+1. \quad (\text{D}_7)$$

As  $M_k = N^{a^3-1}$ ,

$$P_k = \frac{N^{a^4} N^{a^3}}{N^a N}.$$

Using  $N = \zeta^2 + \zeta + \omega$  and (D<sub>7</sub>), direct Frobenius expansion gives

$$N^a = \omega^2 \zeta^{-2} + \omega \zeta^{-1} + \omega^2,$$

$$N^{a^3} = \zeta^{-2} + \zeta^{-1} + \omega^2,$$

$$N^{a^4} = \omega \zeta^2 + \omega^2 \zeta + \omega.$$

Multiplying  $N^{a^3}$  and  $N^a$  by the same cancelling factor  $\zeta^2$ , and using  $1 + \omega + \omega^2 = 0$ , yields

$$\begin{aligned} N^{a^4}(\zeta^2 N^{a^3}) &= \zeta^4 + \zeta + \omega, \\ (\zeta^2 N^a)N &= \omega^2 \zeta^4 + \zeta^3 + 1, \end{aligned}$$

which proves (D<sub>2</sub>).

The minimal polynomial of  $\zeta$  over  $\mathbb{F}_2$  is  $X^{2h} + X^h + 1$ . Write the numerator and denominator in (D<sub>2</sub>) as  $C, D$ . Since  $P_k^q = P_k^{-1}$ , membership in  $\mathbb{F}_q$  would force  $P_k = 1$ . But  $C = D$ , after reducing  $\zeta^{2h} = \zeta^h + 1$ , would give

$$\zeta^{h+4} + \zeta^h + \zeta^3 + \zeta + 1 = 0, \quad (\text{D}_8)$$

a nonzero polynomial of degree  $h+4 < 2h$ .

If  $P_k \in \mathbb{F}_{a^2}$ , its order divides  $\gcd(\Lambda_k, a^2-1) = 3$ , so  $P_k \in \{1, \omega, \omega^2\}$ . The first case was excluded; the other two respectively give

$$\zeta^{h+3} + \zeta = 0$$

and

$$\zeta^{h+4} + \zeta^{h+3} + \zeta^4 + \zeta^3 + \zeta + 1 = 0.$$

Both are again nonzero polynomials of degree below  $2h$ . Every proper divisor of  $2h = 2 \cdot 3^k$  divides either  $h$  or  $2h/3$ ; hence every proper subfield of  $\mathbb{F}_{2^{2h}}$  lies in either  $\mathbb{F}_q$  or  $\mathbb{F}_{q^2}$ . This proves (D<sub>3</sub>).

For the 3-part, observe that

$$P_k^{\Lambda_k/3} = M_k^{(q+1)/3} = N^{(q^2-1)/3} = N_{\mathbb{F}_{2^{2h}}/\mathbb{F}_4}(N). \quad (\text{D}_9)$$

Choose  $u \in \mathbb{F}_{16}$  with  $u^2 + u = \omega$ . Then  $u^4 = u + 1$ ,  $u^5 = \omega$ , and  $u$  has order 15. The extensions  $\mathbb{F}_{2^{2h}}/\mathbb{F}_4$  and  $\mathbb{F}_{16}/\mathbb{F}_4$  are linearly disjoint, and over  $\mathbb{F}_{16}$  the minimal polynomial of  $\zeta$  is  $X^h + \omega$ , irreducible because  $\text{ord}_{3h}(16) = h$ . Since

$$N = (\zeta + u)(\zeta + u + 1),$$

base change in (D<sub>9</sub>) gives

$$N_{\mathbb{F}_{2^{2h}}/\mathbb{F}_4}(N) = (u^h + \omega)((u + 1)^h + \omega).$$

The residues of  $h = 3^k$  modulo 15 are 3, 9, 12, 6 as  $k \equiv 1, 2, 3, 0 \pmod{4}$ ; reduction with  $u^4 = u + 1$  gives

$$P_k^{\Lambda_k/3} = \begin{cases} \omega^2, & k \text{ odd}, \\ 1, & k \text{ even}. \end{cases}$$

Because  $v_3(\Lambda_k) = 1$ , this proves (D<sub>4</sub>).

Let  $e = \text{ord}(P_k)$ . From (D<sub>3</sub>),  $\text{ord}_e(2) = 2h$ . Since  $e \mid \Lambda_k$  and  $e \notin \{1, 3\}$ , some prime  $r \mid \Psi_k$  divides  $e$ ; every such  $r$  has  $\text{ord}_r(2) = 2h$ , so  $r \equiv 1 \pmod{2h}$  and  $r \geq 2h + 1$ . When  $k$  is odd, (D<sub>4</sub>) also gives  $3r \mid e$ , which proves (D<sub>5</sub>).

If the full  $\ell$ -primary part of  $\Lambda_k$  were missing from  $e$ , then  $e \mid \Lambda_k/\ell$ . This contradicts (D<sub>5</sub>) under (D<sub>6</sub>). Equation (D<sub>1</sub>) transfers the conclusion back to  $M_k$ , and the last assertions follow.  $\square$

**Proposition 48** (fibotomic form of the current power-residue test). *[proved] Retain Proposition 47, put*

$$U = \{x \in \mathbb{F}_{q^2}^\times : x^{q+1} = 1\}, \quad \pi(x) = \frac{x}{(x+1)^2} = \frac{1}{x+x^{-1}},$$

and set  $Z = \pi(M_k)$ . The map  $\pi$  is two-to-one from  $U \setminus \{1\}$  onto

$$\mathcal{T}_q = \{z \in \mathbb{F}_q : \text{Tr}_{\mathbb{F}_q/\mathbb{F}_2}(z) = 1\},$$

with fibres  $\{x, x^{-1}\}$ .

For an odd prime  $\ell$ , define

$$H_\ell(W) = \sum_{j=0}^{(\ell-1)/2} \binom{\ell-j-1}{j} W^j \in \mathbb{F}_2[W].$$

If  $\ell \mid \Psi_k$ , then

$$M_k \in U^\ell \iff \text{there is } W \in \mathcal{T}_q \text{ with } W^\ell + ZH_\ell(W)^2 = 0. \quad (\text{F})$$

The polynomial

$$\mathcal{F}_{\ell,Z}(W) = W^\ell + ZH_\ell(W)^2$$

either has no root in  $\mathbb{F}_q$ , or splits there into  $\ell$  distinct linear factors. Finally  $H_\ell$  is squarefree and is a product of exactly  $(\ell-1)/(2h)$  irreducible polynomials over  $\mathbb{F}_2$ , each of degree  $h$ . Thus  $D'_k$  is equivalently the assertion that  $\mathcal{F}_{\ell,Z}$  is rootless over  $\mathbb{F}_q$  for every prime  $\ell \mid \Psi_k$ .

*Proof.* For  $x \in U$ , one has  $x^q = x^{-1}$ , so  $\pi(x) \in \mathbb{F}_q$ . The equation  $\pi(x) = z$  is

$$X^2 + z^{-1}X + 1 = 0.$$

The characteristic-two quadratic criterion says that this polynomial is irreducible over  $\mathbb{F}_q$  exactly when  $\text{Tr}(z) = 1$ . Its roots are inverse and have norm one. Counting, or the same quadratic argument in reverse, proves the stated two-to-one description.

The defining polynomial identity is

$$\frac{X^\ell + 1}{(X + 1)^\ell} = H_\ell\left(\frac{X}{(X + 1)^2}\right). \quad (\text{H})$$

If  $M_k = T^\ell$  with  $T \in U$ , put  $W = \pi(T)$ . Identity (H) gives

$$Z = \pi(T^\ell) = \frac{W^\ell}{H_\ell(W)^2},$$

so  $W$  satisfies (F). Conversely, lift a trace-one solution  $W$  to  $T \in U$ . Then (H) gives  $\pi(T^\ell) = \pi(M_k)$ , hence  $T^\ell = M_k$  or  $M_k^{-1}$ . Replacing  $T$  by  $T^{-1}$  in the second case proves  $M_k \in U^\ell$ .

In fact every  $\mathbb{F}_q$ -root of  $\mathcal{F}_{\ell,Z}$  has trace one. Indeed,  $H_\ell(W) \neq 0$  at a root, since otherwise the equation would force  $W = 0$ , whereas  $H_\ell(0) = 1$ . Lift a nonzero root through  $\pi(T) = W$ . If  $\text{Tr}(W) = 0$ , the quadratic criterion puts  $T$  in  $\mathbb{F}_q$ . Identity (H) then gives  $\pi(T^\ell) = Z$ , but for  $y \in \mathbb{F}_q \setminus \{1\}$ ,

$$\pi(y) = \left(\frac{y}{y+1}\right)^2 + \frac{y}{y+1}$$

has absolute trace zero, whereas  $Z = \pi(M_k)$  has trace one. This is a contradiction.

If one root exists, therefore lift it to  $T \in U$  and orient  $T$  so that  $T^\ell = M_k$ . Then the  $\ell$  elements

$$\pi(T\xi), \quad \xi \in \mu_\ell,$$

are roots in  $\mathbb{F}_q$ . They are distinct: an inverse collision would give  $T^2 \in \mu_\ell$ , hence  $M_k^2 = 1$ , whereas  $M_k \neq 1$  because  $N \notin \mathbb{F}_q$ . Since  $\mathcal{F}_{\ell,Z}$  has degree  $\ell$ , it splits completely. Its derivative is  $W^{\ell-1}$ , and trace-one roots are nonzero, so they are simple.

Finally, the roots of  $H_\ell$  are

$$\pi(\xi), \quad \xi \in \mu_\ell \setminus \{1\},$$

with  $\xi$  and  $\xi^{-1}$  identified. Their Frobenius-orbit length is the least  $d > 0$  with  $2^d \equiv \pm 1 \pmod{\ell}$ . Since  $\text{ord}_\ell(2) = 2h$ , this least  $d$  is  $h$ . The number and degrees of the factors follow, and the roots are distinct, proving squarefreeness.  $\square$

**Proposition 49** (translated-binomial form of  $D'_k$ ). *[proved] Retain  $k \geq 2$ ,  $h = 3^k$ , and choose  $u \in \mathbb{F}_{16}$  with  $u^2 + u = \omega$ . Put*

$$L = \mathbb{F}_{16^h} = \mathbb{F}_{2^{4h}}, \quad \beta = \zeta + u.$$

*Then the minimal polynomial of  $\beta$  over  $\mathbb{F}_{16}$  is the irreducible degree- $h$  polynomial*

$$B_k(Y) = (Y + u)^h + \omega.$$

*For every prime  $\ell \mid \Psi_k$ , the following are equivalent:*

- (i) *the full  $\ell$ -primary part occurs in  $\text{ord}(M_k)$ ;*

(ii)  $\beta \notin (L^\times)^\ell$ ;

(iii)  $B_k(X^\ell) = (X^\ell + u)^h + \omega$  is irreducible over  $\mathbb{F}_{16}$ .

More precisely,  $B_k(X^\ell)$  is either irreducible of degree  $h\ell$ , or is a product of exactly  $\ell$  irreducibles of degree  $h$ .

*Proof.* Since

$$\text{ord}_{3h}(16) = \frac{\text{ord}_{3h}(2)}{\gcd(\text{ord}_{3h}(2), 4)} = h,$$

the element  $\zeta$  has degree  $h$  over  $\mathbb{F}_{16}$ . As  $\zeta^h = \omega$ , its minimal polynomial there is  $X^h + \omega$ ; translation by  $u$  gives  $B_k$ .

Let  $K = \mathbb{F}_{2^{2h}}$ . The corrected norm is

$$N = N_{L/K}(\beta) = \beta^{2^{2h}+1}.$$

Because  $2^{2h} \equiv 1 \pmod{\ell}$ , this norm induces multiplication by two on  $L^\times/(L^\times)^\ell$ . Hence it first gives

$$\beta \in (L^\times)^\ell \iff N \in (L^\times)^\ell.$$

The natural map

$$K^\times/(K^\times)^\ell \longrightarrow L^\times/(L^\times)^\ell$$

is injective: restriction followed by the degree-two norm is multiplication by two, which is invertible modulo the odd prime  $\ell$ . Since  $N \in K$ , this sharpens the equivalence to

$$\beta \in (L^\times)^\ell \iff N \in (K^\times)^\ell,$$

and Proposition 46 identifies the negation with the full  $\ell$ -primary assertion. Kummer theory and Capelli's lemma give the irreducibility equivalence.

If  $\beta = \alpha^\ell$  with  $\alpha \in L$ , every root of  $B_k(X^\ell)$  lies in  $L$ , because  $\mu_\ell \subset L$ . Its  $\ell$ -th power is a conjugate of  $\beta$ , of degree  $h$  over  $\mathbb{F}_{16}$ ; the root therefore has degree both at least and at most  $h$ . All irreducible factors consequently have degree  $h$ , and there are exactly  $\ell$  of them.  $\square$

**Corollary 3** (short translated cyclotomic period). *[proved] Let  $\chi : L^\times \rightarrow \mu_\ell$  have exact order  $\ell$ , realize its values as complex roots of unity, and write  $\chi(\beta) = \xi_\ell^r$ . Then*

$$\mathcal{J}_{k,\ell} := \sum_{\theta^h = \omega} \chi(\theta + u) = \sum_{j=0}^{h-1} \xi_\ell^{r16^j},$$

and

$$\beta \in (L^\times)^\ell \iff \mathcal{J}_{k,\ell} = h.$$

*Proof.* The roots of  $X^h + \omega$  are  $\zeta^{16^j}$ ,  $0 \leq j < h$ , and  $u \in \mathbb{F}_{16}$ . Thus

$$\chi(\zeta^{16^j} + u) = \chi((\zeta + u)^{16^j}) = \xi_\ell^{r16^j}.$$

If  $r = 0$ , every summand is one. Conversely, equality with the positive real number  $h$  forces equality in the complex triangle inequality and hence forces every summand, in particular the first, to be one.  $\square$

**Proposition 50** (full  $\mathbb{F}_{16}$ -line saturation). *[proved] Retain Proposition 49, put  $s = 3^{k-1}$ , and choose  $u$  so that it has order fifteen and  $\omega = u^5$ . For a prime  $\ell \mid \Psi_k$ , write*

$$\mathcal{C} = L^\times / (L^\times)^\ell, \quad v(c) = [\zeta + c] \quad (c \in \mathbb{F}_{16}), \quad A = 2^s \pmod{\ell},$$

*using additive notation for  $\mathcal{C}$ . Then  $A$  has order six,  $A^2 - A + 1 = 0$ , and  $A^3 = -1$ . The four base-field classes vanish,*

$$v(c) = 0 \quad (c \in \mathbb{F}_4), \tag{L_{16,1}}$$

*while for  $c \in \mathbb{F}_{16}^\times$ ,*

$$v(\omega c^{-2^s}) = Av(c), \quad v(c^{-2^h}) = -v(c). \tag{L_{16,2}}$$

*For the twelve exponents not divisible by five one has*

| $j$ | $\{m : v(u^m) = A^j v(u)\}$ |            |
|-----|-----------------------------|------------|
| 0   | $\{1, 4\}$                  |            |
| 1   | $\{3, 12\}$                 |            |
| 2   | $\{11, 14\}$                |            |
| 3   | $\{7, 13\}$                 |            |
| 4   | $\{6, 9\}$                  |            |
| 5   | $\{2, 8\}$ .                | (L_{16,3}) |

*Consequently*

$$\zeta + u \in (L^\times)^\ell \iff \zeta + c \in (L^\times)^\ell \quad \text{for every } c \in \mathbb{F}_{16}. \tag{L_{16,4}}$$

*If  $D'_k$  holds for  $\ell$ , none of the twelve translates with  $c \notin \mathbb{F}_4$  is an  $\ell$ -th power. Equivalently, for an exact-order- $\ell$  complex character  $\chi$  of  $L^\times$  and  $t = \chi(\zeta + u)$ ,*

$$\sum_{c \in \mathbb{F}_{16}} \chi(\zeta + c) = 4 + 2 \sum_{j=0}^5 t^{A^j}, \tag{L_{16}}$$

*and failure of  $D'_k$  is exactly the extremal value sixteen. Thus Frobenius, conjugation, and all  $\mathbb{F}_4$ -translates generate only scalar multiples of the one unresolved Kummer class  $v(u)$ .*

*Proof.* One has  $\text{ord}_\ell(2) = 2h = 6s$ , proving the assertions about  $A$ . For  $c = 0$ , the element  $\zeta$  has  $\ell$ -coprime order  $3h$ . For  $c \in \{1, \omega, \omega^2\}$ , Propositions 10 and 11 give order dividing  $3h(2^h - 1)$ , again coprime to  $\ell$ . This proves (L\_{16,1}); constants in  $\mathbb{F}_{16}^\times$  and  $\zeta$  also have zero class.

The congruence  $2^s \equiv h - 1 \pmod{3h}$  gives  $\zeta^{2^s} = \omega \zeta^{-1}$ . Discarding scalar factors with zero class,

$$Av(c) = [(\zeta + c)^{2^s}] = [\omega \zeta^{-1} + c^{2^s}] = [\zeta + \omega c^{-2^s}] = v(\omega c^{-2^s}).$$

Likewise  $2^h \equiv -1 \pmod{3h}$  and modulo  $\ell$ , so

$$-v(c) = [(\zeta + c)^{2^h}] = [\zeta^{-1} + c^{2^h}] = [\zeta + c^{-2^h}].$$

On exponents modulo fifteen the two transformations are

$$R(m) = 5 - (2^s \bmod 15)m, \quad S(m) = -(2^h \bmod 15)m.$$

For odd  $k$  they are  $(5 - 2m, 7m)$ , and for even  $k$  they are  $(5 - 8m, 13m)$ . Iterating either pair on the twelve residues outside  $5\mathbb{Z}/15\mathbb{Z}$  gives the table  $(L_{16,3})$ . Since every  $A^j$  is nonzero modulo  $\ell$ , all twelve classes vanish exactly when  $v(u)$  vanishes, proving  $(L_{16,4})$  and its nonvanishing counterpart.

Applying  $\chi$  to the table gives each  $t^{A^j}$  twice, while the four  $\mathbb{F}_4$ -terms are one. If  $t = 1$ , the sum is sixteen; conversely equality with the positive real number sixteen forces every unit-modulus summand to be one by equality in the triangle inequality.  $\square$

**Proposition 51** (semiprimitive exhaustion and the saturated planes). *[proved] Retain Proposition 50, put*

$$h = 3^k, \quad q = 2^h, \quad K = \mathbb{F}_{q^2}, \quad L = \mathbb{F}_{q^4},$$

and, for a prime  $\ell \mid \Psi_k$ , set

$$H_K = (K^\times)^\ell, \quad H_L = (L^\times)^\ell.$$

Every nontrivial order- $\ell$  character  $\chi$  on  $K^\times$  has, for the canonical additive character,

$$G_K(\chi^j) = q \quad (1 \leq j < \ell). \quad (\text{SP}_1)$$

Every order- $\ell$  character on  $L^\times$  is the norm lift of one on  $K^\times$ , and quadratic Hasse–Davenport gives

$$G_L(\tilde{\chi}^j) = -q^2 \quad (1 \leq j < \ell). \quad (\text{SP}_2)$$

Consequently, whenever  $a, b, a + b \not\equiv 0 \pmod{\ell}$ ,

$$J_K(\chi^a, \chi^b) = q, \quad J_L(\tilde{\chi}^a, \tilde{\chi}^b) = -q^2, \quad (\text{SP}_3)$$

while  $J(\chi^a, \chi^{-a}) = -1$  in either field.

Failure of  $D'_k$  at  $\ell$  has the following two equivalent additive forms:

$$W := \mathbb{F}_{16} + \mathbb{F}_{16}\zeta, \quad W \setminus \{0\} \subset H_L, \quad (\text{SP}_4)$$

and

$$V := \mathbb{F}_4 + \mathbb{F}_4\zeta + \mathbb{F}_4\zeta^2, \quad V \setminus \{0\} \subset H_K. \quad (\text{SP}_5)$$

Thus a bad translate saturates an eight-dimensional  $\mathbb{F}_2$ -space in  $L$ , and its norm-down form saturates a six-dimensional  $\mathbb{F}_2$ -space in  $K$ , not merely the original affine line.

The exact semiprimitive Fourier equations nevertheless permit both planes. If  $W^\perp$  is the additive annihilator of  $W$ , put

$$A_L = |(W^\perp \setminus \{0\}) \cap H_L|, \quad R = q^2, \quad w = 256.$$

Then orthogonality and  $(\text{SP}_2)$  force

$$A_L = \frac{(R-1)(R+w) - \ell R(w-1)}{\ell w}, \quad |W^\perp \setminus (H_L \cup \{0\})| = \frac{(R-1)(R+w)(\ell-1)}{\ell w}. \quad (\text{SP}_6)$$

Likewise, for  $w_0 = 64$ ,

$$A_K := |(V^\perp \setminus \{0\}) \cap H_K| = \frac{(q+1)(q-w_0) + \ell q(w_0-1)}{\ell w_0}, \quad (\text{SP}_7)$$

and

$$|V^\perp \setminus (H_K \cup \{0\})| = \frac{(\ell-1)(q+1)(q-w_0)}{\ell w_0}. \quad (\text{SP}_8)$$

All four quantities are nonnegative integers with the required total sizes. Hence the complete semiprimitive Gauss and Jacobi table, even together with the dimensions of the saturated spaces, gives no contradiction. A proof of  $D'_k$  must use the distinguished cyclotomic placement of  $W$  or  $V$ , not only the semiprimitive character table or additive-subspace incidence.

*Proof.* Since  $\text{ord}_\ell(2) = 2h$ , one has  $q \equiv -1 \pmod{\ell}$ . Thus every order- $\ell$  character of  $K^\times$  is trivial on  $\mathbb{F}_q^\times$ . Decompose  $K^\times$  into its  $q+1$  projective  $\mathbb{F}_q$ -lines. On the trace-zero line  $\mathbb{F}_q$ , the additive sum is  $q-1$ ; on every other line it is  $-1$ . The induced nontrivial projective character sums to zero, giving  $G_K(\chi) = q$ . The same argument applies to every nontrivial power. The norm induces an isomorphism on the order- $\ell$  character groups, so Hasse–Davenport proves  $(\text{SP}_2)$ . The standard quotient formula for Jacobi sums proves  $(\text{SP}_3)$ , with the inverse case evaluated separately.

If  $D'_k$  fails, Proposition 50 puts every  $\zeta + c$ ,  $c \in \mathbb{F}_{16}$ , in  $H_L$ . Since  $\mathbb{F}_{16}^\times \subset H_L$ , every nonzero  $a\zeta + b$  is an  $\ell$ -th power, proving  $(\text{SP}_4)$ ; the converse follows by taking  $a = 1, b = u$ .

For  $(\text{SP}_5)$ , every polynomial of degree at most two over  $\mathbb{F}_4$  is a product of linear factors or is an irreducible quadratic. In the first case its value at  $\zeta$  is an  $\ell$ -th power by  $(\text{SP}_4)$ . In the second, choose a root  $c \in \mathbb{F}_{16} \setminus \mathbb{F}_4$ ; its value is

$$(\zeta + c)(\zeta + c^4) = N_{L/K}(\zeta + c),$$

again an  $\ell$ -th power. Conversely the corrected norm  $N = \zeta^2 + \zeta + \omega$  belongs to  $V$ ; membership in  $H_K$  is exactly failure of  $D'_k$ .

Finally the two additive Fourier values of  $H_L$  are

$$\frac{-1 - R(\ell-1)}{\ell} \quad \text{on } H_L, \quad \frac{R-1}{\ell} \quad \text{off } H_L,$$

by  $(\text{SP}_2)$ . Fourier inversion applied to  $|W \cap H_L| = w-1$  gives  $(\text{SP}_6)$ . The corresponding values for  $H_K$  are

$$\frac{-1 + q(\ell-1)}{\ell} \quad \text{on } H_K, \quad \frac{-1 - q}{\ell} \quad \text{off } H_K,$$

and  $|V \cap H_K| = w_0 - 1$  gives  $(\text{SP}_7)$ – $(\text{SP}_8)$ . Integrality follows from  $\ell \mid q+1$ ,  $w \mid q^2$ , and  $w_0 \mid q$ ; nonnegativity is immediate from  $\ell \leq q+1$  and  $q \geq 2^9$ .  $\square$

**Proposition 52** (cyclotomic Dirichlet coefficient and subspace-polynomial exhaustion). *[proved]* Retain  $k \geq 2$ , put  $h = 3^k$ , and write

$$K = \mathbb{F}_{4^h} = \mathbb{F}_4(\zeta), \quad \zeta^h = \omega.$$

Let  $\ell \mid \Psi_k$ , let  $\chi : K^\times \rightarrow \mu_\ell$  have exact order  $\ell$ , realized complexly, and let  $\tilde{\chi} = \chi \circ \text{Norm}_{\mathbb{F}_{16^h}/K}$ . Put

$$s = h/3, \quad A = 2^s \pmod{\ell},$$

choose  $u \in \mathbb{F}_{16}$  of order fifteen with  $u^5 = \omega$ , and set  $t = \widetilde{\chi}(\zeta + u)$ . For  $0 \leq r < h$ , define

$$\mathcal{A}_r(\chi) = \sum_{\substack{f \in \mathbb{F}_4[X] \text{ monic} \\ \deg f = r}} \chi(f(\zeta)).$$

Then

$$\mathcal{A}_0 = 1, \quad \mathcal{A}_1 = 4, \quad \mathcal{A}_2 = 10 + \sum_{j=0}^5 t^{A_j}. \quad (\text{DP}_1)$$

Failure of  $D'_k$  at  $\ell$  is equivalent to the single extremal coefficient

$$\mathcal{A}_2 = 16. \quad (\text{DP}_2)$$

If  $\mathcal{A}'_1$  is the degree-one coefficient after the quadratic constant-field extension to  $\mathbb{F}_{16}$ , then

$$\mathcal{A}'_1 = 2\mathcal{A}_2 - \mathcal{A}_1^2 = 2\mathcal{A}_2 - 16. \quad (\text{DP}_3)$$

Thus the sixteen-term  $\mathbb{F}_{16}$ -line extremum and the six-term irreducible-quadratic extremum are the same constraint under constant-field base change, not two independent conditions.

More precisely, let

$$L_\chi(T) = \sum_{r=0}^{h-1} \mathcal{A}_r T^r = (1-T)L_\chi^*(T), \quad L_\chi^*(T) = \sum_{r=0}^d b_r T^r, \quad d = h-2,$$

so  $b_r = \sum_{j=0}^r \mathcal{A}_j$ . The functional equation has root number  $+1$  and gives

$$b_{d-r} = 2^{d-2r} \overline{b_r}. \quad (\text{DP}_4)$$

If  $D'_k$  fails, then

$$(b_0, b_1, b_2) = (1, 5, 21), \quad b_{d-2} = 21 \cdot 2^{h-6}. \quad (\text{DP}_5)$$

For

$$V = \langle 1, \zeta, \zeta^2 \rangle_{\mathbb{F}_4},$$

the distinguished trace annihilator is

$$V^\perp = \zeta \{f(\zeta) : f \in \mathbb{F}_4[X], \deg f \leq h-4\}. \quad (\text{DP}_6)$$

Consequently, in the bad case,

$$\sum_{x \in V^\perp} \chi(x) = 3b_{d-2} = 63 \cdot 2^{h-6}, \quad (\text{DP}_7)$$

exactly the value forced independently by additive Poisson summation and  $G_K(\chi) = 2^h$ . Thus the first two Dirichlet coefficients, the exact cyclotomic trace-dual placement, the semiprimitive Gauss sign, the functional equation, and the Riemann hypothesis are mutually compatible with failure.

Finally put

$$D = \zeta^4 + \zeta, \quad E = D^5(D^3 + 1).$$

The  $\mathbb{F}_4$ -subspace polynomial of  $V$  is

$$\begin{aligned} P_V(Z) &= Z^{64} + ((1 + D^3)^4 + E^3)Z^{16} \\ &\quad + (D^{12} + E^3(1 + D^3))Z^4 + E^3D^3Z. \end{aligned} \quad (\text{DP}_8)$$

Both  $D$  and  $E$  are  $\ell$ -th powers unconditionally. Hence the derivative coefficient  $E^3D^3$  supplies no missing nonresidue; a subspace-polynomial proof would have to use an additive relation among the  $Z^{16}$ - and  $Z^4$ -coefficients rather than merely products of the saturated elements.



*Proof.* The four monic linears  $X + c$ ,  $c \in \mathbb{F}_4$ , evaluate to  $\ell$ -th powers, so  $\mathcal{A}_1 = 4$ . Of the sixteen monic quadratics over  $\mathbb{F}_4$ , ten are reducible and therefore also have character value one. The six irreducibles are

$$f_c(X) = (X + c)(X + c^4),$$

indexed by the six Frobenius pairs in  $\mathbb{F}_{16} \setminus \mathbb{F}_4$ . Since

$$\tilde{\chi}(\zeta + c) = \chi(f_c(\zeta)),$$

the orbit table of Proposition 50 gives their six values exactly  $t^{A_j}$ ,  $0 \leq j < 6$ . This proves (DP<sub>1</sub>). The six summands have modulus one, so  $\mathcal{A}_2 = 16$  if and only if all are one; the  $j = 0$  term then gives  $t = 1$ , which is exactly failure of  $D'_k$ .

The reciprocal roots of  $L_\chi$  are squared under quadratic constant-field extension. Newton's identity therefore gives

$$\mathcal{A}'_1 = 2\mathcal{A}_2 - \mathcal{A}_1^2,$$

proving (DP<sub>3</sub>).

The character is primitive modulo  $X^h + \omega$  and even because it is trivial on  $\mathbb{F}_4^\times$ . Standard function-field Dirichlet theory gives  $L_\chi = (1 - T)L_\chi^*$ , with  $\deg L_\chi^* = d = h - 2$ , and the coefficient relation in (DP<sub>4</sub>) up to its root number. We determine that sign directly. Let  $U$  be the  $\mathbb{F}_4$ -space of polynomials in  $\zeta$  of degree at most  $h - 2$ . Newton sums for  $X^h + \omega$  give

$$\mathrm{Tr}(\zeta^r) = 0 \quad (1 \leq r < h),$$

so  $U^\perp = \mathbb{F}_4\zeta$ . Additive Poisson summation and  $\chi(\mathbb{F}_4^\times \zeta) = 1$  give

$$3b_d = \sum_{x \in U} \chi(x) = \frac{3}{4}G_K(\chi).$$

Proposition 51 gives  $G_K(\chi) = 2^h$ , so

$$b_d = 2^{h-2} = 2^d.$$

The root number is therefore  $+1$ . Under failure,  $b_2 = 1 + 4 + 16 = 21$ , and (DP<sub>4</sub>) at  $r = 2$  gives (DP<sub>5</sub>).

For  $x = \sum_{j=0}^{h-1} x_j \zeta^j$ , the same Newton-sum calculation gives

$$\mathrm{Tr}(x) = x_0, \quad \mathrm{Tr}(x\zeta) = \omega x_{h-1}, \quad \mathrm{Tr}(x\zeta^2) = \omega x_{h-2}.$$

This proves (DP<sub>6</sub>). Every nonzero polynomial of degree at most  $h - 4$  is a unique  $\mathbb{F}_4^\times$ -multiple of a monic one, while  $\chi(\zeta) = 1$ . Hence

$$\sum_{x \in V^\perp} \chi(x) = 3b_{h-4} = 3b_{d-2}.$$

Direct Poisson summation under  $V \setminus \{0\} \subset \ker \chi$  gives  $(63/64)G_K(\chi)$ , the same value  $63 \cdot 2^{h-6}$ . This proves (DP<sub>7</sub>).

It remains to derive the subspace polynomial. First put  $U_1 = \langle 1, \zeta \rangle_{\mathbb{F}_4}$ . Since  $\prod_{c \in \mathbb{F}_4} (Z + c) = Z^4 + Z$ ,

$$\begin{aligned} P_{U_1}(Z) &= (Z^4 + Z)^4 + D^3(Z^4 + Z) \\ &= Z^{16} + (1 + D^3)Z^4 + D^3Z. \end{aligned}$$

Evaluating at  $\zeta^2$  gives

$$E = P_{U_1}(\zeta^2) = D^8 + D^5 = D^5(D^3 + 1).$$

Because  $P_{U_1}$  is  $\mathbb{F}_4$ -linearized,

$$P_V(Z) = P_{U_1}(Z)^4 + E^3 P_{U_1}(Z),$$

which expands to  $(\text{DP}_8)$ .

Finally,

$$D = \prod_{c \in \mathbb{F}_4} (\zeta + c),$$

so  $\chi(D) = 1$ . Also  $E$  is the product of all sixteen monic-quadratic values. The ten reducible values have character one, and the other six multiply to

$$t^{\sum_{j=0}^5 A^j} = 1$$

because  $A$  has order six modulo  $\ell$ . Thus  $\chi(E) = 1$  without assuming failure, which finishes the proof. □

**Proposition 53** (relative Dickson form and its degree ceiling). *[proved] Retain*

$$s = 3^{k-1}, \quad a = 2^s, \quad E = \mathbb{F}_a, \quad K = \mathbb{F}_{a^6}, \quad \Lambda_k = a^2 - a + 1,$$

and set  $T_k = \{x \in K^\times : x^{\Lambda_k} = 1\}$ . The element  $P_k \in T_k$  has degree six over  $E$ . Put

$$Y_k = P_k + P_k^{-1} \in \mathbb{F}_{a^3}, \quad G_k(T) = \prod_{j=0}^2 (T + Y_k^{a^j}) \in E[T].$$

Then  $G_k$  is irreducible of degree three and  $X^3 G_k(X + X^{-1})$  is the irreducible reciprocal minimal polynomial of  $P_k$  over  $E$ . If  $\mathcal{D}_\ell$  is characterized by

$$\mathcal{D}_\ell(v + v^{-1}) = v^\ell + v^{-\ell},$$

then, for every prime  $\ell \mid \Psi_k$ ,

$$P_k \notin T_k^\ell \iff G_k(\mathcal{D}_\ell(X)) \text{ is irreducible over } E. \quad (\text{RD})$$

In the bad case the composition splits over  $\mathbb{F}_{a^3}$ , and all its irreducible factors over  $E$  have degree dividing three. Moreover

$$T_k \cap \mathbb{F}_{a^2}^\times = \mu_3, \quad T_k \cap \mathbb{F}_{a^3}^\times = \{1\}.$$

Thus the elements of  $T_k$  of relative degree below six are exactly  $\mu_3$ , while the bad subgroup  $T_k^\ell$  contains exactly

$$\frac{\Lambda_k}{\ell} - 3$$

elements of exact degree six. Except in the already settled case  $\Lambda_k/\ell = 3$ , full relative degree and reciprocal-torus structure cannot prove  $(D'_k)$ .

*Proof.* Proposition 47 gives  $[\mathbb{F}_2(P_k) : \mathbb{F}_2] = 6s$ , so  $E(P_k) = K$ . Since  $a^3 + 1 = (a + 1)\Lambda_k$ , one has  $P_k^{a^3} = P_k^{-1}$ , whence  $Y_k \in \mathbb{F}_{a^3}$ . The element  $P_k$  is quadratic over  $E(Y_k)$ ; its relative degree six forces  $[E(Y_k) : E] = 3$ , proving the minimal-polynomial assertions.

Suppose  $P_k \notin T_k^\ell$ . The whole current  $\ell$ -primary part of  $K^\times$  lies in  $T_k$ , so this is equivalent to  $P_k \notin (K^\times)^\ell$ . Kummer theory and Capelli make

$$X^{3\ell}G_k(X^\ell + X^{-\ell})$$

irreducible of degree  $6\ell$ . If  $\alpha^\ell = P_k$  and  $t = \alpha + \alpha^{-1}$ , then  $\mathcal{D}_\ell(t) = Y_k$ . Since  $\alpha$  is at most quadratic over  $E(t)$ , the degree of  $t$  is at least  $3\ell$ ; it is a root of the degree- $3\ell$  polynomial  $G_k(\mathcal{D}_\ell(X))$ , proving the forward implication in (RD).

Conversely, if  $P_k = \alpha^\ell$  with  $\alpha \in T_k$ , then  $\alpha^{a^3} = \alpha^{-1}$ , so  $t \in \mathbb{F}_{a^3}$ . The same holds after multiplying  $\alpha$  by any  $\ell$ -th root of unity and after applying the three  $E$ -conjugates. Hence every root of the Dickson composition lies in  $\mathbb{F}_{a^3}$ , proving the bad-case factorization statement and the reverse implication.

Finally the two displayed intersections have orders

$$\gcd(\Lambda_k, a^2 - 1) = 3, \quad \gcd(\Lambda_k, a^3 - 1) = 1.$$

Every proper intermediate field of  $K/E$  lies in  $\mathbb{F}_{a^2}$  or  $\mathbb{F}_{a^3}$ , so only  $\mu_3$  has relative degree below six. The cyclic group  $T_k^\ell$  has order  $\Lambda_k/\ell$  and contains  $\mu_3$ , because  $\ell \neq 3$ . Removing those three elements gives the asserted exact count.  $\square$

**Proposition 54** (trace-degree ceiling for the fibotomic criterion). *[proved] Retain Proposition 48, and let*

$$\mathcal{I}_\ell := \pi(U^\ell \setminus \{1\}) \subseteq \mathbb{F}_q \quad (\ell \text{ a prime divisor of } \Psi_k).$$

*Every element of  $\mathcal{I}_\ell$  has absolute trace one, and*

$$|\mathcal{I}_\ell| = \frac{(q+1)/\ell - 1}{2} \geq \frac{3a+2}{2} > a. \quad (\text{T})$$

*At least  $a/2 + 1$  elements of  $\mathcal{I}_\ell$  have exact degree  $h$  over  $\mathbb{F}_2$ . Consequently, absolute trace one together with full degree  $h$  cannot prove that the distinguished  $Z$  lies outside  $\pi(U^\ell)$ .*

*Proof.* The two-to-one trace-one parametrization in Proposition 48 restricts to  $U^\ell$ . This cyclic subgroup has order  $(q+1)/\ell$ , which is odd, so its only self-inverse element is 1. This proves the equality in (T). Since  $\ell \leq \Psi_k = \Lambda_k/3$  and  $q+1 = (a+1)\Lambda_k$ , it also gives

$$|\mathcal{I}_\ell| \geq \frac{3(a+1) - 1}{2} = \frac{3a+2}{2}.$$

Every proper subfield of  $\mathbb{F}_q$  lies in  $\mathbb{F}_{2^{h/3}} = \mathbb{F}_a$ , because  $h = 3^k$ . Hence at most  $a$  elements of  $\mathcal{I}_\ell$  have degree smaller than  $h$ , leaving at least  $(3a+2)/2 - a = a/2 + 1$  of exact degree  $h$ .

For completeness, the distinguished  $Z$  itself has exact degree  $h$ . Indeed,  $(D_3)$  and  $P_k \in \mathbb{F}_2(M_k)$  force  $[\mathbb{F}_2(M_k) : \mathbb{F}_2] = 2h$ . The element  $M_k$  is quadratic over  $\mathbb{F}_2(Z)$ , satisfying

$$X^2 + Z^{-1}X + 1 = 0,$$

while  $Z \in \mathbb{F}_q$ . Therefore  $[\mathbb{F}_2(Z) : \mathbb{F}_2] = h$ .  $\square$

**Proposition 55** (all proper-subfield norms are blind). *[proved] Retain  $k \geq 2$  and the notation of Proposition 46. Let  $\ell$  be a prime divisor of  $\Psi_k$ , and let  $d$  be any proper divisor of  $2h$ . Then*

$$N_{\mathbb{F}_{2^{2h}}/\mathbb{F}_{2^d}}$$

*annihilates the entire  $\ell$ -Sylow subgroup of  $\mathbb{F}_{2^{2h}}^\times$ . Equivalently, multiplying an element by any current  $\ell$ -primary component leaves every multiplicative norm to a proper subfield unchanged.*

*Proof.* The norm exponent is

$$E_d = \frac{2^{2h} - 1}{2^d - 1}.$$

Since  $\text{ord}_\ell(2) = 2h$  and  $d < 2h$ , one has  $\ell \nmid 2^d - 1$ . Therefore

$$v_\ell(E_d) = v_\ell(2^{2h} - 1),$$

so exponentiation by  $E_d$  kills the full  $\ell$ -Sylow subgroup. □

**Remark 9** (erratum: current-level versus cumulative  $D_k$ ). *[proved] One has*

$$\frac{2^{3^k} + 1}{3^{k+1}} = \prod_{j=1}^k \frac{\Phi_{2 \cdot 3^j}(2)}{3} = \prod_{j=1}^k \Psi_j.$$

The earlier version of this note called divisibility by the left-hand side  $D_k$  and claimed it equivalent to the current-level  $m = 4$  statement. It additionally requires every *old* exception factor to divide  $\text{ord}(M_k)$ , so it is strictly stronger unless an old-factor propagation theorem is supplied. Proposition 56 gives no such theorem. The exact conjecture is  $D'_k$  above.

For  $k = 1$ ,  $\Psi_1 = 1$  and  $D'_1$  is vacuous. **[certified  $k \leq 6$ ]**  $D'_k$  holds for  $k = 2, \dots, 6$ : every prime factor of the fully factored  $\Phi_{2 \cdot 3^k}(2)$  divides  $\text{ord}(M_k)$ . Hence  $m_p = 4$  exactly — not merely  $\geq 4$  — for every prime  $p$  with  $f(p) \in \{18, 54, 162, 486, 1458\}$ :

| $f$  | primes $p$ with $m_p = 4$                                                             |
|------|---------------------------------------------------------------------------------------|
| 18   | 19                                                                                    |
| 54   | 87211                                                                                 |
| 162  | 163, 135433, 272010961                                                                |
| 486  | 1459, 139483, 10429407431911334611,<br>918125051602568899753                          |
| 1458 | 227862073, 3110690934667, 216892513252489863991753,<br>1102099161075964924744009, P78 |

(P78 is the certified 78-digit prime of  $\Phi_{2 \cdot 3^6}(2)$ ; PRP-local here, factordb marks it proven.) The anchor rows  $m_{19} = m_{163} = m_{1459} = 4$  (DiMuro / calculator) are reproduced, never assumed; the remaining eleven rows are new. Independent cross-checks: the term algebra of **experiments/ordinal\_excess\_probe.py** re-derives the  $p = 19$  root pattern ( $m = 0, \dots, 3$  root,  $m = 4$  no root) and  $m_{87211} = 4$ ; an explicit Artin–Schreier compositum model  $\mathbb{F}[y]/(y^2 + y + \omega)$  verifies  $\sigma(4) = 5$ , the norm identity, and the full direct power test  $(\kappa + 4)^{(2^{4h}-1)/p} \neq 1$  on the smallest prime of each level  $k \leq 6$ .

**[consistent]**  $k = 7, 8$ : every *known* prime factor of  $\Phi_{2 \cdot 3^7}(2)$  and  $\Phi_{2 \cdot 3^8}(2)$  — seven and five primes respectively; factordb CF entries verified locally by divisibility, primality, and order tests,

the small ones re-derived by sieve, and the list cross-audited against the raw factordb API after an adversarial review caught a dropped 43-digit  $k = 7$  prime — also has  $m_p = 4$ . An  $m_p \geq 5$  example, if one exists, now hides strictly inside the unfactored cofactors; their prime multiplicities are likewise unknown until factorization.

**Proposition 56** (twisted norm tower). *[proved] With  $\eta = \zeta^3 = \kappa_{3^{k-1}}$ ,*

$$N_{\mathbb{F}_{2^{2 \cdot 3^k}}/\mathbb{F}_{2^{2 \cdot 3^{k-1}}}}(N_k) = \eta^2 + \omega^2 \eta + 1,$$

*which is neither  $N_{k-1} = \eta^2 + \eta + \omega$ , nor one of its Frobenius conjugates (their exponent pattern is  $(2^{j+1}, 2^j)$ ), nor a scaling  $N_{k-1}(\lambda\eta)$  with  $\lambda \in \mu_3$ .*

*Proof.* The generator  $\tau$  of the degree-3 step sends  $\zeta \mapsto \zeta^c$  with  $c = 2^{2 \cdot 3^{k-1}}$  a primitive cube root of unity modulo  $3^{k+1}$ ; the cube roots are  $1 + a3^k$ , so  $\tau(\zeta) = \omega^a \zeta$  ( $a \neq 0$ ) while  $\tau(\omega) = \omega$ . Hence the norm is  $\prod_{r^3=\eta} (r^2 + r + \omega)$  over the three cube roots  $r$  of  $\eta$ . Writing  $\alpha, \beta$  for the two roots of  $T^2 + T + \omega$  (so  $\alpha + \beta = 1$ ,  $\alpha\beta = \omega$ ), each factor is  $(r + \alpha)(r + \beta)$ , and the product collapses by the resolvent

$$\prod_{r^3=\eta} (r^2 + r + \omega) = (\alpha^3 + \eta)(\beta^3 + \eta) = \eta^2 + (\alpha^3 + \beta^3)\eta + (\alpha\beta)^3 = \eta^2 + \omega^2 \eta + 1,$$

since  $\alpha^3 + \beta^3 = (\alpha + \beta)^3 + \alpha\beta(\alpha + \beta) = 1 + \omega = \omega^2$  and  $(\alpha\beta)^3 = \omega^3 = 1$ . Machine-checked at  $k = 2, 3, 4$ .  $\square$

**Proposition 57** (trace form and norm blindness). *[proved] Retain the notation of Proposition 46, and put*

$$\gamma_k := \zeta + \zeta^{-1}, \quad A_k := \gamma_k^2 + \gamma_k + 1, \quad Q := 2^{2 \cdot 3^{k-1}} = 2^{2h/3}.$$

*Set  $\gamma_0 = A_0 = 1$ . Then, with  $\overline{N} = N^{2^h}$ ,*

$$N + \overline{N} = A_k, \quad N\overline{N} = A_k^Q, \quad M_k + M_k^{-1} = A_k^{2-Q}, \quad Z = A_k^{Q-2}. \quad (4)$$

*Moreover*

$$N_{\mathbb{F}_{2^{3k}}/\mathbb{F}_{2^{3^{k-1}}}}(A_k) = A_{k-1}, \quad N_{\mathbb{F}_{2^{3k}}/\mathbb{F}_{2^{3^{k-1}}}}(Z) = A_{k-1}^{-1}. \quad (5)$$

*For every prime  $\ell \mid \Psi_k$ , however, this relative norm is blind to the required  $\ell$ -power residue class.*

*Proof.* Since  $2^h \equiv -1 \pmod{3h}$  and  $\zeta^h + \zeta^{2h} = 1$ , direct expansion gives

$$N + \overline{N} = \zeta^2 + \zeta^{-2} + \zeta + \zeta^{-1} + 1 = A_k.$$

The congruence  $Q \equiv h + 1 \pmod{3h}$  follows inductively by cubing  $1 + 3^r$  modulo  $3^{r+2}$ . Hence  $\gamma_k^Q = \zeta^{h+1} + \zeta^{-(h+1)}$ , and another direct expansion yields  $N\overline{N} = A_k^Q$ . Therefore

$$M_k + M_k^{-1} = \frac{(N + \overline{N})^2}{N\overline{N}} = A_k^{2-Q},$$

and  $Z = \pi(M_k) = 1/(M_k + M_k^{-1})$ , which proves (4).

Over  $B = \mathbb{F}_{2^{h/3}}$ , the element  $\gamma_k$  has minimal polynomial  $X^3 + X + \gamma_{k-1}$ . Base-change this cubic extension by  $B_2 = B\mathbb{F}_4 = \mathbb{F}_{2^{2h/3}}$ ; its compositum is  $\mathbb{F}_{2^{2h}}$ , and the degree remains three. With  $\omega = \zeta^h \in \mathbb{F}_4$  one has  $A_k = (\gamma_k + \omega)(\gamma_k + \omega^2)$ . Evaluating that minimal polynomial at  $\omega$  and  $\omega^2$  gives

$$N_{\mathbb{F}_{2^{2h}}/B_2}(\gamma_k + \omega) = \gamma_{k-1} + \omega^2, \quad N_{\mathbb{F}_{2^{2h}}/B_2}(\gamma_k + \omega^2) = \gamma_{k-1} + \omega.$$

Their product is  $A_{k-1}$ . Since norm commutes with this scalar extension, this is precisely  $N_{\mathbb{F}_{2^h}/B}(A_k)$ , proving the first identity in (5). Furthermore  $Z = A_k^{Q-2}$ , so

$$N_{\mathbb{F}_{2^h}/B}(Z) = A_{k-1}^{Q-2} = A_{k-1}^{-1},$$

because  $Q - 2 \equiv -1 \pmod{2^{h/3} - 1}$ . This proves the second identity.

If  $\ell \mid \Psi_k$ , then  $\ell \nmid 2^h - 1$  and  $\ell \nmid 2^{h/3} - 1$ . Thus both  $A_k$  and  $A_{k-1}$  lie in multiplicative groups on which raising to the  $\ell$ -th power is an automorphism, so (5) is tautological on  $\ell$ -power classes. Equivalently,  $\text{ord}_\ell(Q) = 3$  and  $1 + Q + Q^2 \equiv 0 \pmod{\ell}$ : the degree-three norm annihilates the relevant character. The surviving current-level condition is the Capelli test  $R_k(X^\ell)$  irreducible from Proposition 46.  $\square$

So, unlike the  $\gamma$  tower of Proposition 13, old-prime full-order parts do *not propagate by that argument*, and  $D'_k$  is genuinely per-level for the present method: the certification at each  $k$  stands on its own. Structurally, at the degree-3 step with  $Q = 2^{2 \cdot 3^{k-1}}$ , the relative norm kernel has order

$$Q^2 + Q + 1 = \Phi_{3^k}(2)\Phi_{2 \cdot 3^k}(2).$$

Its gcd with  $Q - 1$  is 3, so there is no direct-product splitting at the 3-part as in Proposition 14; nevertheless, for every  $\ell \neq 3$  the new  $\ell$ -Sylow subgroups of both columns lie entirely in the norm kernel. This is the prime-to-3 analogue of Proposition 14 and explains why a lower-level norm identity cannot determine  $D'_k$ .

**Remark 10** (the 3-part). **[proved]** Proposition 47 and  $v_3(a + 1) = k$  imply

$$v_3(\text{ord } M_k) = k + 1 \quad (k \text{ odd}), \quad v_3(\text{ord } M_k) \leq k \quad (k \text{ even}).$$

Thus the odd-level full 3-part is now explained analytically. **[consistent]** At even levels  $2 \leq k \leq 6$ , the sharper observed value is  $v_3(\text{ord } M_k) = k - 1$ ; the remaining one-step deficiency is not determined by the projection theorem.

## 5 Boundedness

**[conjectured]** (Lenstra [11])  $m_p$  is globally bounded. Lenstra gave unconditional lower-bound rules (singleton odd  $Q$  forces positive excess;  $f(p) = 2 \cdot 3^k$  forces excess at least 4) but left absolute boundedness open.

**[proved]** If the 0/1/4 candidate holds, then  $m_p \leq 4$  for all  $p$ ; proving Conjecture 2 would settle boundedness.

**[proved]** No proof for arbitrary finite-field elements can come from a fixed initial translate window: Proposition 7 constructs elements of arbitrarily large exact degree for which every translate in any prescribed finite window is a  $p$ -th power. A boundedness proof must therefore use the distinguished Conway-tower identities.

**[proved]** The 3-power column cannot produce  $m \in \{2, 3\}$  (Proposition 12); any  $C_k$  failure jumps directly to  $m_r \geq 4$ .

**[proved]** (conditional on  $C_k$ ) For every prime  $r$  with  $f(r) = 3^j$ ,  $j \leq k$ :  $C_k$  implies  $m_r = 1$ . With  $C_k$  certified for  $k \leq 6$ , all primes in the 3-power column with  $f(r) \leq 6561$  have  $m_r = 1$  unconditionally.

**[certified  $k \leq 6$ ]** (June 2026 pass) On the  $2 \cdot 3^k$  exception column the matching upper bound is now certified at every visible prime:  $m_p = 4$  exactly for every prime with  $f(p) = 2 \cdot 3^k$ ,  $k \leq 6$ , and

for every known prime at  $k = 7, 8$  (Section 4.3). Boundedness on the exception column therefore holds unconditionally through  $f = 1458$  and is equivalent at each new level to  $D'_k$  (Conjecture 3) beyond.

**[open]** Boundedness outside the 3-power and  $2 \cdot 3^k$  columns. The non-cyclotomic singleton chains (e.g. the 11-chain and the 23, 29, 47 components) have  $m = 1$  rows in the calculator data and OEIS, but no order-criterion proof covers them the way  $C_k$  covers the 3-power column.

**[open]** Whether any prime  $p$  has  $m_p \geq 5$ . No counterexample is known; the question is not known to be decidable by any current structural argument outside the 3-adic columns.

**[proved]** A proof of the full 0/1/4 candidate would in particular prove both the Conway–Fermat maximality statement  $\delta_n = F_n$  at every quadratic level (Proposition 30) and the Gaussian-period primitivity statement  $C_k$  at every cubic level (Proposition 43). These are independent new-factor problems; neither follows from the existing norm recursions.

## 6 Ranked next moves

1. *Conway–Fermat quotient lemma (highest analytical priority)*. Prove or refute  $\delta_n = F_n$  in Proposition 30. Proposition 31 reduces it to the explicit nonvanishing tests  $S_{F_n/\ell}(a_{n-1}) \neq 0$ , while Proposition 32 proves that degree, trace, norm, and the top Artin–Schreier equation cannot suffice. The missing step is to show that the resultant recursion of Proposition 33 always selects the full-order fibotomic stratum. Proposition 34 proves that predecessor maximality and the exact one-step operator are not enough: an alternative full-order  $A_3$ -factor maps to the proper  $F_5/641$ -stratum. Proposition 35 now gives a necessary-and-sufficient trace/norm ladder for the distinguished factor’s entire earlier ancestry. It also proves that every actual trace fibre contains many normal full-degree  $\ell$ -th powers, and that imposing the actual relative norm compresses exactly to the original test  $S_{(q+1)/\ell}(a_{n-1}) = 0$ . Thus the missing input is not another coarse ancestry invariant. Equivalently, Proposition 37 asks for the nontrivial Kummer symbol of  $c_n$  at the distinguished point of the supersingular correspondence; its quarter-turn Frobenius symmetry is proved character-neutral and, by Proposition 39, does not lift to the ramified Kummer cover. Proposition 40 places the surviving Miller-unit class in the toric part of the three-point generalized Jacobian. Proposition 41 exhausts the extension class and non-rational elliptic torsion, while Proposition 42 gives both its exact Rosenlicht/Miller coordinate and a rational five-torsion splitting. Their evident reciprocity collapse is proved circular. The next step is therefore to prove that this now-explicit distinguished value of  $x + 2y$  is nonzero, not to seek another coordinate model or ordinary elliptic pairing.
2. *New-prime lemma for the 3-power spine*. For every  $\ell^a \mid \Phi_{3^k}(2)$ , prove  $v_\ell(\text{ord } \gamma_k) = a$ . The norm tower already carries all old factors, so this single per-level statement is exactly  $C_k$ . Equivalently, prove that the distinguished root  $\eta_k$  of  $X^3 + \eta_{k-1}X^2 + (\eta_{k-1} + 1)X + 1$  generates the norm-one torus  $U_k$ . The sharp analytical languages are therefore either the  $\ell$ -power residue symbol of the cyclotomic unit  $1 + \zeta$  or the character theory of this explicit cubic trace/norm fibre, not another order census. Proposition 23 removes every prime-index subgroup below its partition bound, while Proposition 25 identifies failure with the extremal value of one short cyclotomic period; a completion must control the remaining characters. Proposition 26 shows that failure saturates all cyclotomic secants, while the associated Kummer cover is rational; the missing input is therefore the exact Artin value of the distinguished cyclotomic unit rather than a generic Weil bound. Proposition 20 identifies the conditional Galois obstruction as a locally admissible  $C_\ell \rtimes_2 C_{3^k}$  ray extension, so local reciprocity alone cannot exclude it. Proposition 19 sharpens this to an exact exceptional-zero statement: norm operators, the plus Stickelberger element, and  $2 - \sigma$  all

vanish on the one-dimensional 2-eigenspace, while Hilbert reciprocity multiplies the complete conjugate orbit of the desired local symbol to an automatic identity. The next arithmetic object is therefore a first-order unit regulator in that simple exceptional eigenspace.

3. *General transverse reciprocity.* Evaluate the power-residue symbol of  $\Theta_{q,s}$  from Proposition 6. A theorem forcing its full exact-order support would prove the singleton-one arm simultaneously for the 11-chain and the 23, 29, 47, 179, ... components. The easy Kummer norm is provably the wrong object.
4. *Exception-column current-factor lemma.* Prove  $D'_k$ , equivalently prove  $R_k(X^\ell)$  irreducible for every  $\ell \mid \Psi_k$  in Proposition 46, or prove the polynomial in Proposition 48 rootless for its distinguished  $Z$ . Proposition 47 already forces one current prime and every factor above an explicit threshold. The twisted norm formula in Proposition 56 and the norm blindness in Proposition 57 rule out direct induction on  $k$ ; a residue-symbol argument for  $N_k = \zeta^2 + \zeta + \omega$  must account for the twist. Proposition 49 and Proposition 53 give equivalent explicit translated-binomial and relative Dickson irreducibility tests, and prove that full relative degree is still too weak. Proposition 50 further shows that failure saturates the complete  $\mathbb{F}_{16}$ -affine line; a completion must make its explicit sixteen-term character sum non-extremal by information beyond Frobenius and base-field translation. Proposition 51 strengthens the bad case to complete six- and eight-dimensional additive spaces and proves that the exact semiprimitive Gauss–Jacobi/Fourier incidence table permits them. Generic semiprimitive estimates therefore cannot supply the missing identity. Proposition 52 uses the distinguished cyclotomic placement and identifies failure with the extremal second Dirichlet coefficient  $\mathcal{A}_2 = 16$ . Its functional equation, trace dual, root number, constant-field base change, and multiplicative subspace-polynomial coefficient all reproduce identities compatible with that extremum. A completion must evaluate this explicit six-term period non-extremally or introduce an independent relation involving higher Dirichlet coefficients.
5. *Numerical work only as falsification and audit.* The  $m_{359}/m_{719}$  transverse norm, complete factorizations at  $k = 7, 8$ , and exact-order censuses remain useful counterexample searches and checks of any proposed lemma, but they no longer lead the proof program.

## 7 Provenance and validation

### 7.1 Independent oracle

`experiments/ordinal_excess_probe.py` is a small local term-algebra oracle – not a replacement for CGSuite [17] or the C++ calculator [18], but a way to verify the first subtle cases without using the Rust production tower as its own oracle. It implements the impartial term algebra used by the calculators, a multiplicative-order test for small component fields, and a fixed-base exponentiation path (ported from the calculator’s strategy) for targeted root tests where full order factorization is unnecessary. Current probe output:



| $p$ | $m$ | $Q$  | root? |          |
|-----|-----|------|-------|----------|
| 7   | 0   | {3}  | yes   |          |
| 7   | 1   | {3}  | no    |          |
| 19  | 1   | {9}  | yes   |          |
| 19  | 4   | {9}  | no    |          |
| 73  | 1   | {9}  | no    |          |
| 89  | 1   | {11} | no    |          |
| 47  | 1   | {23} | no    |          |
| 179 | 1   | {89} | no    | (--deep) |

The  $p = 47$  line certifies  $m_{47} = 1$  using only lower verified rows; since  $f(47) = 23$  and  $Q(23) = \{23\}$ , the carry is  $\alpha_{47} = \kappa_{23} + 1 = \omega^{(\omega^7)} + 1$ , the value now implemented in `tower.rs`. The  $p = 179$  line is the first locally certified dependency row beyond the quick probe path; it still uses only the already-certified lower row  $m_{89} = 1$ .

## 7.2 The $\alpha_{47}$ promotion

*Superseded 2026-06-13:* this pass moved the boundary to  $\alpha_{53}$  by promoting a single locally verified row; the finite excess table is now the full OEIS A380496 b-file (126 rows, odd primes  $3 \leq p \leq 709$ ), so the refusal boundary sits at  $\alpha_{719}$  and the boundary test is `boundary_returns_none_past_prime_709`. The original pass is recorded below for provenance.

The shipped state at the time: `src/scalar/big/ordinal/tower.rs` carries  $\alpha_{47}$  with the landmark test `locally_verified_alpha_47_landmark` and moves the refusal boundary to  $\alpha_{53}$ ; the documented boundaries in `nim.rs`, `mod.rs`, `src/games/nimber_game.rs`, `README.md`, and `docs/OPEN.md` were updated in step. The full gate ran clean: `cargo fmt --check`, `cargo test`, `cargo check/clippy` with and without the `python` feature (warnings denied), and the probe under `py_compile`.

## 7.3 Supporting probes

The  $3^k$ -family certification chain is `experiments/cyclotomic_3k_family.py` (committed, maintained), joined 2026-06-12 by `experiments/exception_column_m4.py` (the  $2 \cdot 3^k$  exception column, Section 4.3; committed, maintained, stdlib-only,  $\sim 2$  minutes), with the research-run probes under `experiments/excess/` (machine-generated, rescued from ephemeral storage; triage before citing). The analysis in Sections 2–5 synthesizes the 2026-06-10 parallel research run plus the 2026-06-12 exception-column pass; each claim above carries its own tag, and no claim depends on an unverified solver.

## Acknowledgements

The `ogdoad` library, the experiments, and this draft were developed with LLM assistance; the analysis consolidates a structured parallel research run (2026-06-10) and an independent adversarial proof audit (2026-07-20). Mathematical direction, framing, and responsibility for claims remain the author’s.

## References

- [1] O. Ahmadi, I. E. Shparlinski, J. F. Voloch, *Multiplicative order of Gauss periods*, Int. J. Number Theory **6** (2010), 877–882; arXiv:0707.4034.
- [2] F. Breuer, *Multiplicative orders of Gauss periods and the arithmetic of real quadratic fields*, Finite Fields Appl. **73** (2021), 101848.
- [3] L. Carlitz, *Distribution of primitive roots in a finite field*, Quart. J. Math. Oxford (2) **4** (1953), 4–10.
- [4] L. Cagliero, A. Herman, and F. Szechtman, *Artin–Schreier towers of finite fields*, Finite Fields Appl. **106** (2025), 102606; arXiv:2405.10159.
- [5] J. H. Conway, *On Numbers and Games*, Academic Press, 1976.
- [6] L. De Feo, H. Randriam, É. Rousseau, *Standard lattices of compatibly embedded finite fields*, Proc. ISSAC 2019; arXiv:1906.00870.
- [7] J. DiMuro, *On  $\text{On}_p$* , arXiv:1108.0962.
- [8] S. Gao, S. A. Vanstone, *On orders of optimal basis generators*, Math. Comp. **64** (1995), 1227–1233.
- [9] K. Iwasawa, *On explicit formulas for the norm residue symbol*, J. Math. Soc. Japan **20** (1968), 151–165.
- [10] E. Kaltofen, V. Shoup, *Subquadratic-time factoring of polynomials over finite fields*, Math. Comp. **67** (1998), 1179–1197.
- [11] H. W. Lenstra, Jr., *On the algebraic closure of two*, Indag. Math. **39** (1977), 389–396; <https://pub.math.leidenuniv.nl/~lenstrahw/PUBLICATIONS/1977e/art.pdf>.
- [12] H. W. Lenstra, Jr., *Nim multiplication*, Séminaire de Théorie des Nombres de Bordeaux, 1977–1978.
- [13] R. C. Mullin, A. Mahalanobis, *Dickson bases and finite fields*, CACR Technical Report 2005–03, University of Waterloo, 2005.
- [14] T. Mészáros, L. Rónyai, T. Szabó, *Singer difference sets and the projective norm graph*, arXiv:1908.05591 (2019).
- [15] G. Meinardus, *Asymptotische Aussagen über Partitionen*, Math. Z. **59** (1954), 388–398.
- [16] OEIS Foundation Inc., entry A380496, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A380496>.
- [17] A. N. Siegel, *CGSuite* (software), NimFieldCalculator, <https://github.com/aaron-siegel/cgsuite>.
- [18] D. Peeters, *transfinite-nim-calculator* (software), <https://github.com/DjangoPeeters/transfinite-nim-calculator>.
- [19] R. Popovych, *Multiplicative orders of elements in Conway’s towers of finite fields*, Algebra Discrete Math. **25** (2018), 137–146; arXiv:1509.01958.

- [20] J. von zur Gathen, V. Shoup, *Computing Frobenius maps and factoring polynomials*, Comput. Complexity **2** (1992), 187–224.
- [21] D. Wiedemann, *An iterated quadratic extension of  $GF(2)$* , Fibonacci Quart. **26** (1988), 290–295.
- [22] L. C. Washington, *Introduction to Cyclotomic Fields*, 2nd ed., Graduate Texts in Mathematics 83, Springer, 1997.